

Section C - Description/Specifications/Statement of Work

GENERAL SPECIFICATIONS

HOPEWELL LAKE SITE IMPROVEMENTS

1.1 SCOPE OF CONTRACT

- A. This project includes construction the following site improvements at Hopewell Lake:
 - 1. New Site Entrance Roundabout
 - 2. New Campground Entrance Road
 - 3. New Trailhead Parking Lot
 - 4. Modified Parking Lot at Group Area
 - 5. New Waterline and Hydrants for site
 - 6. New Campground Host Site
 - 7. New Site Furnishings
- B. The work includes clearing and grubbing of the areas to receive the new roads and parking areas, installation of new bulletin boards, construction of aggregate surfaced roadways and other pads, and installation of signs, barrier rocks, curb stops, culvert and other work as described in the plans and specifications.
- C. Quantities associated with these options are identified in the Schedule of Items.

1.2 PROJECT LOCATION

- A. The project is located on the Tres Piedras Ranger District of the Carson National Forest, Rio Arriba County, New Mexico. It is located approximately 19.8 miles Northeast of Tres Piedras, New Mexico. The project may be accessed from State Highway 64.

1.3 SITE INFORMATION AND LIMITATIONS

- A. The following site conditions are considered incidental to the contract and the contractor will not be paid directly for any of the following items:
 - 1. Construction sites will be closed to the public during construction. The Forest Service will issue a closure order to the public. The Contractor will be responsible for signing and limiting public access.
 - 2. Contractor will provide all materials and labor necessary to protect streams, wet or dry, during construction activities. The Contractor will be required to submit an erosion control plan and receive approval from the COR, in writing, prior to the beginning of construction.
 - 3. All construction equipment shall be pressure washed before entering National Forest System lands. The removal of mud and debris from treads, tracks and undercarriage, with emphasis on axles, frame, cross-members, motor mounts, and underneath steps, running boards, and front bumper/brushguard assemblies will be required. The purpose is to reduce or eliminate the transportation of noxious weeds, which is required by Federal and State regulations.
 - 4. Utilities are not available at the site for construction purposes.
 - 5. The Contractor shall provide temporary toilet facilities (porta-potty) at the site

during all construction work.

6. The Contractor shall locate all existing utilities at the site prior to any ground disturbing activities.
7. The Contractor is responsible for sourcing construction materials for aggregate, rock and riprap materials that blend with surrounding landscape (colors, and shape) and meet contract requirements. The contractor shall certify in submittal that this responsibility is being met. Contracting Officer will consult with Forest Recreation Staff to ensure that materials will blend with surrounding landscape.

1.4 TRAFFIC CONTROL AND CONSTRUCTION SIGNING

- A. No work that endangers, interferes, or conflicts with traffic or access to work sites shall be performed until a plan for satisfactory warning and handling of traffic has been submitted by the contractor and approved by the COR and **New Mexico** Department of Transportation. Obtain necessary permits for roadway activities along New Mexico state highways from the New Mexico Department of Transportation. Construction signing for traffic control shall conform to the Manual of Uniform Traffic Control Devices (MUTCD). All traffic control signs will be placed in areas adequate for a truck pulling a fifth wheel trailer to be turned around. Contractor shall not be paid directly for this item, rather it will be considered incidental to other items of work listed in the Schedule of Items.
- B. Traffic Control Plan shall include temporary signage and other features to re-route pedestrian traffic using the Continental Divide Trail. Refer to drawings for location of the trail and coordinate re-routes with the Contracting Officer prior to submission of the proposed re-routes.

1.5 DESIGN REFERENCES

- A. This project was designed in accordance with the following documents (the contractor shall have a minimum one (1) copy of the document onsite to reference as needed):
 1. Forest Service Outdoor Recreation Accessibility Guidelines (FSORAG).
https://www.fs.usda.gov/recreation/programs/accessibility/FSORAG_2013%20Update.1.pdf
 2. Forest Service Trail Accessibility Guidelines (FSTAG).
https://www.fs.usda.gov/recreation/programs/accessibility/FSTAG_2013%20Update.pdf
 3. Architectural Barriers Act (ABA). <https://www.access-board.gov/aba/#aba-502>
- B. Contractor is required to comply with these documents during construction. Any field adjustment proposed by the Contractor shall be in accordance with these documents and approved by the Contracting Officer.

1.6 WORK CAMPS, STAGING AND STORAGE AREAS

- A. Areas for staging operations and storage of materials shall be approved by the CO. The Contractor must request in writing for approval from the CO to stage trailers (work or housing) on site.

1.7 INSPECTION OF WORKSITE

- A. The contractor acknowledges they have taken the necessary steps to ascertain the nature and location of work and have investigated and satisfied themselves as to the general and local conditions that can affect the work or its cost. Any failure of the contractor to take the actions described and acknowledged in this paragraph will not relieve the Contractor from the responsibility of estimating properly the difficulty and cost of successfully performing the work, or for proceeding to successfully perform the work without additional expenses to the government.

1.8 START DATE

- A. Summer 2026

1.9 CONTRACT TIME

- A. Base Bid: 365 Calendar Days
- B. Option Bid: No additional days, to be incorporated into Base Bid Calendar Days

1.10 SPECIFICATIONS

- A. The following specifications are attached. Some sections in the schedule of items refer to other sections not listed and are subsidiary to or are included in payment for other pay items in this contract. These items are considered incidental, and no additional compensation will be made.

010250	CONSTRUCTION STAKING
011250	MEASUREMENT AND PAYMENT
011900	MOBILIZATION
013300	SUBMITTAL PROCEDURES
014100	QUALITY CONTROL
023701	SEDIMENT AND EROSION CONTROL MEASURES
024100	WASTE MATERIAL DISPOSAL
024102	REMOVAL AND DISPOSAL OF EXISTING STRUCTURES
033000	CAST IN PLACE CONCRETE
033020	CONCRETE FROM PACKAGED DRY MIX FOR MINOR STRUCTURES
101110	BULLETIN BOARD
101400	SIGNS
101415	MINOR SITE ID SIGN
129300	SITE FURNISHINGS
129700	FEE TUBE
133402	UTILITY BOXES
221100	WATER DISTRIBUTION SYSTEMS
221102	CURB VALVES AND BOXES
221103	PLUMBING APPURTENANCES
221105	WATER HYDRANTS
221300	ONSITE WASTEWATER TREATMENT SYSTEMS

260500	COMMON WORK RESULTS FOR ELECTRICAL
260519	LOW VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES
260526	GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS
260533	RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS
260553	IDENTIFICATION FOR ELECTRICAL SYSTEMS
262416	OTHER ELECTRICAL TESTS AND CALIBRATIONS
262716	WALK-IN ENCLOSURE
311000	CLEARING AND GRUBBING
312000	EARTHWORK
312100	PROJECT SITE PREPARATION AND GRADING
312225	EXCAVATION AND EMBANKMENT
321204	CRUSHED AGGREGATE BASE OR SURFACE COURSE
321206	LOOSE ROCK RIPRAP
321460	TRAILS AND PATHS
322100	FAMILY UNIT AND TENT PAD
322205	BARRIER ROCK
322622	PRECAST CONCRETE CURB STOPS
323116	GATES FOR ROADS
329220	SEEDING
332228	CULVERT PIPE AND PIPE ARCHES

END OF SECTION C

April 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 010250 - CONSTRUCTION STAKING

PART 1 - GENERAL

1.1 SUMMARY

- A. This work shall consist of the construction staking of roads, parking lots, parking spurs, and culverts in accordance with the Drawings and Specifications. The work includes the furnishing of all labor, equipment, instruments, materials, transportation, and other incidentals necessary to complete the construction staking in accordance with these specifications and acceptable engineering practice.

1.2 QUALITY CONTROL

- A. Construction staking shall be accomplished under the direction of a Civil Engineer, Engineering Technician, or Land Surveyor closely associated and familiar with construction staking; periodic visits to the project site are required.

1.3 MEASUREMENT AND PAYMENT

- A. Measurement shall be lump sum for staking the project through all phases of construction.

PART 2 - PRODUCTS

2.1 GOVERNMENT-FURNISHED SURVEY CONTROL SHEETS

- A. The Government will provide the contractor with a set of survey control sheets showing horizontal and vertical control for site to be staked. The Government will provide the Contractor with layout information that will include:
 - 1. Control Points - Northing, Easting, Existing Elevations.
 - 2. Roads, Parking Areas - Northing, Easting, and Finish Elevations for subgrade centerline and edge of roads, and parking areas.

2.2 STAKES

- A. Identification stakes and hubs shall be of sufficient length and width to provide and solid set in the ground and to provide space for marking above ground when applicable. Other dimensions and materials may be used, such as steel reinforcing bars, wire flagging and markers, and metal pins, if approved in writing by the Contracting Officer (CO). The top 2 inches of all slope, guard, reference, clearing, and structure stakes shall be painted or marked with plastic flagging. Colors used on stakes or for flagging shall be as follows:

- | | |
|---|---------------------|
| 1. Roads, parking lots, and parking spurs | -Orange |
| 2. Culverts | -Blue/White Striped |
| 3. Waterlines and Other Utilities | -Blue |
| 4. Family Unit Pads and Paths | -Red |

2.3 SURVEY NOTE PAPER AND BOOKS

- A. Paper for survey notes shall be moisture-resistant paper. Notes shall be contained in books with covers that will protect the contents and retain the pages in numerical sequence during field use. Field notebooks or note paper shall be furnished by the contractor.

PART 3 - EXECUTION

3.1 GENERAL

- A. The Contractor shall be required to provide all necessary staking and control for all phases of construction. The contractor shall use the data and information shown on the project drawings to establish the location of all facilities to be constructed under this contract. The Contractor may choose any method of establishing horizontal and vertical controls provided the following minimum precision and accuracy requirements are met.
- B. Accuracy requirements:
1. Minimum position close; 1/1000
 2. Vertical closure: 0.05 ft (vertical)/station (100 feet horizontal)
 3. Cross section and slope stake precision:
- C. Allowable deviation of cross section line projection from a true perpendicular to tangents, a true bisector of angle points, or a true radius of curves: +/-3 degrees.
- D. Cross section topography measurements shall be taken so that variations in ground form a straight line connecting the cross section points will not exceed: 1.0 ft.
- E. Horizontal and vertical accuracy for slope stake, slope stake references, and clearing limits. In feet or percentage of horizontal distance measured from centerline or reference stake, whichever is greater. Slope reference stakes and slope stakes: 0.15 ft or 0.6%. Clearing limits: 1.0 ft.

3.2 ESTABLISHING AND REFERENCING IMPROVEMENT LOCATIONS

- A. Access Roads, Parking Spurs, and Parking lots:
1. Slope stakes (cut and fill catch points), clearing limits, and slope stake references shall be established on both sides of the centerline at each station established. Slope stakes and slope stake references shall be located on a line at right angles to tangents and on radial lines or curves. Slope stake catch points shall be located by using the grading plans and/or profiles, typical road section templates, as shown on the drawings and the information provided with staking points.

2. Slope stakes shall be set at 50-foot intervals, at significant breaks in the ground profile, culvert locations, and widening transition points. Slope stakes shall be set at 25-foot intervals around curves having a radius less than 100 feet.

B. Group Units, Family Units and Tent Pads

1. Slope stakes shall be established at the corners of family units. Slope stakes shall be written to reference the sub-grade grade of the unit or pad being staked.

C. Paths and Trails

1. Slope stakes (cut and fill catch points), clearing limits, and slope stake references shall be established on both sides of the centerline at each station established. Slope stakes and slope stake references shall be located on a line at right angles to tangents and on radial lines or curves. Slope stake catch points shall be located by using the grading plans and/or profiles as shown on the drawings.
2. Slope stakes shall be set at 25-foot intervals, at significant breaks in the ground profile, culvert locations, and widening transition points.

D. Culverts

1. Slope stakes and slope reference stakes shall be set at all culvert locations. A culvert reference stake and hub shall be set on the centerline of the culvert 10 feet from each end or beyond the clearing limit, whichever is greater. The following shall be recorded on these stakes:
 - a. The actual field measured length
 - b. The vertical and horizontal distance from hubs to the invert at the ends of the culvert.

E. Waterline, Sewerline, & other Utilities

1. Waterline, Sewerline, & other Utilities shall be staked at the locations shown on the drawings with the specified minimum burial depth. Waterline and sewerline elevations shall be accurately staked to provide for drainage as indicated on the drawings. Grade stakes shall be established as necessary to provide the control for the construction work.
 - a. Where shown on the drawings, local LIDAR survey was used to design waterline alignment elevations. The contractor shall verify the elevations provided in the drawings, and adjust in the field to alignment, grade and elevation in coordination with the Contracting Officer. Work shall be considered incidental to construction stakeout.

3.3 DISCREPANCIES

- A. The contractor shall compare the staked centerline horizontal and vertical alignment with the design data. Differences between previously recorded and observed elevations of benchmarks shall be referred to the Contracting Officer. Differences in centerline profile elevations exceeding 1 foot at any two or more consecutive points shall be

reported to the contracting Officer for evaluation and possible revision. Staking of these areas shall be deferred until the Contracting Officer resolves these differences.

3.4 VERTICAL CONTROL

- A. Vertical control for construction shall be as referenced on the Drawings.

3.5 MARKING STAKES

- A. All stakes shall be legibly marked, in the format agreed upon with the Contracting Officer, with a stake pencil that leaves an imprinted or with waterproof ink. Marking shall conform to the nomenclature below.

PI	Point of Intersection of tangents
PC	Point of curvature
POC	Point on curve
Pt	Point of tangency
POT	Point on tangent
RP	Reference point
P	P-line (preliminary location line)
L	L-line (final location line)
BM	Benchmark
TBM	Temporary benchmark
BT	Begin taper (any)
ET	End taper (any)
C	Cut
F	Fill
L	Centerline
D	Ditch
W	Width

3.6 SURVEY NOTES

- A. All survey data shall be neatly recorded in survey books. All survey notes shall become the property of the Forest Service. Errors shall be deleted by lining out. Date, crew names and positions, instrumentation, and weather shall be recorded in the notes at the beginning of each day's work. The party chief shall sign or initial each page of the notes immediately after the last entry for each day's work.
- B. Electronically recorded survey notes shall be consecutively numbered and headed to identify the contents. The notes shall be supported and accompanied by a bound book that records the project name and for each day identifies date, crew names and positions, instrumentation, weather, type of survey, stationing of sections between which survey was performed, and survey data or sketched that cannot be electronically recorded. The party chief shall sign or initial the electronically recorded notes and Day Book immediately after the last entry for each day's work.

END OF SECTION 010250

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 011250 - MEASUREMENT AND PAYMENT

PART 1 - GENERAL

1.1 SUMMARY

- A. Measurement and payment for contract work will be made only for and under those pay items included in the Schedule of Items. All other work, labor, materials, equipment, and incidentals necessary to successfully complete the project will be considered as included in the payment for items shown. This section defines the method of measurements and basis of payment for work items listed in the Schedule of Items.
- B. When more than one class, size, type, thickness, etc. is specified in the Schedule of Items for any pay item, suffixes will be added to the item number to differentiate between the pay items.

1.2 DETERMINATION OF QUANTITIES

- A. The following measurements and calculations shall be used to determine contract quantities for payment.
 - 1. For individual construction items, longitudinal and lateral measurements for area computations shall be made horizontally or corrected to horizontal measurement unless otherwise specified. Measurements for geotextiles, netting and erosion control blankets shall be along slope lines.
 - 2. For excavation or embankment volumes, the average end area method shall be used to compute volumes. However, if in the judgment of the Contracting Officer (CO), the average end area method is impractical, measurement shall be made by volume in hauling vehicles or by other three-dimensional methods.
 - 3. For Structures, they shall be measured according to neat lines shown on the drawings or as altered by the CO, in writing, to fit field conditions.
 - 4. For items that are measured by the linear foot, such as pipe culverts, fencing, guardrail, piping, utilities, and underdrains, measurements shall be made parallel to the base or foundation upon which the structures are placed.
 - 5. For aggregates weighed for payment, the tonnage shall not be adjusted for moisture content, unless otherwise provided for.
 - 6. For standard manufactured items (such as fence, wire, plates, rolled shapes, pipe conduits) identified by gauge, weight, section dimensions, and so forth, such identifications shall be considered the nominal weights or dimensions. Unless controlled by tolerances in cited specifications, manufacturer's tolerances shall be accepted.
- B. Earthwork Tolerances - Adjustments of horizontal or vertical alignment, within the tolerances specified in this contract, or shifts of balance points up to 100 feet shall be

made by the contractor as necessary to produce the designed sections and to balance earthwork. Such adjustments shall not be considered as "Changes."

1.3 UNITS OF MEASUREMENT

- A. Payment shall be by units defined and determined according to U.S. Standard measure and by the following:
1. Acre: Make longitudinal and transverse measurements for area computations horizontally.
 2. Cubic Yard (CY): A measurement computed by one of the following methods:
 - a. Excavation, Embankment, or Borrow. The measurement computed by the average end area method from measurements made longitudinally along a centerline or reference line.
 - b. Material in Place or Stockpile. The measurement computed using the dimensions of the in-place material.
 - c. Material in the Delivery Vehicle. The measurement computed using measurements of material in the hauling vehicles at the point of delivery. Vehicles shall be loaded to at least their water level capacity. Leveling of the loads may be required when vehicles arrive at the delivery point.
 3. Each (EA): One complete unit, which may consist of one or more parts.
 4. Linear Foot (LF): Measurement of work along its length from point-to-point; parallel to the base or foundation. Do not measure overlaps.
 5. Lump Sum (LS): One complete unit.

1.4 METHOD OF MEASUREMENT

- A. One of the following methods of measurement for determining final payment is designated on the Schedule of Items for each pay item:
1. **ACTUAL QUANTITIES (AQ)** - These quantities are determined from actual measurements of completed work.
 2. **DESIGNED QUANTITIES (DQ)** - These quantities denote the final number or units to be paid for under the terms of the contract. They are based upon the original design data available prior to advertising the project. Original design data include the preliminary survey information, design assumptions, calculations, drawings, and the presentation in the contract. Changes in the number of units shown in the Schedule of Items may be authorized under any of the following conditions:
 - a. As a result of changes in the work authorized by the CO.
 - b. As a result of the CO determining that errors exist in the original design that cause a pay item quantity to change by 15 percent or more.
 - c. As a result of the Contractor submitting to the CO a written request showing evidence of errors in the original design that cause a pay item quantity to change by 15 percent or more. The evidence must be verifiable and consist

of calculations, drawings, or other data that show how the designed quantity is believed to be in error.

3. LUMP SUM QUANTITIES (LSQ) - These quantities denote one complete unit of work as required by or described in the contract, including necessary materials, equipment, and labor to complete the job. They shall not be measured.
4. STAKED QUANTITIES (SQ) - These quantities are determined from staked measurements prior to construction.
5. VEHICLE QUANTITIES (VQ) - These quantities are measured or weighed in hauling vehicles.

PART 2 - PRODUCTS (NOT APPLICABLE)

PART 3 - EXECUTION (NOT APPLICABLE)

END OF SECTION 011250

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 011900 - MOBILIZATION

PART 1 - GENERAL

1.1 SUMMARY

- A. This item is intended to compensate the Contractor for operations including, but not limited to, those necessary for the movement of personnel, equipment, supplies, and incidentals to the project site; for payment of premiums for bonds and insurance for the project; and for any other work and operations which must be performed or costs that must be incurred incident to the initiation of meaningful work at the site and for which payment is not otherwise provided for under the contract.

1.2 MEASUREMENT AND PAYMENT

- A. The measurement shall be lump sum for mobilization. Payment shall be as follows:
 - 1. Bond premiums will be reimbursed after receipt of the evidence of payment.
 - 2. 50% of the lump sum, not to exceed 5% of the original contract amount, will be paid following completion of 5% of the original contract amount not including mobilization and bond premiums.
 - 3. Payment of the remaining portion of the lump sum, up to 10% of the original contract amount, will be paid following completion of 10% of the original contract amount not including mobilization and bond premiums.
 - 4. Any portion of the lump sum in excess of 10% of the original contract amount will be paid after final acceptance.
 - 5. Progress payments for mobilization and preparatory work shall be subject to retainage.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION 011900

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 013300 - SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other miscellaneous submittals. See Table 013300-1 for a summary of required submittals.
- B. See other specification section within this package for additional requirements on submittal.

1.2 SUBMITTAL PROCEDURES

- A. Coordination: Coordinate preparation and processing of submittals with performance of construction activities.
 - 1. Coordinate transmittal of different types of submittals for related parts of the Work so processing will not be delayed because of need to review submittals concurrently for coordination.
 - a. The Contracting Officer (CO) reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.
- B. Processing Time: Allow enough time for submittal review, including time for re-submittals, as follows. Time for review shall commence on CO's receipt of submittal.
 - 1. Initial Review: Allow 14 days for initial review of each submittal. Allow additional time if processing must be delayed, to permit coordination with subsequent submittals. CO will advise Contractor when a submittal being processed must be delayed for coordination.
 - 2. If intermediate submittal is necessary, process it in same manner as initial submittal.
 - 3. Allow 14 days for processing each re-submittal.
 - 4. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing.
- C. Identification: Place a permanent label or title block on each submittal for identification.
 - 1. Indicate name of firm or entity that prepared each submittal on label or title block.
 - 2. Provide a space on label or beside title block to record Contractor's review and approval markings and action taken by CO.
 - 3. Include the following information on label for processing and recording action taken:

- a. Project name.
 - b. Date.
 - c. Name and address of Contractor.
 - d. Name of manufacturer.
 - e. Unique identifier, including revision number.
 - f. Number and title of appropriate Specification Section.
 - g. Drawing number and detail references, as appropriate.
 - h. If more than one item is shown on submittal sheet, identify item.
- D. Deviations: Highlight, encircle, or otherwise identify deviations from the Contract Documents on submittals.
- E. Additional Copies: Unless additional copies are required for final submittal, and unless CO observes noncompliance with provisions of the Contract Documents, initial submittal may serve as final submittal.
- F. Use for Construction: Use only final submittals with mark indicating action taken by CO in connection with construction.

1.3 MEASUREMENT AND PAYMENT

- A. No separate measurement and/or payment will be made for this section. Payment shall be included with work shown in the schedule of items.

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS – (Submittals requiring CO approval)

- A. General: Prepare and submit Action Submittals required by individual Specification Sections.
- 1. Number of Copies: Submit three copies of each submittal, unless otherwise indicated. CO will return two copies. Mark up and retain one returned copy as a Project Record Document.
- B. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.
- 1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
 - 2. Mark each copy of each submittal to show which products and options are applicable.
 - 3. Include the following information, as applicable:
 - a. Manufacturer's written recommendations.
 - b. Manufacturer's product specifications.
 - c. Manufacturer's installation instructions.
 - d. Manufacturer's catalog cuts.
 - e. Wiring diagrams showing factory-installed wiring.
 - f. Compliance with recognized trade association standards.

- g. Compliance with recognized testing agency standards.
- C. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data.
 - 1. Preparation: Include the following information, as applicable:
 - a. Dimensions.
 - b. Identification of products.
 - c. Fabrication and installation drawings.
 - d. Roughing-in and setting diagrams.
 - e. Wiring diagrams showing field-installed wiring, including power, signal, and control wiring.
 - f. Notation of dimensions established by field measurement.
 - 2. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring.
- D. Contractor's Construction Schedule: The contractor shall submit a Construction Schedule, for approval by CO, in accordance with the contract provisions within 5 day of commencement of work.
- E. Samples: Prepare physical units of materials or products, including the following:
 - 1. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available.

2.2 INFORMATIONAL SUBMITTALS – (Submittals NOT requiring CO approval)

- A. General: Prepare and submit Informational Submittals required by other Specification Sections.
 - 1. Number of Copies: Submit three copies of each submittal, unless otherwise indicated. CO will not return copies.
 - 2. Certificates and Certifications: Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of that entity.
 - 3. Test and Inspection Reports: Comply with requirements in Section 014100 "Quality Control."
- B. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements.
- C. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements.

- D. Maintenance Data: Prepare written and graphic instructions and procedures for operation and normal maintenance of products and equipment.
- E. Manufacturer's Instructions: Prepare written or published information that documents manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer.

PART 3 - EXECUTION

3.1 GENERAL

- A. Review each submittal and check for compliance with the Contract Documents. Note corrections and field dimensions. Mark with approval stamp before submitting to CO.
- B. Approval Stamp: Stamp each submittal with a uniform, approval stamp. Include Project name and location, submittal number, Specification Section title and number, name of reviewer, date of Contractor's approval, and statement certifying that submittal has been reviewed, checked, and approved for compliance with the Contract Documents.
- C. CO will not review submittals that do not bear Contractor's approval stamp and will return them without action.
- D. Submittals not required by the Contract Documents will not be reviewed and may be discarded.
- E. Substitutions – Whenever materials, products, and equipment are listed by name or brand in the specifications and/or on the drawings, it is used as a measure of quality, utility, or standard. If the Contractor prefers to use any other brand or manufacturer of same quality, appearance, and utility to that specified, he shall request substitution as provided below, not less than 30 days before the planned installation of the item. The Contracting Officer will approve or disapprove the request for substitution.
- F. Requests for substitutions will only be considered if contractor submits the following:
 - 1. Complete technical data including drawings, complete performance specifications, test data, samples and performance tests of the article proposed for substitution. Submit additional information if required by Contracting Officer. All items in the above information shall be circled, tagged, or marked in some way to indicate all deviations or differences which the proposed item differs from the originally specified item.
 - 2. Similar data as above for item originally specified. All items shall be marked to identify where/how the proposed substitution will differ.
 - 3. A statement by the Contractor that the proposed substitution is in full compliance with the contract documents, applicable codes, and laws.
 - 4. The Contractor shall be responsible for any effect upon related work in the project for any substitution and shall pay any additional costs generated by any substitutions.

- 3.2 SUBMITTAL SCHEDULE – Submittals shall be made as required by and called for in the drawings and specifications. The following table is a summary of the required submittals for the project - the table is to assist the Contractor and may not be all inclusive – additional submittals may be required by specific specifications:

TABLE 013000-1

Spec. Section	Section Title	Subsec-tion	Required Submittal
C	General Specifications	1.4A	Traffic Control Plan
010250	Construction Staking	3.6A	Survey Notes
013300	Submittal Procedures	2.1D	Construction Schedule
014100	Quality Control	1.3A	Contractor quality control plan
014100	Quality Control	1.3B	Permits, Licenses, and Certificates
014100	Quality Control	1.3C	Test and inspection reports
014100	Quality Control	1.3D	As-Built drawings
023701	Sediment & Erosion Control Measures	1.2A	Sediment & Erosion Control Plan
033000	Cast In-Place Concrete	1.3A	Product Data
033000	Cast In-Place Concrete	1.3B	Design Mixtures
033000	Cast In-Place Concrete	1.3C	Shop Drawings
033000	Cast In-Place Concrete	1.3D	Field quality-control test reports
101400	Signs	1.2A	Sign List Submittal
101415	Minor Site I.D. Sign	1.3A	Sample Sign Mock-up
129300	Site Furnishings	1.2A	Product Data
133402	Utility Boxes	1.3A	Product Data
133402	Utility Boxes	1.3B	Shop Drawings
133402	Utility Boxes	1.3C	Product Certifications
221100	Water Distribution Systems	1.3A	Product Data
221100	Water Distribution Systems	1.3B	Plan for Disposal of Water Used for Testing and Disinfection
221100	Water Distribution Systems	1.3C	Operation and Maintenance Manual
221100	Water Distribution Systems	1.3D	Disinfection Report
221103	Plumbing Appurtenances	1.3A	Product Data
221103	Plumbing Appurtenances	1.3B	Operation and Maintenance Data
221105	Water Hydrants	1.3A	Manufacturer's Literature & Maintenance Data
221105	Water Hydrants	1.3B	Certifications
221105	Water Hydrants	1.3C	Maintenance Information
221105	Water Hydrants	1.3D	Stain Color Sample
221300	Onsite Wastewater Treatment Systems	1.3A	Product Data
221300	Onsite Wastewater Treatment Systems	1.3B	Shop Drawings

221300	Onsite Wastewater Treatment Systems	1.3C	Coordination Drawings
Spec. Section	Section Title	Subsection	Required Submittal
221300	Onsite Wastewater Treatment Systems	1.3D	Certificates
260500	Common Work Results for Electrical	1.2A	Product Data
260519	Low-Voltage Electrical Power Conductors and Cables	1.3A	Product Data
260519	Low-Voltage Electrical Power Conductors and Cables	1.3B	Field Quality Control Reports
262716	Walk-In Enclosure	1.3A	Shop Drawings
262716	Walk-In Enclosure	1.3B	Manufacturer's Installation Instructions
262716	Walk-In Enclosure	1.3C	Manufacturer's Operation and Maintenance Instructions
312000	Earthwork	1.3A	Approval Source of Aggregates and Backfill Materials and Certified Sieve Results
321204	Crushed Aggregate Base or Surface Course	1.2A	Aggregate Source, Gradation and Material Properties
321204	Crushed Aggregate Base or Surface Course	1.2B	Compaction Density Test Results and Proctor
332228	Culvert Pipe and Pipe Arches	1.3A	Fabricator's Certification
332228	Culvert Pipe and Pipe Arches	1.4A	Culvert Locations and Lengths

END OF SECTION 013300
April 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 014100 – QUALITY CONTROL

PART 1 - GENERAL

1.1 This work shall consist of providing quality control in conformance with the inspection, testing, and product certification requirements of this contract to ensure compliance with the drawings and specifications. The Contractor shall provide all personnel, equipment, tests, and reports necessary to meet the requirements of the contract.

1.2 QUALITY CONTROL

- A. The Contractor shall provide and maintain a quality control system that will ensure all services, supplies, and construction work required under this contract conforms to the contract requirements. The Contractor shall perform, or cause to be performed, the sampling, inspection, and testing required to substantiate that all services, supplies, and construction conform to the contract requirements.
- B. Special Tests and Inspections: Contractor will engage a testing agency to conduct required special tests and inspections. The Contractor shall authorize the testing agency to perform the required testing and inspections on the work completed. The authority shall include:
 - 1. Testing agency will interpret tests and inspections and state in each report whether tested and inspected work complies with or deviates from the Contract Documents.
 - 2. Testing agency will re-test and re-inspect corrected work.
- C. Retesting/Reinspecting: Contractor shall provide quality-control services for retesting and reinspection for replaced construction work or for work that failed to comply with the requirements under the contract.

1.3 SUBMITTALS

- A. Contractor Quality Control Plan
- B. Permits, Licenses, and Certificates
- C. Test and Inspection Reports
- D. As-Built Drawings

1.4 MEASUREMENT AND PAYMENT

- A. No separate payment will be made for the work included under this section; rather payment shall be included in the items of work listed in the Schedule of Items.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION

3.1 QUALITY CONTROL SYSTEM

- A. General: Perform required testing, inspections, sampling, and similar services per direction specified in the contract drawings and specifications and in accordance with established industry standards.

3.2 CONTRACTOR QUALITY CONTROL PLAN

- A. At the time of the preconstruction conference, the Contractor shall submit for approval a written Contractor Quality Control Plan.
 - 1. If the plan requires any revisions or corrections, the Contractor shall resubmit the plan within 10 days.
 - 2. The Government reserves the right to require changes in the plan during the contract period as necessary.
 - 3. No change in the approved plan may be made without written concurrence by the Contracting Officer.
 - 4. At a minimum, the plan shall include the following:
 - a. A list of personnel responsible for quality control and assigned duties. Include each person's qualifications.
 - b. A copy of a letter of direction to the Contractor's Quality Control Supervisor outlining assigned duties.
 - c. Names, qualifications, and descriptions of laboratories to perform sampling and testing, and samples of proposed report forms.
 - d. Methods of performing, documenting, and enforcing quality control of all work.
 - e. Methods of monitoring and controlling environmental pollution and contamination as required by all applicable regulations and laws.

3.3 TEST AND INSPECTION REPORTS

- A. Submit three copies of complete test results no later than three calendar days after the test was performed.
- B. Submit failing test results and proposed remedial actions within four hours of noted deficiency.
- C. Testing and Inspection Reports shall include the following:
 - 1. Date of issue.
 - 2. Project title and number.
 - 3. Name, address, and telephone number of testing agency.
 - 4. Dates and locations of samples, tests, or inspections.
 - 5. Names of individuals performing tests and inspections.

6. Reference Specification Section(s).
7. Complete test or inspection data.
8. Test and inspection results and an interpretation of test results.
9. Ambient conditions at time sample was taken, tested, or inspected.
10. Comments or professional opinion on whether tested or inspected work complies with the Contract Document requirements.
11. Name and signature of laboratory inspector.
12. Recommendations on retesting and reinspecting.

3.4 PERMITS, LICENSES, AND CERTIFICATES

- A. For Contracting Officer's records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations relevant to the on performance of the work.

3.5 AS-BUILT DRAWINGS

- A. The Contractor shall maintain a set of the contract drawings depicting as-built conditions. These drawings shall be maintained in a current condition and shall be available for review. All variations from the original contract drawings shall be indicated in red on the drawings. Upon completion of the contract work, as-built drawings shall be submitted to the Contracting Officer.

3.6 SAMPLING, TESTING, AND CERTIFICATION REQUIREMENTS

- A. Sampling, testing, and Certification requirements and frequency for specific items shall be as specified in the drawings and specification. The following table is a summary of the required sampling, testing, and certification for the project - the table is to assist the Contractor, but may not be all inclusive – additional submittals may be required by specific specification section:

B.

TABLE 014100-1			
Item	Subsection	Certification or Test Required	Frequency
033000	2.12	Mixing and Delivery	Each Truck
033000	3.12 B	Concrete – Slump, Air, Temperature	1 composite every two (2) truck loads delivered
033000	3.12 B	Concrete - Compression Test Specimens	at 7 days and 28 days (see spec)
221100	3.2O	HDPE Pipe - Bent Strap Test	Once Daily
221100	3.5A	Hydrostatic Testing of Water System	See Spec.
221100	3.5B	Pressure and Leakage Test – Water Line	See Spec.

TABLE 014100-1			
Item	Subsection	Certification or Test Required	Frequency
221100	3.6A	Bacteriological Examination	After disinfection, before use
221300	3.6 A	Hydrostatic Testing	See Spec
221300	3.6 A	Flushing Pipelines	See Spec
312000	3.14C & 3.19C	Compaction Test – at Pipe Zone	One per 600 Feet of Trench
312000	3.14D & 3.19C	Compaction Test – above Pipe Zone	One per 600 Feet of Trench
312000	3.14I	Tracer Wire Conductivity Test	Until Passing
312000	3.19C	Moisture-Density Relationship	One test every soil type encountered
312225	3.12B	Compaction – Road Embankment (Method 4)	One test every 300 sq. yd. of embank- ment per lift
321204	3.4 A	Compaction Tests – Road Base and Surface Course	One test for every 300 sq. yd. of aggregate
321204	3.4A	Compaction Tests – Aggregate Base for Concrete Slabs	One test for each family unit or con- crete slab
332228	3.7C	Compaction Tests	One test for every 12 inches of lift placed

END OF SECTION 014100
April 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 023701 - SEDIMENT AND EROSION CONTROL MEASURES

PART 1 - GENERAL

1.1 The work under this section consists of furnishing all necessary labor, equipment, materials, and performing all operations in connection with construction sediment and control measures.

A. General

1. All erosion and sediment control measures are to be placed prior to any disturbance caused by grading and or excavation and shall conform to the requirements of the appropriate regulatory agency for the State.
2. The Contractor shall be solely responsible for ensuring that erosion and sediment control measures are implemented and maintained at the site.
3. Soil disturbing activities include but are not limited to: Clearing and grubbing, excavation for utilities and foundations, roadway and parking lot construction, construction or modification of site drainage, grading, and preparation for final seeding.

1.2 SUBMITTALS

A. The Contractor shall be required to submit a stormwater pollution prevention plan (SWPPP) in accordance with this specification and state/local requirements. Submit plan to the Contracting Officer no less than two (2) prior to the planned start of work for approval (note: work cannot commence until permits have been obtained and SWPPP has been approved). The plan shall demonstrate an understanding of all laws, regulations, permits and orders that govern stormwater, including but not limited to the following:

1. New Mexico Environmental Department (NMED)
2. Applicable National Pollutant Discharge Elimination System (NPDES) permits.

1.3 PERMITS

A. The Contractor shall apply for and pay for a National Pollutant Discharge Elimination System (NPDES) permit.

1.4 MEASUREMENT AND PAYMENT

A. No separate measurement and/or payment will be made for this section. Payment for work in associated with this section shall be included under Project Site Prep and Grading under the schedule of items.

PART 2 - PRODUCTS

2.1 SYNTHETIC FILTER FABRIC FOR SILT FENCES.

- A. Pervious sheet of polypropylene, nylon, or polyethylene fabric conforming to the following physical and hydraulic characteristics:

• Physical Properties (Min.)	Requirement	Test Method
• Grab Tensile, lbs.	W120/F100	ASTM-D-4632
• Grab Elongation, %	15	ASTM-D-4632
• Mullen Burst, psi	275	ASTM-D-3786
• Puncture, lbs.	65	ASTM-D-4833
• Trapezoidal Tear, lbs.	50	ASTM-D-4533
• UV Resistance, %	80	ASTM-D-4355
• AOS, US Sieve #	30/40	ASTM-D-4751
• Permittivity gal/min-sq. ft.	90	ASTM-D-4491

- B. Filter fabric should contain ultraviolet ray inhibitors and stabilizers to provide a minimum of 6 months of expected usable construction life at a temperature range of 0 to 120 F.
- C. Support Posts: 4 foot - 2 x 2 wood.

2.2 STRAW BALES

- A. Oat or wheat straw, free from weeds, viable weed seeds, foreign matter detrimental to plant life, and dry.
1. Grass hay bales are not acceptable.
 2. In order to prevent deterioration of the bindings, all bales shall be either wire-bound or string-tied so that bindings are oriented around the sides rather than along the tops and bottoms of the bales.

2.3 STRAW WATTLES OR ROLLS

- A. Furnish straw wattles or rolls that are manufactured from weed free straw and wrapped in a tubular photodegradable plastic netting made from 85% high density polyethylene, 14% ethyl vinyl acetate and 1% color for UV inhibition. Conform to the following:
1. Diameter 9 inches min.
 2. Netting strand thickness 0.030 inches
 3. Netting knot thickness 0.055 inches
 4. Mass of netting 0.315 to 0.385 ounces per foot

PART 3 - EXECUTION

3.1 CONSTRUCTION

- A. Install straw bales at local drainage ways to prevent silt intrusion upon adjacent drainage courses. Remove straw bales following establishment of vegetation cover and utilize as mulch at swales or on steep slopes.
- B. Prior to construction, install silt fence along the downhill construction limits to prevent

silt intrusion upon adjacent land.

- C. Install sediment and erosion control measures on the down slope toe of all topsoil stockpiles.
- D. Maintain and remove all erosion controls as specified.
- E. Temporary seeding shall be placed on exposed surfaces that will not be brought to final grading or permanent cover treatment within 30 days of the exposure to reduce erosion and sedimentation by stabilizing exposed soils. Seeded areas shall be checked regularly for bare spots, washouts, and healthy growth to assure that a good stand of grass is being maintained. Reseed areas that fail to establish vegetation cover as soon as such areas are identified.

3.2 DUST CONTROL

- A. In areas subject to surface and air movement of dust, where on-site or off-site damage is likely to occur, one or more of the following preventive measures shall be taken for dust control:
 - 1. Minimize the period of soil exposure by utilizing temporary ground cover and other temporary stabilization practices.
 - 2. Sprinkle the site with water until surface is wet. Repeat as needed.

3.3 SILT FENCE

- A. Silt fences are appropriate for the following general locations:
 - 1. Immediately upstream of the point(s) of runoff discharge from a site before flow becomes concentrated. Below disturbed areas where runoff may occur in the form of overland flow.
 - 2. Along the down slope toe of all topsoil stockpiles.
- B. Materials.
 - 1. Utilize standard strength synthetic filter fabric for sediment barriers. The filter fabric shall be purchased in a continuous roll cut to the length of the barrier to avoid the use of joints. When joints are necessary, filter cloth shall be spliced together only at a support post, with a minimum 6-inch overlap, and securely sealed.
 - 2. The standard strength filter fabric shall be stapled or wired to the fence and 6 inches of the fabric shall be extended into the ground. Filter fabric shall not be stapled to existing trees.
 - 3. Support posts shall be spaced at a maximum 6 feet and driven securely into the ground a minimum of 24 inches.
 - 4. Filter fabric shall be buried a minimum of 12 inches.
 - 5. The height of a silt fence shall not exceed 36 inches. Higher fences may impound volumes of water sufficient to cause failure of the structure.
- C. Maintenance.
 - 1. Silt fences and filter barriers shall be inspected immediately after each rainfall and at least daily during prolonged rainfall.

2. Silt fences shall be inspected for depth of sediment, tears, and to see if the fabric is securely attached to the fence posts, and to see that the fence posts are firmly in the ground. Any deficiencies shall be repaired immediately.
3. Should the fabric on a silt fence or filter barrier decompose or become ineffective prior to the end of the expected usable life and the barrier still be necessary, the fabric shall be replaced promptly.
4. Sediment deposits should be removed after each storm event and/or when deposits reach approximately 1/3 the height of the barrier or when the sediments limit or prevent the flow of water through the fabric hydraulic.
5. Any sediment deposits remaining in place after the silt fence or filter barrier is no longer required shall be dressed to conform with the existing grade, prepared, and seeded.

3.4 STRAW BALE EROSION CONTROL FENCE

- A. Straw bale erosion control fences are appropriate for the following general locations:
 1. Sheet flow applications: Straw bales shall be placed in a single row, lengthwise on the contour with ends of adjacent bales tightly abutting one another.
 2. Channel flow applications: Straw bales shall be placed in a single row, lengthwise and oriented perpendicular to the direction of flow with ends of adjacent bales tightly abutting one another. The barrier shall be extended to such a length that the bottoms of the end bales are higher in elevation than the top of the lowest middle bale to assure that sediment laden runoff will flow either through or over the barrier but not around it.
- B. The barrier shall be entrenched and backfilled. A trench shall be excavated the width of a bale and the length of the proposed barrier to a minimum depth of 4 inches. After the bales are staked and chinked, the excavated soil shall be backfilled against the barrier. Backfill shall conform to the ground level of the downhill side and shall be built up to 4 inches against the uphill side of the barrier.
- C. Each bale shall be securely anchored by at least 2 stakes or rebar driven through the bale. The first stake in each bale shall be driven toward the previously laid bale to force the bales together. Stakes or rebar shall be driven a minimum of 12 inches into the ground or deep enough into the ground to securely anchor the bales, whichever is greater.
- D. The gaps between bales shall be chinked (filled by wedging with straw to prevent water from escaping between the bales). Loose straw scattered over the area immediately uphill from a straw bale barrier tends to increase barrier efficiency.

3.5 STRAW WATTLE OR ROLL

- A. Straw wattles shall be installed in accordance to manufacturer's installation guidelines.
- B. At a minimum:
 1. The wattle shall be entrenched and backfilled. A trench shall be excavated the width of the straw waddle and the length of the proposed barrier to a depth of 2-3 inches.
 2. Each wattle shall be securely anchored by at least one 18-24 inch stake every 3-4

feet and with a stake on each end. Stakes shall be driven perpendicular to slope face through the middle of the wattle until 2-3 inches remains exposed above the waddle.

3. After the wattles are staked, compact excavated soil against the uphill side of the barrier.
4. Adjacent wattles should tightly abut.

3.6 MAINTENANCE

- A. Inspection shall be continuous, and repair or replacement shall be made promptly as needed. Straw bale carriers shall be removed when they have served their usefulness, but not before the upslope areas have been permanently stabilized.

END OF SECTION 023701

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 024100 - WASTE MATERIAL DISPOSAL

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the loading, handling, hauling, and placing of excess excavation material, unsuitable excavation material, clearing and grubbing debris, and construction and demolition debris.

1.2 MEASUREMENT AND PAYMENT

- A. There will be no separate measurement or payment for work in this Section. Waste material disposal is considered incidental to other items of work shown in the Schedule of Items. Refer to the drawings for items to be removed and disposed.

PART 2 - PRODUCTS – NOT APPLICABLE

PART 3 - EXECUTION

3.1 WASTE MATERIAL TO BE HAULED TO A DISPOSAL AREA

- A. All excavated material not used in the construction of embankments or backfilling of trenches, or other excess material resulting from the excavation and embankment operation shall be hauled to a disposal area.
- B. All unsuitable excavated material and oversize boulders shall be hauled to a disposal area.
- C. All stumps, slash and other clearing and grubbing debris shall be hauled to a disposal area.
- D. Disposal Area: All waste material above shall be hauled to the disposal area on Government property within 10 miles of the project, as designated by the Contracting Officer.
 - 1. Waste material shall be piled and compacted to form a dense layer.
 - 2. The size and shape of the piled waste material shall be designated by the Contracting Officer.
 - 3. The piled material shall be covered with soil to a uniform depth of six inches minimum and sloped to 2:1 or flatter.
 - 4. The disposal site shall be left suitable for seeding.
 - 5. Where shown on the drawings, excess excavation may be disposed onsite. Disposal quantity shall be indicated on the drawings. Additional excess excavation disposal onsite may be permitted, if finished slopes are clean, set to flatter slope (4:1 or flatter) and do not encroach upon existing site amenities or features. Coordinate with the Contracting Officer and receive approval for excess excavation onsite prior to undertaking effort.

3.2 WASTE MATERIAL TO BE HAULED TO A LANDFILL

- A. All demolition materials, garbage, and other refuse generated shall be removed from the project site and legally disposed off Government property in an approved landfill.
- B. The contractor is responsible for all costs and permits associated with landfill disposal.
- C. The Government is not responsible for waste material upon its departure from the project site.

END OF SECTION 024100

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 024102 - REMOVAL AND DISPOSAL OF EXISTING STRUCTURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Includes removal disposal, and backfilling of existing structures, site improvements/features and utilities where shown on the drawings. This item also includes scarifying and re-grading the area around features and structures removed.
- B. Related Sections Include the Following:
 - 1. Section 024100 "Waste Material Disposal."
 - 2. Section 329206 "Seeding."

1.2 MEASUREMENT AND PAYMENT

- A. Payment will be lump sum (LS) for removal and disposal of structures, site features and obstructions, including demolition, excavation, backfill, removal, hauling and all incidentals and work required to complete this item.

PART 2 - PRODUCTS

2.1 SELECT BACKFILL

- A. Select backfill shall be native or imported soil, sand, or granular soil free of stones larger than 6 inches in greatest dimension. Refer to Section 024100 "Waste Material Disposal" for details.

PART 3 - EXECUTION

3.1 SALVAGE

- A. Items to be salvaged include:
 - 1. Refer to drawings for details.

3.2 GENERAL

- A. Topsoil stripping and storage, if required, shall comply with the applicable specification.
- B. The Contractor may use any means to remove the structures or items shown on the Drawings to be removed, if the safety of the public is provided, and the surrounding site and structures are preserved.
- C. The entire substructure shall be demolished to a depth of 1 foot below the present ground level. All wood, concrete, and metal materials to be removed shall be broken into sufficiently small pieces to enable handling and disposal. All other structures shall

be backfilled, and the surrounding ground restored to as near normal conditions as practicable in accordance with these specifications.

- D. Backfill for removed items may be excess material from other excavation or borrow from an approved source.
 - 1. Backfill shall be brought up in 12-inch compacted layers.
 - 2. Compaction shall be by mechanical tampers, with at least three passes over the entire surface.
 - 3. Maximum size of rock in backfill shall be 8 inches.
 - 4. Backfill shall be compacted and rounded 6 inches above the natural ground.
 - 5. If the backfill is not in an existing roadway or parking area, and at least 20 feet away from any proposed structure, the backfill shall be as specified above, except backfill shall be graded to blend with the natural ground and each layer compacted to **85 percent** of the optimum dry density determined by AASHTO T-99, Method C.
 - 6. In areas where backfill is in graveled or paved surface roadways, or within 20 feet of a proposed structure, the backfill shall be as specified above, except it shall be terminated at the grade shown on the drawings and each layer compacted to **95 percent** of the optimum dry density determined by AASHTO T-99, Method C. Completion of backfill shall be in accordance with the applicable surfacing specifications.
 - 7. When topsoil replacement is required, backfill and compaction shall be as specified above, except the backfill shall terminate 6 inches below the natural ground. Placement of topsoil, when required, shall comply with the applicable specification.
- E. Excessive settlement or other evidence of improper backfill shall be corrected by reopening the excavation to the depth required for proper refilling and compaction.

3.3 DISPOSAL

- A. Disposal of debris shall comply with Section 024100 "Waste Material Disposal."

3.4 FINISHING AND CLEANUP

- A. After the area has been backfilled, the disturbed area shall be finish graded to present a natural appearance and cleaned up by removing all debris and materials left from the removal of items in this Section.

3.5 SEEDING

- A. Seeding, if required, shall comply with Section 329206 "Seeding."

END OF SECTION 024102

NOVEMBER 2022

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 033000 - CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section specifies cast-in place concrete, including formwork, reinforcement, concrete materials, mixture design, placement procedures, and finishes.

1.2 DEFINITIONS

- A. Cementitious Materials: Portland cement alone or in combination with one or more of the following: blended hydraulic cement, fly ash and other pozzolans, ground granulated blast-furnace slag, and silica fume; subject to compliance with requirements.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Design Mixtures: For each concrete mixture. Indicate amounts of mixing water to be withheld for later addition at Project site.
- C. Steel Reinforcement Shop Drawings when project design requires special consideration.
- D. Field quality-control test reports.

1.4 MEASUREMENT AND PAYMENT

- A. Measurement of concrete will be one or more of the following methods as indicated in the Schedule of Items.
 - 1. Other Items: Other cast-in-place concrete required for this project is considered incidental to other items of work shown in the Schedule of Items.
 - 2. Crushed Aggregate Base will be paid for under Section 321204 "Crushed Aggregate Base or Surface Course."

PART 2 - PRODUCTS

2.1 FORM-FACING MATERIALS

- A. Smooth-Formed Finished Concrete: Form-facing panels that will provide continuous, true, and smooth concrete surfaces. Furnish in largest practicable sizes to minimize number of joints.
 - 1. Plywood, metal, or other approved panel materials.

2. Exterior-grade plywood panels, suitable for concrete forms, complying with DOC PS 1, and as follows:
 - a. Structural 1, B-B or better; mill oiled and edge sealed.
 - b. B-B (Concrete Form), Class 1 or better; mill oiled and edge sealed.
- B. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit.
- C. Forms for Cylindrical Columns, Pedestals, and Supports: Metal, glass-fiber-reinforced plastic, paper, or fiber tubes that will produce surfaces with gradual or abrupt irregularities not exceeding specified formwork surface class. Provide units with sufficient wall thickness to resist plastic concrete loads without detrimental deformation.
- D. Form-Release Agent: Commercially formulated form-release agent that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent treatments of concrete surfaces.
- E. Form Ties: Factory-fabricated, removable or snap-off metal or glass-fiber-reinforced plastic form ties designed to resist lateral pressure of fresh concrete on forms and to prevent spalling of concrete on removal.
 1. Furnish units that will leave no corrodible metal closer than 1 inch (25 mm) to the plane of exposed concrete surface.
 2. Furnish ties with integral water-barrier plates to walls indicated to receive dampproofing or waterproofing.

2.2 STEEL REINFORCEMENT

- A. Reinforcing Bars: ASTM A 615/A 615M, Grade 60, deformed.
- B. Plain-Steel Wire: ASTM A 82, as drawn.
- C. Deformed-Steel Wire: ASTM A 496.

2.3 REINFORCEMENT ACCESSORIES

- A. Joint Dowel Bars: ASTM A 615/A 615M, Grade 60, plain-steel bars, cut bars true to length with ends square and free of burrs.
- B. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded wire reinforcement in place. Manufacture bar supports from steel wire, plastic, or precast concrete according to CRSI's "Manual of Standard Practice," of greater compressive strength than concrete and as follows:
 1. For concrete surfaces exposed to view where legs of wire bar supports contact forms, use CRSI Class 1 plastic-protected steel wire or CRSI Class 2 stainless-steel bar supports.

2.4 CONCRETE MATERIALS

- A. Cementitious Material: Use the following cementitious materials, of the same type, brand, and source, throughout Project:
 - 1. Portland Cement: ASTM C 150, Type II.
- B. Normal-Weight Aggregates: ASTM C 33, graded, from a single source.
 - 1. Maximum Coarse-Aggregate Size: 1 inch (25 mm) nominal.
 - 2. Fine Aggregate: Free of materials with deleterious reactivity to alkali in cement.
- C. Water: ASTM C 94/C 94M and potable.
- D. Fly ash, ground iron blast-furnace slag, or silica fume may partially replace cement in any mix as follows:
 - 1. Fly Ash:
 - a. Class F – Not more than 20 percent of the minimum mass of portland cement may be replaced with class F fly ash.
 - b. Class C – Not more than 25 percent of the minimum mass of portland cement may be replaced with class C fly ash.
 - 2. Ground Iron Blast-Furnace Slag: Not more than 25 percent of the minimum mass of portland cement may be replaced with ground iron blast-furnace slag.
 - 3. Silica Fume (microsilica): Not more than 10 percent of the minimum mass of portland cement may be replaced with silica fume.
 - 4. Additionally, fly ash, slag, and silica fume will constitute no more than 50 percent of the total replacement weight.

2.5 ADMIXTURES

- A. Air-Entraining Admixture: ASTM C 260.
- B. Chemical Admixtures: Provide admixtures certified by manufacturer to be compatible with other admixtures and that will not contribute water-soluble chloride ions exceeding those permitted in hardened concrete. Do not use calcium chloride or admixtures containing calcium chloride.
 - 1. Retarding Admixture: ASTM C 494/C 494M, Type B.
- C. Synthetic Fiber: Monofilament or fibrillated polypropylene fibers engineered and designed for use in concrete pavement, complying with ASTM C 1116, Type III, 1/2 to 1-1/2 inches long.

2.6 CURING MATERIALS

- A. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. when dry.
- B. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.

- C. Water: Potable.
- D. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C 309, Type 1, Class B, dissipating or nondissipating. Liquid Membrane-Forming Compounds. Material shall be certified by curing compound manufacturer to not interfere with bonding of floor covering.

2.7 RELATED MATERIALS

- A. Expansion- and Isolation-Joint-Filler Strips: ASTM D 1751, asphalt-saturated cellulosic fiber or ASTM D 1752, cork or self-expanding cork.
- B. Semirigid Joint Filler: Two-component, semirigid, 100 percent solids, epoxy resin with a Type A shore durometer hardness of 80 per ASTM D 2240.
- C. Bonding Agent: ASTM C 1059, Type II, non-redispersible, acrylic emulsion or styrene butadiene.
- D. Epoxy Bonding Adhesive: ASTM C 881, two-component epoxy resin, capable of humid curing and bonding to damp surfaces, of class suitable for application temperature and of grade to suit requirements, and as follows:
 - 1. Types IV and V, load bearing, for bonding hardened or freshly mixed concrete to hardened concrete.

2.8 REPAIR MATERIALS

- A. Repair Underlayment: Cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 1/8 inch (3.2 mm) and that can be feathered at edges to match adjacent floor elevations.
 - 1. Cement Binder: ASTM C 150, portland cement or hydraulic or blended hydraulic cement as defined in ASTM C 219.
 - 2. Primer: Product of underlayment manufacturer recommended for substrate, conditions, and application.
 - 3. Aggregate: Well-graded, washed gravel, 1/8 to 1/4 inch (3.2 to 6 mm) or coarse sand as recommended by underlayment manufacturer.
 - 4. Compressive Strength: Not less than 4100 psi (29 MPa) at 28 days when tested according to ASTM C 109/C 109M.
- B. Repair Overlayment: Cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 1/8 inch (3.2 mm) and that can be feathered at edges to match adjacent floor elevations.
 - 1. Cement Binder: ASTM C 150, portland cement or hydraulic or blended hydraulic cement as defined in ASTM C 219.
 - 2. Primer: Product of topping manufacturer recommended for substrate, conditions, and application.
 - 3. Aggregate: Well-graded, washed gravel, 1/8 to 1/4 inch (3.2 to 6 mm) or coarse sand as recommended by topping manufacturer.

4. Compressive Strength: Not less than 5000 psi (34.5 MPa) at 28 days when tested according to ASTM C 109/C 109M.

2.9 CONCRETE MIXTURES, GENERAL

- A. Prepare design mixtures for each type and strength of concrete, proportioned on the basis of laboratory trial mixture or field test data, or both, according to ACI 301.
 1. Use a qualified independent testing agency for preparing and reporting proposed mixture designs based on laboratory trial mixtures.
- B. Limit water-soluble, chloride-ion content in hardened concrete to 0.06 percent by weight of cement.
- C. Admixtures: Use admixtures according to manufacturer's written instructions.
- D. Color Pigment: Add color pigment to concrete mixture according to manufacturer's written instructions and to result in hardened concrete color consistent with approved mockup.

2.10 CONCRETE MIXTURES FOR BUILDING ELEMENTS

- A. Proportion normal-weight concrete mixture as follows:
 1. Minimum Compressive Strength: 3500 psi (24.1 MPa) at 28 days.
 2. Minimum Cementitious Materials Content: 540 lb/cu. yd. (320 kg/cu. m).
 3. Slump Limit: 4 inches (100 mm), plus or minus 1 inch (25 mm).
 4. Air Content: 5 percent, plus or minus 1 percent at point of delivery for 1-inch (25-mm) nominal maximum aggregate size.
 5. Air Content: Do not allow air content of troweled finished floors to exceed 3 percent.
 6. Synthetic Fiber: For all slab-on-grade applications, uniformly disperse in concrete mixture at manufacturer's recommended rate, but not less than 1.0 lb/cu. yd. (0.60 kg/cu. m).

2.11 FABRICATING REINFORCEMENT

- A. Fabricate steel reinforcement according to CRSI's "Manual of Standard Practice."

2.12 CONCRETE MIXING

- A. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C 94/C 94M and ASTM C 1116, and furnish batch ticket information.
 1. When air temperature is between 85 and 90 deg F (30 and 32 deg C), reduce mixing and delivery time from 1-1/2 hours to 75 minutes; when air temperature is above 90 deg F (32 deg C), reduce mixing and delivery time to 60 minutes.

- B. Project-Site Mixing: Measure, batch, and mix concrete materials and concrete according to ASTM C 94/C 94M. Mix concrete materials in appropriate drum-type batch machine mixer.
 - 1. For mixer capacity of 1 cu. yd. (0.76 cu. m) or smaller, continue mixing at least 1-1/2 minutes, but not more than 5 minutes after ingredients are in mixer, before any part of batch is released.
 - 2. For mixer capacity larger than 1 cu. yd. (0.76 cu. m), increase mixing time by 15 seconds for each additional 1 cu. yd. (0.76 cu. m).
 - 3. Provide batch ticket for each batch discharged and used in the Work, indicating Project identification name and number, date, mixture type, mixture time, quantity, and amount of water added. Record approximate location of final deposit in structure.

PART 3 - EXECUTION

3.1 FORMWORK

- A. Design, erect, shore, brace, and maintain formwork, according to ACI 301, to support vertical, lateral, static, and dynamic loads, and construction loads that might be applied, until structure can support such loads.
- B. Construct formwork so concrete members and structures are of size, shape, alignment, elevation, and position indicated, within tolerance limits of ACI 117.
- C. Limit concrete surface irregularities, designated by ACI 347R as abrupt or gradual, as follows:
 - 1. Class A, 1/8 inch (3.2 mm) for smooth-formed finished surfaces.
 - 2. Class B, 1/4 inch (6 mm) for rough-formed finished surfaces.
- D. Construct forms tight enough to prevent loss of concrete mortar.
- E. Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush or wrecking plates where stripping may damage cast concrete surfaces. Provide top forms for inclined surfaces steeper than 1.5 horizontal to 1 vertical.
 - 1. Install keyways, reglets, recesses, and the like, for easy removal.
 - 2. Do not use rust-stained steel form-facing material.
- F. Set edge forms, bulkheads, and intermediate screed strips for slabs to achieve required elevations and slopes in finished concrete surfaces. Provide and secure units to support screed strips; use strike-off templates or compacting-type screeds.
- G. Provide temporary openings for cleanouts and inspection ports where interior area of formwork is inaccessible. Close openings with panels tightly fitted to forms and securely braced to prevent loss of concrete mortar. Locate temporary openings in forms at inconspicuous locations.
- H. Chamfer or round exterior corners and edges of permanently exposed concrete.

- I. Form openings, chases, offsets, sinkages, keyways, reglets, blocking, screeds, and bulkheads required in the Work. Determine sizes and locations from trades providing such items.
- J. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.
- K. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.
- L. Coat contact surfaces of forms with form-release agent, according to manufacturer's written instructions, before placing reinforcement.

3.2 EMBEDDED ITEMS

- A. Place and secure anchorage devices and other embedded items required for adjoining work that is attached to or supported by cast-in-place concrete. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
 - 1. Install anchor rods, accurately located, to elevations required and complying with tolerances in Section 7.5 of AISC's "Code of Standard Practice for Steel Buildings and Bridges."

3.3 REMOVING AND REUSING FORMS

- A. General: Formwork for sides of beams, walls, columns, and similar parts of the Work that does not support weight of concrete may be removed after cumulatively curing at not less than 50 deg F (10 deg C) for 48 hours after placing concrete, if concrete is hard enough to not be damaged by form-removal operations and curing and protection operations are maintained.
 - 1. Leave formwork for beam soffits, joists, slabs, and other structural elements that supports weight of concrete in place until concrete has achieved at least 70 percent of its 28-day design compressive strength.
 - 2. Remove forms only if shores have been arranged to permit removal of forms without loosening or disturbing shores.
- B. Clean and repair surfaces of forms to be reused in the Work. Split, frayed, delaminated, or otherwise damaged form-facing material will not be acceptable for exposed surfaces. Apply new form-release agent.
- C. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for exposed concrete surfaces unless approved by Contracting Officer.

3.4 STEEL REINFORCEMENT

- A. General: Comply with CRSI's "Manual of Standard Practice" for placing reinforcement.

1. Do not cut or puncture vapor retarder. Repair damage and reseal vapor retarder before placing concrete.
- B. Clean reinforcement of loose rust and mill scale, earth, ice, and other foreign materials that would reduce bond to concrete.
- C. Accurately position, support, and secure reinforcement against displacement. Locate and support reinforcement with bar supports to maintain minimum concrete cover. Do not tack weld crossing reinforcing bars.
- D. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.

3.5 JOINTS

- A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.
- B. Construction Joints: Install so strength and appearance of concrete are not impaired, at locations indicated or as approved by the Contracting Officer.
 1. Place joints perpendicular to main reinforcement. Continue reinforcement across construction joints, unless otherwise indicated. Do not continue reinforcement through sides of strip placements of floors and slabs.
 2. Use a bonding agent at locations where fresh concrete is placed against hardened or partially hardened concrete surfaces.
- C. Contraction Joints in Slabs-on-Grade: Form weakened-plane contraction joints, sectioning concrete into areas as indicated. Construct contraction joints for a depth equal to at least one-fourth of concrete thickness as follows:
 1. Grooved Joints: Form contraction joints after initial floating by grooving and finishing each edge of joint to a radius of 1/8 inch (3.2 mm). Repeat grooving of contraction joints after applying surface finishes. Eliminate groover tool marks on concrete surfaces. Fore Sidewalks joint spacing shall not exceed 5 feet.
 2. Sawed Joints: Form contraction joints with power saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut 1/8-inch- (3.2-mm-) wide joints into concrete when cutting action will not tear, abrade, or otherwise damage surface and before concrete develops random contraction cracks. **No Sawed Joints are allowed on concrete exposed to freezing.**
- D. Expansion Joints in Slabs-on-Grade: After removing formwork, install joint-filler strips at slab junctions with vertical surfaces, such as column pedestals, foundation walls, grade beams, and other locations, as indicated.
 1. Extend joint-filler strips full width and depth of joint, terminating flush with finished concrete surface, unless otherwise indicated.
 2. Install joint-filler strips in lengths as long as practicable. Where more than one length is required, lace or clip sections together.

3.6 CONCRETE PLACEMENT

- A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed.
- B. Do not add water to concrete during delivery, at Project site, or during placement unless approved by the Contracting Officer.
- C. Before test sampling and placing concrete, water may be added at Project site, subject to limitations of ACI 301.
- D. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete will be placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.
 - 1. Deposit concrete in horizontal layers of depth to not exceed formwork design pressures and in a manner to avoid inclined construction joints.
 - 2. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.
 - 3. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 inches (150 mm) into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.
- E. Deposit and consolidate concrete for floors and slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.
 - 1. Consolidate concrete during placement operations so concrete is thoroughly worked around reinforcement and other embedded items and into corners.
 - 2. Maintain reinforcement in position on chairs during concrete placement.
 - 3. Screed slab surfaces with a straightedge and strike off to correct elevations.
 - 4. Slope surfaces uniformly to drains where required.
 - 5. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, before excess bleedwater appears on the surface. Do not further disturb slab surfaces before starting finishing operations.
- F. Cold-Weather Placement: Comply with ACI 306.1 and as follows. Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.
 - 1. When average high and low temperature is expected to fall below 40 deg F (4.4 deg C) for three successive days, maintain delivered concrete mixture temperature within the temperature range required by ACI 301.
 - 2. Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.
 - 3. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in mixture designs.

- G. Hot-Weather Placement: Comply with ACI 301 and as follows:
1. Maintain concrete temperature below 90 deg F (32 deg C) at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is Contractor's option.
 2. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade uniformly moist without standing water, soft spots, or dry areas.

3.7 FINISHING FORMED SURFACES

- A. As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams. Repair and patch tie holes and defects. Remove fins and other projections that exceed specified limits on formed-surface irregularities.
- B. Rubbed Finish: Apply the following to smooth-formed finished as-cast concrete where indicated:
1. Grout-Cleaned Finish: Wet concrete surfaces and apply grout of a consistency of thick paint to coat surfaces and fill small holes. Mix one-part portland cement to one and one-half parts fine sand with a 1:1 mixture of bonding admixture and water. Add white portland cement in amounts determined by trial patches so color of dry grout will match adjacent surfaces. Scrub grout into voids and remove excess grout. When grout whitens, rub surface with clean burlap and keep surface damp by fog spray for at least 36 hours.
- C. Related Unformed Surfaces: At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a texture matching adjacent formed surfaces. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces, unless otherwise indicated.

3.8 FINISHING FLOORS AND SLABS

- A. General: Comply with ACI 302.1R recommendations for screeding, restraighening, and finishing operations for concrete surfaces. Do not wet concrete surfaces.
- B. Float Finish: Consolidate surface with power-driven floats or by hand floating if area is small or inaccessible to power driven floats. Restraighten, cut down high spots, and fill low spots. Repeat float passes and restraighening until surface is left with a uniform, smooth, granular texture.
1. Apply float finish to surfaces to receive trowel finish and to be covered with fluid-applied or sheet waterproofing.
- C. Trowel Finish: After applying float finish, apply first troweling and consolidate concrete by hand or power-driven trowel. Continue troweling passes and restraighten until surface is free of trowel marks and uniform in texture and appearance. Grind smooth any surface defects that would telegraph through applied coatings or floor coverings.

1. Apply a trowel finish to surfaces to be covered with resilient flooring, carpet, ceramic or quarry tile set over a cleavage membrane.
 2. Finish and measure surface so gap at any point between concrete surface and an unlevelled, freestanding, 10-foot- (3.05-m-) long straightedge resting on 2 high spots and placed anywhere on the surface does not exceed 1/8 inch (3.2 mm)
- D. Trowel and Fine-Broom Finish: Apply a first trowel finish to surfaces indicated. While concrete is still plastic, slightly scarify surface with a fine broom.
1. Comply with flatness and levelness tolerances for trowel finished floor surfaces.

3.9 MISCELLANEOUS CONCRETE ITEMS

- A. Filling In: Fill in holes and openings left in concrete structures, unless otherwise indicated, after work of other trades is in place. Mix, place, and cure concrete, as specified, to blend with in-place construction. Provide other miscellaneous concrete filling indicated or required to complete the Work.

3.10 CONCRETE PROTECTING AND CURING

- A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and ACI 301 for hot-weather protection during curing.
- B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft. x h (1 kg/sq. m x h) before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.
- C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for the remainder of the curing period.
- D. Unformed Surfaces: Begin curing immediately after finishing concrete. Cure unformed surfaces, including floors and slabs, concrete floor toppings, and other surfaces.
- E. Cure concrete according to ACI 308.1, by one or a combination of the following methods:
1. Moisture Curing: Keep surfaces continuously moist for not less than seven days with the following materials:
 - a. Water.
 - b. Continuous water-fog spray.
 - c. Absorptive cover, water saturated, and kept continuously wet. Cover concrete surfaces and edges with 12-inch (300-mm) lap over adjacent absorptive covers.

2. **Moisture-Retaining-Cover Curing:** Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches (300 mm), and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period using cover material and waterproof tape.
 - a. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive floor coverings.
 - b. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive penetrating liquid floor treatments.
 - c. Cure concrete surfaces to receive floor coverings with either a moisture-retaining cover or a curing compound that the manufacturer certifies will not interfere with bonding of floor covering used on Project.
3. **Curing Compound:** Apply uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.
 - a. After curing period has elapsed, remove curing compound without damaging concrete surfaces by method recommended by curing compound manufacturer unless manufacturer certifies curing compound will not interfere with bonding of floor covering used on Project.

3.11 CONCRETE SURFACE REPAIRS

- A. **Defective Concrete:** Repair and patch defective areas when approved by Contracting Officer. Remove and replace concrete that cannot be repaired and patched to Contracting Officer's approval.
- B. **Patching Mortar:** Mix dry-pack patching mortar, consisting of one-part portland cement to two and one-half parts fine aggregate passing a No. 16 (1.18-mm) sieve, using only enough water for handling and placing.
- C. **Repairing Formed Surfaces:** Surface defects include color and texture irregularities, cracks, spalls, air bubbles, honeycombs, rock pockets, fins and other projections on the surface, and stains and other discolorations that cannot be removed by cleaning.
 1. Immediately after form removal, cut out honeycombs, rock pockets, and voids more than 1/2 inch (13 mm) in any dimension in solid concrete, but not less than 1 inch (25 mm) in depth. Make edges of cuts perpendicular to concrete surface. Clean, dampen with water, and brush-coat holes and voids with bonding agent. Fill and compact with patching mortar before bonding agent has dried. Fill form-tie voids with patching mortar or cone plugs secured in place with bonding agent.
 2. Repair defects on surfaces exposed to view by blending white portland cement and standard portland cement so that, when dry, patching mortar will match surrounding color. Patch a test area at inconspicuous locations to verify mixture and color match before proceeding with patching. Compact mortar in place and strike off slightly higher than surrounding surface.

3. Repair defects on concealed formed surfaces that affect concrete's durability and structural performance as determined by Contracting Officer.
- D. Repairing Unformed Surfaces: Test unformed surfaces, such as floors and slabs, for finish and verify surface tolerances specified for each surface. Correct low and high areas. Test surfaces sloped to drain for trueness of slope and smoothness; use a sloped template.
1. Repair finished surfaces containing defects. Surface defects include spalls, popouts, honeycombs, rock pockets, crazing and cracks in excess of 0.01 inch (0.25 mm) wide or that penetrate to reinforcement or completely through unreinforced sections regardless of width, and other objectionable conditions.
 2. After concrete has cured at least 14 days, correct high areas by grinding.
 3. Correct localized low areas during or immediately after completing surface finishing operations by cutting out low areas and replacing with patching mortar. Finish repaired areas to blend into adjacent concrete.
 4. Correct other low areas scheduled to receive floor coverings with a repair underlayment. Prepare, mix, and apply repair underlayment and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface. Feather edges to match adjacent floor elevations.
 5. Correct other low areas scheduled to remain exposed with a repair topping. Cut out low areas to ensure a minimum repair topping depth of 1/4 inch (6 mm) to match adjacent floor elevations. Prepare, mix, and apply repair topping and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface.
 6. Repair defective areas, except random cracks and single holes 1 inch (25 mm) or less in diameter, by cutting out and replacing with fresh concrete. Remove defective areas with clean, square cuts and expose steel reinforcement with at least a 3/4-inch (19-mm) clearance all around. Dampen concrete surfaces in contact with patching concrete and apply bonding agent. Mix patching concrete of same materials and mixture as original concrete except without coarse aggregate. Place, compact, and finish to blend with adjacent finished concrete. Cure in same manner as adjacent concrete.
 7. Repair random cracks and single holes 1 inch (25 mm) or less in diameter with patching mortar. Groove top of cracks and cut out holes to sound concrete and clean off dust, dirt, and loose particles. Dampen cleaned concrete surfaces and apply bonding agent. Place patching mortar before bonding agent has dried. Compact patching mortar and finish to match adjacent concrete. Keep patched area continuously moist for at least 72 hours.
- E. Perform structural repairs of concrete, subject to Contracting Officer's approval, using epoxy adhesive and patching mortar.
- F. Repair materials and installation not specified above may be used, subject to Contracting Officer's approval.

3.12 FIELD QUALITY CONTROL

- A. Testing and Inspecting: Contractor will engage a qualified independent testing and inspecting agency to perform field tests and inspections and prepare test reports.
- B. Concrete Tests: Testing of composite samples of fresh concrete obtained according to ASTM C 172 shall be performed according to the following requirements:
 - 1. Testing Frequency: Obtain one composite sample set for every two trucks of concrete (or portion thereof) delivered to the project. A composite sample set consists of three compressive test cylinders and one air entrainment test.
 - 2. Slump: ASTM C 143/C 143M; one test at point of placement for each truck. Perform additional tests when concrete consistency appears to change or water is added.
 - 3. Air Content: ASTM C 231, pressure method, for normal-weight concrete; one test for each composite sample.
 - 4. Concrete Temperature: ASTM C 1064/C 1064M; one test for each truck.
 - 5. Compression Test Specimens: ASTM C 31/C 31M. Cast and laboratory cure one set of three standard cylinder specimens for each composite sample.
 - a. Compressive-Strength Tests: ASTM C 39/C 39M; test one of three laboratory-cured specimens at 7 days and one specimen at 28 days. If either previous tests fail, test third specimen at 28 days.
 - b. Strength of each batch delivered will be satisfactory if 28-day compressive-strength tests equals or exceeds specified compressive strength.
- C. Test results shall be reported in writing to Contracting Officer and Contractor within 48 hours of testing. Reports shall contain project identification name and number, date of concrete placement, name of concrete testing and inspecting agency, location of concrete batch in work, design compressive strength at 28 days, concrete mixture proportions and materials, compressive breaking strength, and type of break for both 7- and 28-day tests.
- D. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.
- E. Correct deficiencies in the work that test reports and inspections indicate does not comply with the Contract Documents.

END OF SECTION 033000
December 2019

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 033020-CONCRETE FROM PACKAGED DRY MIX FOR MINOR STRUCTURES

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes constructing small concrete structures associated with the project.

1.2 MEASUREMENT AND PAYMENT

- A. There will be no separate measurement or payment for work in this Section. Concrete from packaged dry mix is considered incidental to other items of work shown in the Schedule of Items.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Concrete Dry Mix:
 - 1. Design: ASTM C387, made using dry pre-portioned, blended, and bagged mix.
 - 2. Minimum 28-day Compressive Strength: 4000 psi
 - 3. Product: Sakrete Concrete Mix, as manufactured by US Mix Products Company, or an approved equal.
- B. All reinforcement bars shall be deformed billet steel conforming to the requirement of ASTM A615, Grade 60.
- C. Water: Clean and free from alkali and other deleterious materials.

PART 3 - EXECUTION

3.1 CONCRETE INSTALLATION

- A. Mixing:
 - 1. Mix dry concrete and water thoroughly according to the manufacturer.
 - 2. Mixing Equipment: Mechanical mixer or wheelbarrow.
 - 3. Place concrete while fresh; any concrete in which initial set has begun shall be wasted.
- B. Placing: Place the fresh concrete in the forms and work as necessary to remove the air pockets.

- C. Finishing: Apply medium trowel finish to exposed concrete surfaces.
- D. Protection From Cold: Keep the air temperature surrounding the fresh concrete above 40°F for a period of five days after the concrete is placed.
- E. Curing: If determined necessary by the Contracting Officer, cover exposed surfaces with plastic sheeting after the concrete has taken its initial set.

END OF SECTION 033020
APRIL 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 101110 –BULLETIN BOARD

PART 1 - GENERAL

- 1.1 This item shall consist of furnishing and installing a metal bulletin board and concrete footings constructed as shown on the drawings at the location shown on the site plan or as staked in the field in conformity with the lines and grades established, or as directed by the Contracting Officer
- 1.2 RELATED WORK
- A. The work shall be in accordance with the following subsidiary specifications. The subsidiary specifications are referred to in the text by the Section designation only.
1. Section 033020 Concrete from Packaged Dry Mix for Minor Structures
- 1.3 METHOD OF MEASUREMENT
- A. The quantity to be measured shall be the number of individual "Bulletin Board(s)" of type indicated in the Schedule of Items completed in place and accepted. Includes concrete footers and aggregate surface apron around the bulletin board.
1. Single-Panel Bulletin Board.
2. Double-Panel Bulletin Board.
- B. Aggregate Surfaced Gathering Area shown on the drawings shall be evaluated under Section 321204 "Aggregate Base and Surface Course".

PART 2 – PRODUCTS

- 2.1 MATERIALS
- A. Bulletin Board shall be manufactured by Rockart Signs and Markers; 3006 N. Maple, Mesa, Arizona 85215. 480-854-3400, or approved equal:
1. Metal 6x6 posts, 1-panel (48 x 48 Sign panel) Kiosk – No Roof with brown powder-coat finish (Part No. 03-431).
- a. Single-Panel Bulletin Board shall be 3/4" thick tackboard with 3/16" polycarbonate cover (Part No. 03-607).
2. Metal 6x6 posts, 2-Panel (48 x 48 Sign panels) Straight Kiosk – No Roof with brown powder-coat finish (Part No. 03-465).
- a. Left Panel of Double-Panel of Double Panel Bulletin Board shall be 3/4" thick tackboard with 3/16" polycarbonate cover (Part No. 03-607).
- b. Right Panel of Double-Panel Bulletin Board shall be 3/4" thick tackboard only with no polycarbonate cover (Part No. 03-605).

2.2 CONCRETE

- A. Concrete footings shall be in accordance with Section 033000 and as shown on the drawings.

2.3 AGGREGATE SURFACE COURSE

- A. Prepare subgrade to accept aggregate surfacing for apron around bulletin board in accordance to Section 312100 Site Preparation and Grading.
- B. Place Aggregate Surface Course, 1/2-inch minus gradation, with the finished surface having a grade not exceeding 2 percent in any direction. Place in accordance with Section 321204 Aggregate Base and Surface Course.
 - 1. Provide clear space in front of bulletin board, minimum 30" x 48", or larger if shown on the drawings, to meet FSORAG requirements.

PART 3 - EXECUTION

3.1 CONSTRUCTION

- A. Metal bulletin board shall be ordered as a kit and assembled on site with tamper resistant fasteners in accordance with the instructions provided by the manufacturer.

END OF SECTION

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 101400 - SIGNS

PART 1 - GENERAL

- 1.1 This item shall consist of furnishing and installing signs as shown on the Drawings. The location for sign installations shall be as shown on the Drawings and as staked by the Contracting Officer in the field. The contractor shall furnish all posts, concrete, signs, and hardware required to complete the full installation.
- 1.2 SUBMITTAL
- A. Sign List Submittal: The contractor shall submit a list of signs proposed for the project. The list shall include detailed drawings showing the size of the sign, size of letters, colors, materials, sign texts, and location for installation. The Contracting Officer shall approve the submitted list, prior to the Contractor ordering the signs.
- 1.3 METHOD OF MEASUREMENT
- A. The quantity to be measured shall be the number of signs furnished, installed, and accepted as shown in the Schedule of Items.
- B. Double signs on single-post installation will be considered as one installation. Signs that require more than one post to be considered as one installation.

PART 2 - PRODUCTS

- 2.1 SIGN PANELS
- A. General:
1. Sign panels shall be panels of one-piece construction.
 2. Signs shall conform to current editions of the Manual of Uniform Traffic Control Devices (MUTCD) and the Forest Service Manual - EM 7100-15, Sign and Poster Guidelines for the Forest Service.
- B. Sheeting
1. 100% heat activated retro-reflective sheeting material that conforms to AASHTO M 268 and ASTM D4956.
 2. Sheeting shall be High Intensity Type IV.
- C. Aluminum Panels:
1. Shall be fabricated using 100% heat activated retro-reflectorized material.
 2. Shall be furnished with predrilled holes for installation.
 3. All sign sheets and plates shall meet the requirements of ASTM B 209, alloy 6061-T6, or 5052-H38 and shall be of the minimum thickness shown below.
- | Sign Width (inches) | Sheet Aluminum Thickness (inches) |
|---------------------|-----------------------------------|
|---------------------|-----------------------------------|

Less than 8	0.022
8-12	0.040
13-19	0.063
20-30	0.080
31-48	0.100
over 48	0.125

2.2 HDPE Signs

1. Shall be constructed of High-Density Polyethylene (HDPE) containing recycled plastic.
2. The HDPE sheets consist of 1/8" top layer, a 1/2" middle core layer and a 1/8" bottom layer. The combination of layers shall provide a unified 3/4" thick panel free of any post manufacturing jointing.
3. The surfaces of the front and back of the panel will be a matte finish.
4. UV Stabilizers shall be added for longevity.
5. Finished signs will meet the following criteria:

PROPERTY	VALUE	UNITS	ASTM TEST
Density	.96+	g/cc	D-1505
Tensile Yield	4400	psi	D-638
Elongation	>600	%	D-638
Izod Impact	7.0	ft. lb./in. of notch	D-256
Tensile Impact	120	ft. lb. / in. 2	D-1822
Brittleness Temp.	<-76	C.	D-746
Hardness	68	Shore D	D-2240
Flex. Modulus	220,000	psi	D-256

6. HDPE post-consumer recycled plastic sheet shall be prepared and computer-routed. Work shall be performed by a high speed, computer-controlled electronic router with the sheets in a horizontal position so that the route surfaces are uniform in depth and finish.
7. A known vendor for this product is RockArt, Inc. Signs and Markers
3035 North Maple Street
Mesa, AZ 85215
Phone: 480-854-3400
Website: <http://www.rockartsigns.com/>

Or approved equal.

2.3 POSTS

- A. Wood Posts - 4"X 4" and 4"X6" posts shall be rough sawn standard and better hemlock fir or pine. All lumber shall be pressure treated using the "Empty Cell Process" as required by the American Wood Preservers Association (AWPA), with copper naphthenate solution. The copper naphthenate solution shall contain not less than 6% or more than 8% copper in the form of copper naphthenate, in petroleum oil, AWPA P9, conforming to AWPA P8 and C1 for oil-born preservatives.

2.4 FASTENER BOLTS

- A. Vandal-Proof Bolts and hardware shall be galvanized or cadmium plated steel. Bolts shall be 5/16" carriage bolts. Nuts installed with wood posts shall be NTPAFN fluted nuts. Nuts installed with tubular steel posts shall be vandal-proof tufnuts or approved equal. A known supplier is Intermountain Traffic Safety, West Valley City, Utah, phone (801) 972-6515.

2.5 CONCRETE

- A. Concrete for installing sign posts shall conform to the requirement of Section 033020.

2.6 SIGN BRACES

- A. All sign braces shall be made of aluminum alloy 6061-T6 and shall be a minimum 1/4" thick.
- B. Bolts shall be 3/8" X 1" Galvanized steel w/ stainless steel and nylon washers for attaching brace to sign and 3/8" X 3" Galvanized lag bolts with stainless steel and neoprene washers for attaching brace to post.

PART 3 - EXECUTION

3.1 POST SETTING

- A. Post holes not larger than 18-inch diameter shall be excavated to a depth as shown on the drawings. Backfill shall be hand tamped around the wood posts so as to make the post solid and plumb. Any loose posts shall be removed and re-installed. Concrete shall be used to backfill tubular steel posts as shown on the drawings.

3.2 SIGN MOUNTING

- A. Signs shall be mounted as shown on the drawings. Bolt holes shall be neatly drilled in the signs to prevent splintering and drilling through letters or delaminate sheathing. Sign face sheathing or coverings shall not be broken

3.3 GRADING AND CLEANUP

- A. The area around the post shall now be graded to the desired contour. When the installation is completed, the general area shall be cleaned up by removing all debris and material not utilized.

END OF SECTION

May 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 101415 – MINOR SITE ID SIGN (HDPE)

PART 1 - GENERAL

1.1 SUMMARY

- A. This item shall consist of fabricating and installing the Site Identification sign and support base.

1.2 REFERENCE SPECIFICATIONS

- A. The publication listed below form a part of this specification to the extent referenced. The publication is referred to in the text by the basic designation only.
- B. EM 7100-15 Standards for Forest Service Signs, Chapters 7, 8,10 and 14

1.3 SUBMITTAL

- A. Sample Sign Mock-Up: The contractor shall submit a list of signs proposed for the project. The list shall include detailed drawings showing the size of the sign, size of letters, colors, materials, sign texts, and location for installation. The Contracting Officer shall approve the submitted list, prior to the Contractor ordering the signs.

1.4 QUALITY ASSURANCE

- A. The Fabricator shall have sufficient covered shop space, tooling and equipment to properly fabricate, inspect and handle the signs. Raw materials shall be stored in a covered warehouse or similar cool, dry storage area so that it protects the raw materials from environmental factors.
- B. The fabricator shall be regularly engaged in the design and manufacture of plastic signs and shall have at least 5 years' experience. References with contacts shall be provided if requested by the Contracting Officer.

1.5 METHOD OF MEASUREMENT

- A. The quantity to be measured shall be the number of signs and sign bases fabricated, installed and accepted. Signs which require a Forest Service shield and other logos shall be considered as one sign.

PART 2 - PRODUCTS

2.1 REQUIREMENTS:

- A. Sign design shall follow the procedures and methods, utilize the equations and formulas, and incorporate the factors of safety and allowable design stresses and strains as set forth in ASTM D 4097, ASTM D 3299 and ASME RTP-1. Where design conflicts arise between the various standards, the most stringent design shall be used.
- B. The design shall allow for the most severe combination of conditions which may include any or all of the following:
 1. Superimposed loads such as those due to wind and seismic forces.
 2. Loads applied during transportation and erection.
 3. Thermal expansion and contraction.
- C. Prior to construction, the Contractor shall submit a sample sign and Logo mock-up. This sample must include enough detail, including materials, layout, font, color, size and shape to show consistency with the drawings and be in accordance with EM 7100-15. The submittal shall be combined into a single, neatly organized, complete package and submitted to the Contracting Officer for review and approval. Fabrication of the signs shall not begin until the complete submittal package has been approved by the government.
- D. Types of signs to be fabricated and installed include:
 1. Single sided HDPE sign panel
 2. Forest Service Logo
 3. USDA Credit Plaque
- E. A known manufacturer that meets this specification is Wood Products, 4890 Country Road 76, Parlin, CO 81239, Phone 970-641-1675.

2.2 MATERIALS

A. SUBSTRATE

1. The sign panel and logos shall be constructed of High-Density Polyethylene (HDPE) containing recycled plastic. The HDPE sheets consist of 1/8" top layer, a 1/2" middle core layer and a 1/8" bottom layer. The combination of layers shall provide a unified 3/4" thick panel free of any post manufacture jointing, unless requested. The surface of the front and back of the panel will be a matte finish.
2. The finished panel shall meet the following criteria:

a.	Density	.96-.97	g/cc
b.	Tensile Yield	3800-4800	psi
c.	Elongation	>600	%
d.	Tensile Impact	70-120	ft.lb./in.2
e.	Hardness	65-70	Shore D
f.	Flex Modulus	200,000 - 225,000	psi

2.3 COLOR

- A. Color for sign panel, Forest Service logo and USDA credit plaque shall be according to the standards found in EM 7100-15.

2.4 LUMBER

- A. Pressure-treated lumber, dimensioned as shown on the drawings. All lumber above ground shall be painted with two coats of exterior grade paint, Forest Service brown approved by the Contract. Paint below grade lumber with a wood preservative approved by the Contracting Officer.

2.5 HARDWARE

- A. All bolts, threaded rods, nuts and washers shall be galvanized or zinc-plated.
- B. Oly log screws: 3/16-inch shank by 10 inches in length with 5/16-inch hex head, coated for corrosion resistance or approved equal.

PART 3 - EXECUTION

3.1 SIGN FABRICATION

- A. Work shall be pre-formed by a high speed, computer-controlled electric router with the sheets in a horizontal position so that the routed surfaces are uniform in depth and finish.

3.2 ROUTING LETTERS AND LOGO'S

- A. Sign (not Logo's) shall be routed to a depth of 1/4" to 1/2" using Gothic type "D" letters. Do not go through material.

3.3 SIGN INSTALLATION

- A. Sign locations will be approved on the ground by the Contracting Officer, prior to commencement of work. The longitudinal axis of the sign will be perpendicular to the centerline of the main road, plus or minus 5 degrees.
- B. Base and sign supports shall be constructed plumb and at the angles show on the drawings.
- C. Mount signs with zinc coated hardware. Paint bolt heads, washers, nuts and other hardware to match "Rust" powder coated finish.
- D. Care shall be taken when drilling holes that locations avoid routed letters on signs and logos.
- E. Sign base shall be fabricated as shown on the drawings.

END OF SECTION

April 2021

USDA-FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 129300 – SITE FURNISHING

PART 1 - GENERAL

1.1 SUMMARY

- A. Extent of each type of site furnishings is indicated as shown on the drawings. Site furnishings shall be constructed and installed in accordance with FSORAG Section 4.0.
- B. Type of furnishings includes:
 - 1. Picnic Tables
 - 2. Fire Rings

1.2 SUBMITTALS

- A. Product Data: For each product specified.
 - 1. Submit manufacturers data and installation instruction for each type of furnishings, including anchorage and accessory items.

1.3 MEASUREMENT AND PAYMENT

- A. The quantity to be measured shall be the number of furnishings (EA) fabricated and accepted.
 - 1. Fire Rings
 - 2. Picnic Tables

PART 2 - PRODUCTS

2.1 PICNIC TABLES

- A. SIZE:
 - 1. Picnic tables shall be 92 inches long by a minimum of 33 inches wide. Table tops surface shall be 2 feet 9 inches off the finish grade. Picnic table benches shall be 92 inches long by 11 inches wide and be 18 inches from the finish grade. Minimum thickness of all areas to be 4 inches. Minimum weight to be 2,300 lbs.
- B. STYLE
 - 8. Picnic tabletops, bench supports, and table supports cast in one integral part. Multiple pieces glued or bolt together tables are not acceptable.
 - 9. Bottom of legs to extend entire width of the table to supply a large footprint, so that no base slab is required, allowing product to be placed directly on the finished surface.
 - 10. Picnic tabletop shall be 6 inches from inside edge of seat to permit standing once inside. Bench and table support shall not exceed 10 inches to allow easy access to the seating area. Picnic table to have a wheelchair clearance at both ends.

C. COLOR

1. Coloring agents are to be pure mineral oxides and shall be mixed integrally with cement. Color top coating is not acceptable. Color as selected from full range of colors.

D. MATERIALS

1. Table to be made with Portland Cement. Mix shall be minimum 8.5 sack per yard. Aggregates to be free of injurious amounts of organic impurities. Maximum rock size to be 3/4" in aggregate mix. Units to be reinforced with sizes #4 and #5 rebar grid. Fabricated so that all connection points are welded, including leg to tabletop, leg to seat interconnections. Rebar frame set a minimum of 2 inches from outside edges. Cured concrete shall attain minimum compressive strength of 7,000 PSI at 28-days.

E. FINISH

1. All formed surfaces and edges shall be rounded with rock textured finish for exterior surface of support legs, smooth finish for seats, and wood-grained surface for tabletop. Finish of all exposed surfaces to have a concrete graffiti-resistant sealer applied.
2. A known product that meets this specification is Model 104FSR manufactured by Outdoor Creations (hyperlink: [Outdoor Creations-Picnic Tables,Rectangle Tables](#)) or an approved equal.

2.2 ACCESSIBLE FIRE RINGS WITH GRILLS - The complete fixture shall be finished with a high-temperature, flat black paint, such as exhaust manifold paint, rather than the typical factory finish of aluminum paint.

- A. Fire Ring Assembly – The fire ring shall be a 3/16" thick steel plate x 18" side height with a single level hinged grate. The ring shall be provided with three 1/2" diameter "U" anchor pins. The fire ring shall be a No. FS-30/18/PA as manufactured by R.J. Thomas Co. Inc. of Cherokee, Iowa, phone no. 712-225-5115 or an approved equal. Model is also available with optional 8" high heat shield No. HS-F.

PART 3 - EXECUTION

3.1 INSTALLATION – PICNIC TABLES

- A. Placed table on finished surface according to the locations as indicated in the drawings and site plan, and FSORAG Section 4.0.

1. Tabletop surface shall be 2 feet 9 inches off the finish grade. Picnic table benches shall be 92 inches long by 11 inches wide and be 18 inches from the finish grade

3.2 INSTALLATION – FIRE RINGS

- A. Forming shall not be necessary if the walls of the hole are firm and will receive the concrete without sloughing and provided the excavation may be either round or square.
 1. Using bracing as necessary to ensure that fire ring remains plumb during concrete placement and curing.

2. Installation of Fire Rings shall conform with FSORAG Section 4.2.
 3. After the installation is completed, the area around the footing shall be backfilled to the established grade and compacted to the density of the surrounding natural ground.
- B. Apply Adhesive: Secure the concrete ring to the concrete pad with adhesive in accordance with the manufacturer's instructions. Not required if placed on gravel or native material.

END OF SECTION 129300

May 2024

USDA FOREST SERVICE, R3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 129700 - FEE TUBE

PART 1 - GENERAL

- 1.1 This item shall consist of furnishing and installing fee tube as described in the drawings and this section, in conformity with the lines and grades established, or as directed by the Contracting Officer.
- 1.2 METHOD OF MEASUREMENT - The quantity to be measured shall be the number of fee tubes furnished and installed in the location(s) identified and accepted.

PART 2 - PRODUCTS

- 2.1 FEE TUBE
 - A. The fee tube, minimum capacity 120 filler envelopes shall be FS1236 (shell), FS1236A (insert) and EB-1 (envelope box) as manufactured by Iron Rangers Iron Rangers , or an approved equal.
 1. Approved equal shall comply with Forest Service MTDC 931 specifications for vandal resistant fee tubes outlined in the periodical located at the hyperlink provided: <https://www.fs.usda.gov/t-d/pubs/pdfpubs/pdf98232343/pdf98232343.pdf>).
- 2.2 FOREST SERVICE SUPPLIED COMPONENTS
 - A. Padlock shall be supplied by the Forest Service.

PART 3 - EXECUTION

- 3.1 The fee tube shall be installed in accordance with drawings and manufacturer recommendations.
- 3.2 Excavate the new fee tube hole to the size and shape of the fee tube concrete footing in the location as identified by the contracting officer. Backfill around the concrete footing in 6-inch lifts and compact until natural grade.
- 3.3 The complete fee tube shall be installed straight and plumb to the depth AS SHOWN ON THE DRAWINGS.

END OF SECTION 129700
January 2017

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 133402 - UTILITY BOXES

PART 1 - GENERAL

1.1 SUMMARY

- A. This section includes underground utility boxes including plumbing appurtenances, lid and all work within 5 feet of the outside of the box.
- B. Related sections include the following:
 - 1. Section 033020 "Concrete from Package Dry Mix for Minor Structures."
 - 2. Section 221103 "Plumbing Appurtenances."
 - 3. Section 312000 "Earthwork."

1.2 DEFINITIONS

- A. CMP: Corrugated metal pipe
- B. HDPE: High density polyethylene
- C. GSP: Galvanized steel pipe

1.3 SUBMITTALS

- A. Product data for the following:
 - 1. Manhole and lid materials.
 - 2. Box insulation.
- B. Box and Lid Shop Drawings: Include plans and details.
- C. Product Certificates: Signed by manufacturer certifying that products furnished comply with requirements.

1.4 QUALITY ASSURANCE

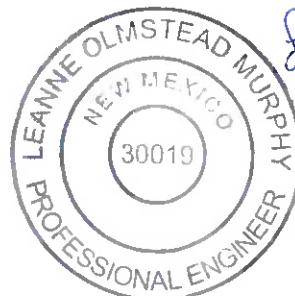
- A. Welding: Certified welder, qualify procedures and personnel according to AWS D1.1, "Structural Welding Code - Steel" or AWS D1.2 "Structural Welding Code - Aluminum."

1.5 MEASUREMENT AND PAYMENT

- A. Utility Boxes: Payment will be for the number of Utility Boxes shall be incidental to work associated with Section 221103 "plumbing appurtenances", including all pipe, gravel, excavation, backfill and all work within 5 feet of the box required to complete the installation.

PART 2 - PRODUCTS

2.1 GALVANIZED STEEL BOX AND LID
UTILITY BOXES



Leanne Murphy
6-24-25

133402-1

- A. Box: Galvanized steel CMP (14-gage), AASHTO M36.
 - 6. Dimensions: As indicated on the Drawings.
- B. Lid: Steel (12-gage), ASTM A283 or A36, hot dip galvanized after fabrication.
 - 6. Dimensions: As indicated on the Drawings.
 - 7. Stiffeners: Steel, ASTM A283 or A36.
- C. Welding: Use E60XX electrode.

2.2 LOCKING MECHANISM

- A. Provision shall be made to secure the lid to the box using a padlock and a locking rod as illustrated on the Drawings. The padlock will be furnished by the Forest Service.

2.3 GRAVEL

- A. Description: Hard, durable screened material varying in size from 1/4 inch to 1 inch, without lumps of soil or clay.
 - 1. "Special Bedding," as specified in Section 312000 "Earthwork" is acceptable.
- B. Gravel shall be placed to the depth indicated on the Drawings.

2.4 CONCRETE SUPPORTS

- A. Plumbing appurtenances shall be supported using concrete blocks as shown on the Drawings. Blocks shall be approved by the Contracting Officer.

2.5 WOOD POST (FOR VENT)

- A. Lumber: S4S, Redwood, construction grade, meeting requirements of Redwood Inspection Service.
- B. Stain: Moorwood solid color stain No. 080-64, as manufactured by Benjamin Moore & Co., or an approved equal.

2.6 PLUMBING APPURTENANCES

- A. According to Section 221103 "Plumbing Appurtenances."

PART 3 - EXECUTION

3.1 BOX INSTALLATION

- A. Excavation and Box Placement: Excavation shall be made with minimum damage to surrounding trees, vegetation, and adjacent slopes. Place the box as indicated on the Drawings. Compact the surface on which the box sets to 97 percent maximum density (AASHTO T99 or ASTM D698).
 - 1. Pipe Clearance: 1/2" to 3" clearance between the box and pipeline at any point.
- B. Backfilling

1. Remove all debris from the excavation before backfilling.
 2. Compaction: Backfill shall be placed in 6-inch maximum thickness layers and compacted by three passes of a mechanical vibrator which has been approved by the Contracting Officer.
 3. No rocks larger than 2-inches in diameter shall be placed within 6 inches of inlet and outlet pipes, or box.
 4. If native material is unsuitable for backfill, "Pipe Zone Material," as specified in Section 312000 "Earthwork" may be used.
- C. Gravel Placement: Smooth and compact the natural earth bottom of the box before placing the gravel layer. Gravel shall be placed in the bottom of the box to the thickness indicated on the Drawings.
- D. Installation of Plumbing Appurtenances: Install plumbing appurtenances as shown on the Drawings and according to Section 221103 "Plumbing Appurtenances." Support all appurtenances and piping near floor and on brick or concrete piers.
- E. Lid: The lid will be installed and secured as indicated on the Drawings.

3.2 MANHOLE INSTALLATION

- A. Install complete with appurtenances and accessories indicated.
1. Install precast concrete manhole sections with joint sealant according to ASTM C891.
 2. Core drill or use hole former for pipe openings in pre-cast base section of shallow manhole.
 3. Connect pipe to structures with specified pipe gasket.
 4. Set grade rings using mastic joint sealant between tank and grade ring and between grade rings. Center grade rings over opening. Grade rings shall be selected so as to locate the manhole cover within 1 inch above specified elevation.
 5. Set frame with mastic joint sealant. Center frame over opening in grade ring. Grout frame to manhole with non-shrink grout. Strike off grout and tool joint to smooth, dense, surface.
 6. Set tops of frames and covers flush with finished surface of manholes that occur in pavements or gravel roadways. Set tops 3 inches above finished surface elsewhere, unless otherwise indicated.

3.3 BOX INSULATION

- A. Insulate the box as indicated on the Drawings. Install and attach to box in manner approved by the Contracting Officer.
- B. Minimum Installed Insulation Thickness: 4 inches at all locations.

3.4 CLEANUP

- A. After backfilling and installation is complete, the general area shall be cleaned by removing all excavated material not used in the backfill and all debris and construction material. Finish-grade the area to blend into the surroundings and create a natural appearance.

END OF SECTION 133402

AUGUST 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 221100 - WATER DISTRIBUTION SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section shall consist of furnishing and installing the following:
 - 1. Potable water distribution pipe and fittings.
 - 2. Drains.
- B. This Section shall also include the flushing, testing and disinfection of waterlines.
- C. Related Sections include the following:
 - 1. Section 024100 "Waste Material Disposal."
 - 2. Section 221102 "Curb Valves and Boxes."
 - 3. Section 312000 "Earthwork."

1.2 DEFINITIONS

- A. GSP: Galvanized Steel Pipe.
- B. HDPE: High Density Polyethylene plastic.
- C. NPWL: Non-potable waterline (includes sewerline).

1.3 SUBMITTALS

- A. Product Data: For the following:
 - 1. Pipe and fittings.
 - 2. Drain and vent screens.
- B. Plan for disposal of water used for testing and disinfection of waterlines.
- C. Water System Operations and Maintenance Manual (including As-Built Drawings).
- D. Disinfection Report

1.4 QUALITY ASSURANCE

- A. All work is to be completed according to applicable Federal, State and Local codes. This work includes, but is not limited to, materials, installation, testing and disinfection
- B. Piping materials shall bear label, stamp, or other markings of specified testing agency.
- C. NSF Compliance:
 - 1. Comply with NSF 14 for plastic potable-water-service piping. Include marking "NSF-pw" on piping.
 - 2. Comply with NSF 61 Annex G for materials for water-service piping and

specialties for domestic water. All pipes, fittings, fixtures, solder and flux shall meet the 0.25% maximum lead content rules for all wetted components.

1.5 MINIMUM SYSTEM REQUIREMENTS

- A. Unless otherwise noted, the minimum working pressure for piping and specialties shall be 200 psig.

1.6 DELIVERY, STORAGE AND HANDLING

- A. Deliver piping with factory-applied end caps. Maintain end caps through shipping, storage, and handling to prevent pipe-end damage and to prevent entrance of dirt, debris, and moisture.
- B. Protect stored piping from moisture and dirt. Elevate above grade.
- C. Protect flanges, fittings, and specialties from moisture and dirt.
- D. Store plastic piping protected from direct sunlight. Support to prevent sagging and bending.

1.7 COORDINATION

- A. Coordinate connection to existing waterlines with Contracting Officer.
- B. Verify piping materials, sizes, entry locations and pressure requirements are compatible with connection point.
- C. Coordinate other utility impacts.

1.8 MEASUREMENT AND PAYMENT

- A. Payment will be as listed for the following items:
 - 1. Pipeline: Linear Feet of pipe of each type and size designated, including fittings, trench, tracer wire, non-potable and sewerline crossings, and connections to existing waterlines.
 - 2. Daylight Drains: Number of Daylight Drains, including excavation, galvanized steel pipe, fittings, screen, suction strainer, riprap, backfill and all work within 10 feet of the drain outlet, as shown on the Drawings.
 - 3. Water System Operations and Maintenance Manual: Shall be considered incidental to work outlined in the Schedule. Operation and maintenance manual includes start up and shut down procedures for the system as well as maintenance costs. Contractor shall provide one hard copy format and a digital format.

PART 2 - PRODUCTS

2.1 GENERAL

- A. All waterlines, pipes, and fittings shall be new and unused, of the type, pressure rating or class, and size specified and as shown on the Drawings.

2.2 PIPES AND FITTINGS

A. HDPE:

1. Pipe, AWWA C901:
 - a. Material Designation: PE 4710 resin
 - b. Material Classification: ASTM D1248
 - c. Minimum Cell Classification: ASTM D 3350, 345434C
 - d. Dimension Ratio and Minimum Pressure Rating: DR11, 200 psig.
2. Fittings:
 - a. Molded HDPE Fittings: ASTM D 3350, PE 4710 resin, butt-fusion type, made to match HDPE pipe dimensions and class.
 - b. Insert Fittings: Not allowed.
 - c. Transition Fittings to non-HDPE Pipe: Approved by HDPE pipe manufacturer. Compression-type fittings are not allowed.
3. Pipe Larger than 2-inch Diameter: Provide in 40-foot lengths rather than in rolls.

B. Galvanized Steel: ASTM A53, Schedule 40, threaded ends meeting ASME B1.20.1.

1. Fittings: Threaded, ASTM A 338.

2.3 PIPING SPECIALTIES

A. Dielectric Fittings: Combination of copper alloy and ferrous; threaded, solder, or plain end types; and matching piping system materials.

1. Dielectric Unions: Factory-fabricated union assembly, designed for 250-psig minimum working pressure at 180 deg F. Include insulating material that isolates dissimilar metals and ends with inside threads according to ASME B1.20.1.
2. Dielectric Flanges: Factory-fabricated companion-flange assembly, for 160- or 300-psig minimum working pressure to suit system pressures.
3. Dielectric-Flange Insulation Kits: Field-assembled companion-flange assembly, full-face or ring type. Components include neoprene or phenolic gasket, phenolic or polyethylene bolt sleeves, phenolic washers, and steel backing washers.
 - a. Provide separate companion flanges and steel bolts and nuts for 150- or 300-psig minimum working pressure to suit system pressures.
4. Dielectric Couplings: Galvanized-steel couplings with inert and noncorrosive thermoplastic lining, with threaded ends and 300-psig minimum working pressure at 225 deg F).
5. Dielectric Nipples: Electroplated steel nipples with inert and noncorrosive thermoplastic lining, with combination of plain, threaded, or grooved end types and 300-psig minimum working pressure at 225 deg F.

B. Daylight Drains and Overflows

1. Suction Strainer
 - a. Design: Galvanized (zinc plated), cold rolled, 16-gage steel strainer with threaded end to fit GSP drain pipe size.

- b. Model: “SSS” Series, Round Hole Strainer, as manufactured by Campbell Manufacturing Inc., or an approved equal.
- 2. Insect Screens
 - a. For Daylight Drains: 4-mesh woven wire cloth, galvanized steel.
- 3. Riprap
 - a. The stones may be obtained from on-site excavated material or a borrow source designated by the Contracting Officer. Stones used shall be such that not more than 10 percent of the individual stones shall have a perimeter less than 12 inches in the smallest section. All of the stones shall have a volume less than a 1/2 cubic foot.
 - b. The stones shall be sound, hard, durable and free from laminations, fractures, or other structural defects. They shall be of such quality that they will not disintegrate on exposure to water or weathering.

2.4 OPERATION AND MAINTENANCE MANUAL

- A. The Contractor shall furnish three sets of instructions for operating and maintaining the complete water system, including start-up and shut-down procedures, operator safety procedures, location of valves and other key system features, parts list and parts order form, instructions for draining the system, etc.

PART 3 - EXECUTION

3.1 EARTHWORK

- A. Refer to Section 312000 “Earthwork” for excavating, trenching, bedding, and backfilling.

3.2 PIPING INSTALLATION

- A. Construct the water system to the lines and grades shown or established in the field.
- B. Stake locations of waterlines as shown on the Drawings. Waterline location shall be approved by the Contracting Officer before installation.
- C. All waterline shall be graded to drain.
- D. All pipe, fittings, and appurtenances shall be handled and installed in strict conformance to the manufacturer’s recommendations.
- E. Drains, valves, and other regulating and controlling devices shall be installed in the line where called for on the drawings.
- F. The interior of all pipe, fittings, and other accessories shall be kept as free as possible from dirt and foreign matter at all times. Each piece of line as it is laid shall be cleaned of all debris. When the pipeline has become dirty on the inside during shipment or storage, it shall be swabbed out by drawing a damp swab through the line before placing in trench. Care shall be exercised to keep all joining surfaces clean. Any pipe with contaminated or damaged joining surfaces, which cannot be satisfactorily cleaned or

repaired, shall be discarded. Under no circumstances shall pipe be installed in water, and no pipe shall be installed when trench conditions or the weather is unsuitable for such work. At all times when work is not in progress, all open ends of the pipe and fittings shall be securely closed to the satisfaction of the Contracting Officer so that no trench water, rodents, earth, or other substance will enter the pipe or fittings.

- G. Any section of pipe already installed and found to be defective shall be removed and replaced with new pipe at the Contractor's expense.
- H. Water and Non-Potable/Sewer Line Crossings
 - 1. Potable and NPWL lines shall never be installed in the same trench.
 - 2. If a new line crosses below an existing line, the existing line shall be supported to prevent settling.
 - 3. Where waterlines intersect NPWLs or are within 10 horizontal feet of NPWLs, the following applies:
 - a. At least one of the pipes (either the waterline or NPWL) shall be constructed or reconstructed with 200 psi pipe or encased in a continuous sleeve with same rating. At crossing, extend 200 psi pipe or sleeve for a distance of 10 feet perpendicular distance from each side of the waterline. The waterline and NPWL pipes or sleeves shall be centered at the crossing so that the joints will be an equal distance and as far as possible from the crossing.
 - 1) Sleeves: Nominal sleeve diameter to be at least 3 inches larger than nominal diameter of pipe to be encased. Install end seals in annular space at each end of sleeves.
 - 2) In lieu of constructing or reconstructing the NPWL with pipe conforming to waterline requirements, the NPWL may be encased in concrete at least 6 inches thick around the pipe measured at the bell, for 10 feet each side of the waterline.
 - b. Construct according to applicable federal, state and local regulations. The Contractor is responsible for obtaining and following these regulations.
 - c. Where applicable, state and local regulations supersede the above requirements.
- I. Culvert Crossings: Where waterline crosses a road culvert, adjust grade of waterline to provide a minimum of 18 inches of separation between the waterline and culvert, maintaining slope of waterline to drain.
- J. A minimum of one-foot separation shall be maintained between electric and waterlines in the same trench. Unless the electric conductors are in conduit the waterline shall be a minimum of one foot below the electric conductors.
- K. Bury piping with depth of cover over top at least 48 inches.
- L. Cutting and Handling of Pipe
 - 1. Cutting of pipe for closure pieces or for other reasons shall be done in a neat manner by methods, which will not damage the pipe. The pipe and fittings shall be handled in such a manner as to ensure delivery and final placement in good,

undamaged condition. Care shall be exercised not to injure the pipe surfaces or coatings. Damaged pipe shall not be used in the work.

M. Laying Pipe

1. No pipe shall be placed in the trench until the Contracting Officer has approved the trench and bedding.
2. Do not drop pipe into trench. If HDPE pipe is fusion welded above ground outside of the trench, gently lay pipe into the trench. Ensure installation procedures meet the manufacturer's requirements.
3. Pipes shall be placed true to the lines and grades shown on the Drawings and as specified. Complete water system shall be constructed to drain by gravity to drains for seasonal shutdown. Waterlines shall be constructed to drain at a minimum grade of **1%**. Waterlines not meeting this requirement shall be removed by the Contactor and replaced at no cost to the Government.
4. Each section of the pipe shall rest upon the pipe bed for the full length of its barrel.
5. Backfill the pipe within 24 hours after installing the pipe in the trench, leaving the joints and one foot of pipe on each side of each joint exposed until testing is complete.
6. When work is not in progress, securely close open ends of pipe and fittings.
7. When high density polyethylene or solvent weld joint pipe is used, it shall be weaved in the trench to provide horizontal slack in the line for expansion and contraction.
8. High density polyethylene pipe shall be allowed to cool to within 20°F of the shaded trench bottom temperature before backfilling.
9. When pipes are installed in a structure or utility box, they shall be adequately supported.
10. Use proper fittings in order to follow the required line and grades and to avoid excessive curvature. Deflections from a straight line or grade, as required by vertical curves, horizontal curves, or offsets, shall not exceed the manufacturer's recommended maximum joint deflection or curvature for their pipe.
11. When more than one pipeline is in the same trench, the pipes shall be separated by at least eight horizontal inches, or as shown on the Drawings.
12. Bends that result in joint tension or permanent pipe deformation are prohibited.
13. Pipelines may be relocated slightly, if necessary, to avoid existing obstacles. However, the required slopes must be maintained.

N. Branch Connections

1. Branch connections to the main shall be made with tees.
2. Horizontal tees for waterline laterals shall not be permitted. The branch side of tees shall be up when drainage is toward the tee, and down when drainage is away from the tee.
3. Waterline tees shall be the same diameter as the largest pipe connected to the tee.

O. Pipe Joints

1. Joints shall be made, using jointing materials and applied with the proper accessories, in accordance with the manufacturer's instructions. Completed joints shall be watertight, and capable of passing the required hydrostatic testing. If

- pipe and fittings are assembled with a lubricant, it must be nontoxic.
2. Dissimilar Joints: Connections between two dissimilar pipes or fittings shall be installed in manner recommended by the pipe manufacturer and approved by the Contracting Officer. Use adapters compatible with both piping materials, outer diameters and system working pressure.
 - a. For HDPE pipelines:
 - 1) Use HDPE/GSP transition fittings for connections to galvanized steel pipe.
 - 2) Use HDPE/PVC transition fittings for connections to PVC pipe.
 - b. For other situations, use fittings to suit the actual conditions.
 - c. Compression-type fittings are not allowed.
 3. Threaded Metal Joints:
 - a. Thread pipes with tapered pipe threads according to ASME B1.20.1, apply tape or joint compound, and apply wrench to fitting and valve ends into which pipes are being threaded.
 - b. Exposed threaded pipe joints on galvanized steel pipe shall be painted with mastic following the manufacturer's application procedures.
 - c. Field cut ends shall be squared off by grinding or filing; ends shall be reamed with all chips and flakes removed.
 - d. Field cut threads or machined joints shall be made with sharp tools to sharp finished ends.
 4. HDPE Heat-Fusion Joints
 - a. According to ASTM D 2657 and piping manufacturer's written instructions.
 - b. The joining method shall be butt fusion method and shall be performed in accordance with the pipe manufacturer's recommendations. Butt fusion equipment used in the joining procedures shall be capable of meeting all conditions recommended by the pipe manufacturer, including but not limited to, temperature requirements, alignment, and fusion pressures.
 - c. Personnel trained in butt fusion techniques by the manufacturer of the pipe, shall supervise or perform butt fusion operations.
 - d. Butt fusion shall be performed between pipe ends or pipe ends and fitting and valve outlets, of like outside diameter and wall thickness, SDR or DR. Butt fusion joining between same diameters but unlike wall thickness shall not be permitted. Transitions between unlike wall thickness shall be made with a transition nipple (a short length of the heavier wall pipe with one end machined to the lighter wall) or by mechanical means.
 - e. Branch connections to the main shall be made with polyethylene fittings, and butt fused.
 - f. Bent Strap Test: On each day that butt fusions are to be made, perform a "Bent Strap Test." The first fusion of the day shall be a trial fusion. The trial fusion shall be allowed to cool completely, and then fusion test straps shall be cut out. The test strap shall be 12" (min) or 30 times the wall thickness in length with the fusion in the center, and 1" (min) or 1.5 times the wall thickness in width. Bend the test strap until the ends of the strap

touch. If the fusion fails at the joint, a new trial fusion shall be made, cooled completely and tested. Butt fusion of pipe to be installed shall not commence until a trial fusion has passed the bent strap test.

- P. Install piping by directional boring, tunneling, jacking, or combination, under streets and other obstructions that cannot be disturbed.

3.3 IDENTIFICATION

- A. According to Section 312000 "Earthwork."
- B. Wrap all buried, non-metallic, nonperforated piping with tracer wire.
- C. Use detectable warning tape over piping and at outside edges of underground structures.

3.4 DAYLIGHT DRAINS

- A. Transition from water system piping to galvanized steel pipe 10-feet from atmospheric outlet of daylight drain.
- B. Install suction strainer and insect screens on the atmospheric outlet of daylight drains. Attach insect screens to pipe with stainless steel hose clamps.
- C. Minimum grade of slope of drain shall be **1%**.
- D. Place riprap securely around outlet with a minimum 12" air gap.

3.5 HYDROSTATIC TESTING

A. General

1. The Contractor shall conduct pressure and leakage tests on all newly constructed portions of the water system, including water mains, service lines, fittings, valves, and hydrants. New waterlines that are extensions or modifications of an existing water system shall be tested at the test pressure specified here unless the Contracting Officer specifies another pressure. Pneumatic testing will not be allowed.
2. The contractor shall ensure that any appurtenance attached to the pipeline at the time of testing is rated to withstand the pressures that will exist during testing.
3. Conduct piping tests before joints are covered and after thrust blocks, if required, have hardened sufficiently. Joints shall remain uncovered until testing is completed satisfactorily.
4. The Contractor shall furnish the pump, pipe, gauge, measuring device, connections, and all other necessary equipment, and shall furnish the necessary personnel to conduct the tests. All equipment, gauges, and attachments shall be subject to approval by the Contracting Officer.
5. The Contracting Officer shall be present during the testing period.
6. Waterlines shall be tested in sections not to exceed 2,000 ft in length or a maximum pressure differential of 30 psi in the test section.
7. No paints, asphalts, tars, enamels, or other types of pipe compounds shall be used to eliminate leaks.

B. Pressure and Leakage Test

1. The test pressure shall:
 - a. Equal at least 125 psi at the highest elevation of the test but not exceed 160 psi at any point being tested.
 - b. Not vary by more than ± 5 psi from the established test pressure.
 - c. Not exceed twice the rated pressure of valves when the test section includes closed valves.
 - d. Be not less than 100 psi for all piping within buildings.
2. Pressure Test Procedure
 - a. Equipment: The test pressure shall be applied by means of a pump connected to the pipe in a manner satisfactory to the Contracting Officer.
 - b. Preparation: Fill pipeline 24 hours before testing. Use only potable water. Each valved section of pipeline shall be slowly filled with water and all air shall be expelled from the pipeline. Apply test pressure to stabilize system.
 - c. Pressurization:
 - 1) Increase pressure in 50-psig increments and inspect each joint between increments.
 - 2) Hold at test pressure for 1 hour.
 - 3) Decrease to 0 psig.
 - 4) Slowly increase again to test pressure and hold for 1 more hour. If the test pressure cannot be sustained within ± 5 psi during this time, then the system will be determined to have leakage.
 - 5) Leakage shall be verified by the Contractor and the Contracting Officer.
 - 6) Locate and repair leaking joints with new materials and repeat test until leakage is within allowed limits. All visible leaks are to be repaired regardless of the amount of leakage.
 - d. Examination: All exposed pipe, fittings, valves, and joints shall be carefully examined. Any cracked or defective pipe, fitting, or valve discovered shall be repaired or replaced by the Contractor at his own expense with new material and the test repeated until satisfactory to the Contracting Officer.
3. Results
 - a. The Contractor shall furnish a written report to the Contracting Officer describing the results of each test. The report must identify the specific portions of the pipeline tested, the pressure, the duration of the tests, and the amount of leakage.

3.6 CLEANING AND DISINFECTION

A. Clean and disinfect water-distribution piping as follows:

1. Purge new water-distribution piping systems and parts of existing systems that have been altered, extended, or repaired before use.
2. Use purging and disinfecting procedure prescribed by authorities having jurisdiction or, if method is not prescribed by authorities having jurisdiction, use

procedure described in AWWA C651 or as described below:

a. Flushing Pipelines

- 1) After the pressure and leakage testing has been completed, all water pipes shall be thoroughly flushed.
- 2) Flushing shall remove all debris from the pipeline, including HDPE pipe shavings, gravel and any other material that may have accidentally entered the pipeline during construction.
- 3) Remove all self-closing faucets and drinking fountains from hydrants before flushing lines. Failure to do so will likely clog the internal parts of the faucet.
- 4) If no hydrants or other outlets are installed at the end of the waterline, an outlet at the end of each lateral shall be provided. Outlets shall be large enough to develop a velocity in the waterlines of at least 2.5 feet per second (see Table 1 for necessary flows). If necessary, use a booster pump or other method to obtain the necessary flows.
- 5) Minimum velocities must be obtained in all mainlines, laterals and drainlines.
- 6) Ensure flushing outlets have adequate drainage and no damage or erosion will occur from the flushing process. Install hoses and/or pipe extensions if necessary.
- 7) As required in AWWA C651 Section 5.1.1.2, provide.

TABLE 1
Required Flow in Gallons per Minute
(GPM) to provide 2.5 feet per second
Flushing Velocity

Type of Pipe			
Nominal Pipe Size (inches)	HDPE (SDR 11)	PVC (160 psi)	Galv. Steel (Schedule 40)
1	7	9	7
1 ¼	11	15	12
1 ½	14	19	16
2	22	30	27
2 ½	-	44	38
3	49	64	58
3 ½	-	84	77
4	81	106	100
5	-	162	156
6	175	229	225
8	274	388	390
10	461	603	615
12	648	848	872

b. Requirement of Chlorination

- 1) Pipelines: After flushing, all new waterlines and repaired portions of,

or extensions to, existing waterlines shall be disinfected by chlorination. Chlorination shall provide a minimum of 25 mg/l residual after 24 hours contact in the system. This may be expected with an application of 50 mg/l, although some conditions may require more. The entire system including the storage tank or spring box shall be filled with the disinfecting solution. All taps, hydrants, and other outlets shall be opened and left open until chlorine is noticeable by odor or testing in the water coming from each. Immediately close taps, hydrants, and other outlets. The chlorinated water shall be retained in the waterlines for at least 24 hours. An alternative of introducing a chlorine solution of 300 mg/l for a period of 3 hours is also approved.

- 2) Reservoirs Only: For large reservoirs where water is not going to be reused to disinfect connecting distribution lines, or where it is not economical to treat the entire contents with 50 mg/l of chlorine, disinfection can be accomplished by swabbing or spraying the walls, floor, and ceiling with a chlorine solution containing 300 mg/l of available chlorine. A minimum of 3 hours shall elapse before the structure is flushed and returned to service.

c. Preparation of Disinfecting Mixture: Chlorine-bearing compounds used for disinfection shall comply with ANSI/NSF 60.

- 1) Calcium hypochlorite must be prepared as a water mixture for introduction into the waterlines. The powder should first be made into a paste and then thinned to either a 1 percent or 5.25 percent chlorine solution.
- 2) Sodium hypochlorite is already a water mixture.
- 3) Table 2 gives the amount of chlorine-bearing compound to mix to obtain either a 1 percent or 5.25 percent solution.
- 4) Table 3 gives the amount of a 1 percent or 5.25 percent solution required to obtain a 50 mg/l solution for various volumes.
- 5) Table 4 gives the volume of water contained in each 100 feet of pipe of various sizes.

TABLE 2

Amount of Chlorine-bearing Compound to Mix to Obtain a 1% or 5.25% Solution				
Product	1% Solution		5.25% Solution	
	Amount of Compound	Gallons of Water	Amount of Compound	Gallons of Water
Calcium Hypochlorite (65% Cl)	1 pound	7.67	1 pound	1.36
Sodium Hypochlorite (5.25% Cl)	1 gallon	4.25	1 gallon	0

TABLE 3

Amount of Chlorine-Water Solution Required to make 50 mg/l Chlorine Solution for Various Volumes		
Volume (Gallons)	5.25% Solution	1.00% Solution
100	1.52 cup	0.50 gal.
500	0.48 gal.	2.50 gal.
1,000	0.95 gal.	5.00 gal.
5,000	4.76 gal.	25.0 gal.
10,000	9.52 gal.	50.0 gal.
50,000	47.6 gal.	250 gal.
100,000	95.2 gal.	500 gal.

TABLE 4

Volume in Gallons per 100 feet of Pipe			
Nominal Pipe Size (inches)	Type of Pipe		
	HDPE (SDR 11)	PVC (160 psi)	Galv. Steel (Schedule 40)
1	4.6	5.8	4.5
1 1/4	7.3	9.6	7.8
1 1/2	9.6	12.6	10.6
2	15.0	19.6	17.4
2 1/2	-	28.8	24.9
3	32.6	42.6	38.4
3 1/2	-	55.6	51.4
4	53.8	70.4	66.1
5	-	107.5	103.9
6	116.7	152.6	150.1
8	182.8	258.6	259.9
10	307.2	401.8	409.6
12	432.2	565.2	581.4

6)

- d. Method of Chlorine Application: The preferred method is to fill the storage tank (for a distribution system) or the spring head box (for a supply system) with chlorinated water of the proper strength. The water shall then be released through the outlet lines to the remainder of the system and retained there for the required period of time. If either of these points of application is unavailable, chlorinated water of the proper strength may be added to the beginning of the pipeline extension, or to any valved section of the extension, through a corporation stop installed by the Contractor in the top of the pipe. Other methods may be approved by the Contracting Officer.
- e. Testing of Chlorine Residual: To ascertain the concentration, the chlorine residual shall be measured by the Contracting Officer in accordance with the procedures described in the current edition of "Standard Methods for the

Examination of Water and Wastewater,” or AWWA M12, or by using an appropriate high-range chlorine test kit

- f. Final Flushing and Testing: Following the retention period, as specified in this Section, all heavily chlorinated water shall be thoroughly flushed from the pipeline until the chlorine concentration in the water leaving any part of the line is no higher than 1 mg/l.
- g. Disposal:
 - 1) Heavily chlorinated water shall not be drained directly into streams, rivers, lakes or drainage ditches which lead to surface water.
 - 2) When emptying the lines or tank after the 24-hour period, whether the system has passed or failed, the chlorine solution shall be neutralized before being discharged. Neutralize system by adding sodium thiosulfate to the storage tank in accordance with Table 5 below. Let the sodium thiosulfate sit in the storage tank for 24-hours and then release the water from the tank and flush the system. Residual chlorine at the outlets shall be minimal.

TABLE 5
Amounts of Chemical Required to Neutralize
Various Residual Chlorine Concentrations in
1,000 gallons of Water

Residual Chlorine Concentration mg/L	Pounds of Sodium Thiosulfate (Na ₂ S ₂ O ₃ – 5H ₂ O)
1	0.012
2	0.024
10	0.12
50	0.60

- h. Bacteriological Examination: After the system has been disinfected as specified, but before the new water piping is put to use or connected to existing piping, the Contractor shall test the water for bacteriological contamination from representative points in the system as approved by the Contracting Officer. The water shall be tested[**in accordance with AWWA C651-05. Two sets of samples, taken 24 hours apart shall be tested**] for presence/absence of coliform organisms in accordance with State drinking water regulations. If coliform bacteria are present, testing shall be repeated in accordance with State drinking water regulations. If the report of this examination is unsatisfactory, the system shall be flushed and the disinfection procedure repeated until the results of bacteriological examinations are satisfactory.
 - 1) Contractor shall provide Disinfection Report or sign a notarized affidavit that the disinfection was completed in conformance with the appropriate AWWA standards to the Contracting Officer (NMAC 20.7.10.201.T(2). Test frequency indicated on the drawings.

END OF SECTION 221100

May 2024

WATER DISTRIBUTION SYSTEMS

221100-17

USDA FOREST SERVICE, R3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 221102 - CURB VALVES AND BOXES

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:
 - 1. Curb Valves and Boxes.
- B. Related Sections include the following:
 - 1. Section 312000 "Earthwork."

1.2 QUALITY ASSURANCE

- A. All work is to be completed according to applicable Federal, State and Local codes.
- B. NSF Compliance:
- C. All products in contact with potable water shall comply with NSF 61 Annex G. All pipes, fittings, fixtures, solder and flux shall meet the 0.25% maximum lead content rules for all wetted components.

1.3 MEASUREMENT AND PAYMENT

- A. Payment will be as listed for the following items:
 - 1. Curb Valves and Boxes: Number of valves of each type and size designated, including boxes, operating rods, lid keys and all work within five feet of the center of the valve box.
 - 2. Exception: Water lines shall be paid for separately under Section 221100 "Water Distribution Systems."

PART 2 - PRODUCTS

2.1 CURB VALVES AND BOXES

- A. Valves: Valves shall be the same nominal size as the line in which they are installed or as indicated on the Drawings.
 - 1. Valve Operating Wrenches: Steel, tee-handle with one pointed end, stem of length to operate deepest buried valve, and end matching curb valve. Furnish one valve operating wrench or shut-off rod with every five (or fraction thereof) curb boxes installed, as indicated on the Drawings.

- a. Valve Operating Extension for Poly Valves: Polyethylene or PVC, with adapter to fit valve, extend to within 4 inches of ground level.
- 2. Poly Valves
 - a. NSF Approved (Potable water systems only)
 - b. Valves 4-Inch and Larger: AWWA C906.
 - c. Valves Smaller Than 4-inch: Similar quality to AWWA C906.
 - d. Opening (Bore): Full.
 - e. Dimension Ratio and Pressure Rating: Same as pipe in which the valve is installed
 - f. Model: Poly-Water Valves, as manufactured by Nordstrom Audco Inc., or an approved equal.
- B. Curb Boxes:
 - 1. Lids: Lockable, with pentagon-type bolt, compatible with valve box; labeled "WATER" or "SEWER" respectively.
 - 2. Pentagon Lid Keys: Furnish one key with every five (or fraction thereof) curb boxes installed.
 - 3. Curb Boxes for Poly Valves:
 - a. Design: Slider or screw type, Buffalo pattern.
 - b. Bottom Sections: Compatible with valve.
 - c. For 1-Inch Valves: 2-1/2-Inch shaft, Series 200 or 250 Plastic Boxes, with #002 bottom sections, as manufactured by Bingham and Taylor or an approved equal.
 - d. For 2-Inch Through 6-Inch Valves: 5-1/4-Inch shaft, Series 500 Plastic Boxes, as manufactured by Bingham and Taylor or an approved equal.
 - 1) Bottom Sections for 2" or 3" Valves: #104

PART 3 - EXECUTION

3.1 EARTHWORK

- A. Excavating and backfilling for valves and boxes is specified in Section 312000 "Earthwork."

3.2 CURB VALVE AND BOX INSTALLATION

- A. Crushed Gravel Bed: Under each valve, 12-inch minimum diameter, 12-inch minimum depth.
- B. Knockouts: Remove knockouts from valve box.
- C. Curb Box Position: Centered over valve bonnet, no part in contact with valve. The curb valve box shall not rest on the pipe but shall be fully supported by the gravel base.
- D. Curb Box Lid: Flush with finished grade, unless indicated otherwise on the Drawings.

END OF SECTION 221102
May 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 221103 - PLUMBING APPURTENANCES

PART 1 - GENERAL

1.1 SUMMARY

- A. This section includes piping, fittings, valves, gauges, backflow preventers, hypochlorinators and other plumbing devices within utility buildings, utility boxes or at other designated locations.
- B. Related sections include the following:
 - 1. Section 133400 "Precast Concrete Utility Building."
 - 2. Section 133402 "Utility Box."
 - 3. Section 260500 "Common Work Results for Electrical."

1.2 SUBMITTALS

- A. Product Data: For the following:
 - 1. Air and vacuum valves.
 - 2. Backflow preventers and assemblies.
- B. Operation and Maintenance Data: To be used in emergency, operation, and maintenance manuals. Include the data for the following:
 - 1. Air and vacuum valves.
 - 2. Backflow preventers.

1.3 MINIMUM SYSTEM REQUIREMENTS

- A. Unless otherwise noted, the minimum working pressure for piping and specialty valves and devices shall be **200** psig.

1.4 QUALITY ASSURANCE

- A. NSF Compliance:
 - 1. All products in contact with potable water shall comply with NSF 61 Annex G. All pipes, fittings, fixtures, solder and flux shall meet the 0.25% maximum lead content rules for all wetted components.

1.5 MEASUREMENT AND PAYMENT

- A. Combination Air Release Valve and Box: Shall be the number of valves and valve boxes (EA) installed in accordance with the drawings.

- B. Chlorinator Building Plumbing: Shall be lump sum (LS) for new plumbing installed as shown on the drawings.

PART 2 - PRODUCTS

2.1 GENERAL

- A. All plumbing appurtenances shall be new and unused, of the type, pressure rating or class, and size specified and as show on the Drawings.

2.2 WATER METER

- A. Description: AWWA C700, displacement-type, low-lead bronze main case, NSF-61 Annex G certified. Register flow in gallons unless cubic feet are indicated.
- B. Model: 3/4" Low Lead SR as manufactured by Sensus Metering Systems, or an approved equal.

2.3 BALL VALVE

- A. Description: Bronze, 400 psi, threaded NPT end connections.
- B. Model: LFB6000 Lead Free Ball Valve as manufactured by Watts Regulator, or an approved equal.

2.4 HOSE BIBB

- A. Description: Rough brass finish, male inlet, 3/4-inch hose thread outlet.
- B. Model: Manufactured by Arrowhead Brass Products, Chicago Faucet Company, or an approved equal.
 - 1. Vacuum Breaker: Model #LF8A Lead Free, ASSE 1011, Non-removable vacuum breaker, as manufactured by Watts Regulator Company, or an approved equal.
 - 2. Hose: Provide 10-foot length of 3/4-inch plastic hose for connection to vacuum breaker.

2.5 CHECK VALVE

- A. Description: Bronze, AWWA C508, swing-check type with 125-psig working-pressure rating, threaded ends, screwed-in cap, renewable brass disc and resilient seat. Include interior coating according to AWWA C550 and ends to match piping.
- B. Model: Figure No. 578 as manufactured by Powell Company, or an approved equal.

2.6 UNION

- A. Description: Female type, malleable iron, 150 pound, with ground brass to iron seat.
- B. Model: Manufactured by Crane Company, or an approved equal.

2.7 PRESSURE GAUGE

- A. Description: 2-inch diameter or larger, 0 to 100 psi range, with brass movement, steel case and bottom mounting with NPS 0.25 threads. Include 1/4" ball valve for gauge isolation.
- B. Model: Ashcroft Model 1000 or 1005 as manufactured by Dresser Industries, or an approved equal

2.8 SAMPLE TAP

- A. Description: Brass, 1/2-inch hose bib with male inlet and smooth-nosed outlet.
- B. Model: P/N 301PE as manufactured by Arrowhead Brass, or an approved equal.

2.9 WYE-STRAINER

- A. Description: 125 psi minimum working pressure, semi-steel or bronze body with threaded ends, with brass plug in blow-off connection and removable stainless steel #30 mesh screen.
- B. Model: Style B or F, as manufactured by O. C. Keckley Company, or Model: P/N LF777S, as manufactured by Watts Regulator Company, or an approved equal.

2.10 PIPE AND FITTINGS

- A. Galvanized Steel: ASTM A53, Schedule 40, threaded ends meeting ASME B1.20.1.
 - 1. Fittings: Threaded, ASTM A 338.

2.11 AIR RELEASE VALVE FOR UTILITY BUILDINGS

- A. Description: AWWA C512, hydromechanical device, capable of automatically releasing accumulated air under pressure, with cast iron body, stainless steel float, 175-psig rated working pressure, and NPS 1 threaded inlet.
- B. Model: No. 50 Air Vent Valve as manufactured by the APCO Valve and Primer Corporation, or an approved equal.

2.12 COMBINATION AIR VALVE FOR PIPELINES

- A. Description: AWWA C512, single body, float-operated, hydromechanical device with two orifices, cast iron body, stainless steel float, **150**-psig rated working pressure, and NPS 1 threaded inlet.
 - 1. Large Orifice Function: Automatically releases large quantities of accumulated air under pressure or admits large quantities of air during filling of piping
 - 2. Small Orifice Function: When large orifice is closed, releases small pockets of air under pressure, automatically and independently of the large orifice.
- B. Acceptable Models:

1. Model: No. 143C Combination Air Valve as manufactured by the APCO Valve and Primer Corporation, or an approved equal.

2.13 PRESSURE RELIEF VALVE FOR UTILITY BUILDINGS

- A. Model: P/N 30L as manufactured by Watts Regulator Company, or an approved equal.
- B. Working Pressure: 100 psi

2.14 PRESSURE RELIEF VALVE FOR USE WITH PRESSURE-REGULATING VALVES

- A. Description: Direct acting, spring loaded, diaphragm type pressure relief valve.
- B. Pressure Range: 20 to 105 psi
- C. Size: 3/4-inch
- D. Model: 55F as manufactured by Cla-Val Company, or an approved equal.

2.15 SUPPORTS FOR PIPES AND APPURTENANCES

- A. Description: Galvanized steel framing system, pipe clamps and hangers.
- B. Manufacturer: Unistrut Corporation or an approved equal.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install piping, fittings, valves, gauges, and other plumbing devices within utility buildings, utility boxes or at other locations according to the manufacturer's written instructions and as indicated on the Drawings. Install so water can be drained from each appurtenance for winterization of the system.

3.2 COMBINATION AIR VALVE FOR PIPELINES INSTALLATION

- A. Install combination air valve as shown on the Drawings. Support the air valve using pipe supports as specified in Part 2. Air valve shall be securely attached to the utility box wall through the supports. The completed installation shall fully support the weight of the air valve and prevent unnecessary stress on the HDPE pipe riser. Installation method subject to Contracting Officer approval.

END OF SECTION 221103

May 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 221105 - WATER HYDRANTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Provides for furnishing and installing hydrants as follows:
 - 1. Type A: A self-closing faucet.
 - 2. Type B: A self-closing faucet and hose bibb with backflow preventer.
 - 3. Type C: Two hose bibbs, upper bibb with vacuum breaker, lower bibb with backflow preventer.
- B. Related Sections include the following:
 - 1. Section 024100 "Waste Material Disposal."
 - 2. Section 033000 "Cast-In-Place Concrete."
 - 3. Section 221100 "Water Distribution System."
 - 4. Section 321204 "Crushed Aggregate Base or Surface Coarse."

1.2 LOCATION

- A. General location and orientation of all hydrants are shown on the drawings.
- B. Exact location and orientation shall be staked and approved by the Contracting Officer.

1.3 SUBMITTALS

- A. Manufacturer's Literature and Maintenance Data: Faucet, hose bibb, vacuum breaker and backflow preventer.
- B. Certification that washed rock meets the gradation requirements of this section.
- C. Maintenance information to be included in maintenance manual.
- D. Stain for Wood Post: Color Sample.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Products shall be kept free from damage during delivery, storage, and handling. Protect from dirt and moisture during storage.

1.5 MEASUREMENT AND PAYMENT

- A. Hydrants: Number of hydrants of each (EA) type designated, including concrete pad, aggregate and all work within ten feet of hydrant post.

PART 2 - PRODUCTS

2.1 GENERAL MATERIALS

- A. Lumber: S4S, Redwood, construction grade, meeting requirements of Redwood Inspection Service.
- B. Stain: Moorwood solid color stain as manufactured by Benjamin Moore & Co. or an approved equal.
 - 1. Color shall be Forest Service Brown as selected by the Contracting Officer.
- C. Framing Anchors: 16 gage, galvanized, 3-1/4 inches high with 1-1/4 inch long flanges
 - 1. Model: GA2 model angle connector, as manufactured by Simpson Strong, or an approved equal.
- D. Galvanized Steel Drain Grate
 - 1. Design: 1/2" x 2" on-center press-locked close mesh, ADA compliant, hot-dipped galvanized steel, 23-3/4" x 23-3/4" square, and 1-1/2" depth with welded lugs.
 - 2. Model: 8-P-2 as manufactured by Alabama Metal Industries Corporation (AMICO), or an approved equal.
- E. Washed Rock for Gravel Sump: 3/4-inch to 1-inch washed and screened material, without lumps of soil, clay or organic material.

2.2 FOREST SERVICE STANDARD HYDRANTS (TYPES A, B & C)

- A. Pipe and Fittings: As specified in Section 221100 "Water Distribution System."
 - 1. Galvanized Steel Pipe and Fittings: Schedule 40.
- B. Plumbing Appurtenances
 - 1. Self-Closing Faucet
 - a. Design: 1/2-inch, plain end, chrome or rough brass finish, self-closing lever handle, ADA compliant.
 - b. Model: 6252EHLF, as manufactured by Haws Corporation, or an approved equal.
 - 2. Hose Bibb
 - a. Design: 1/2-inch FIP, rough brass finish, loose key lockshield, with backflow preventer.
 - b. Model: 253BFPLK "No Kink", as manufactured by Arrowhead Brass Products, Inc., or an approved equal.
 - c. Provide two operating keys for first five hydrants, then one additional key for each additional five hydrants.
 - 3. Stop Valve

- a. Design: 3/4-inch curb valve, with removable key, FIP thread and drain.
- b. Model: Mark II H-10284 as manufactured by the Mueller Company, or an approved equal.
- c. Provide two operating keys for first five hydrants, then one additional key for each additional five hydrants.

4. Vacuum Breaker

- a. Design: ASSE 1011, Brass body, anti-siphon, non-removeable, 3/4-inch HTF inlet, 3/4-inch HTM outlet.
- b. Model: 8A, as manufactured by Watts Regulator, or an approved equal.

2.3 PAD SURFACING

- A. Concrete: As shown on the Drawings and according to Section 033000.

PART 3 - EXECUTION

3.1 FOREST SERVICE STANDARD HYDRANT INSTALLATION

- A. Fabricate and install hydrant post, frame, sump, grate and pad as shown on the Drawings and in compliance with FSORAG 4.7.
- B. Wood Posts: Cut and drill to required lengths and chamfer, as shown on the Drawings. All exposed corners shall be lightly rounded, and the surfaces finished smooth. Apply two coats of stain to portion of post that will be above ground.
- C. Excavate to the bottom of the gravel sump. Below this point, excavate hole for supporting post no larger than necessary for proper backfilling and compacting. Set post plumb, at the elevation shown on the Drawings. Securely brace hydrant and adjacent plumbing in proper position during installation. Backfill 6-inch lifts at optimum moisture content. Compact each lift with three complete passes of a mechanical compaction device.
- D. Stain finishes on the post, if damaged during installation, shall be touched up or repaired. Stain shall not be splattered on the pipe and fittings.

3.2 GENERAL INSTRUCTIONS

- A. Pipe and Fittings: Pipe and fittings shall be watertight. If any appurtenances are not rated for the pressure required for pressure testing the pipelines, ensure hydrant will be isolated from the pipeline during pressure testing.
- B. Backfill and compact area where supply line enters. Backfill with material removed from sump area. Remove stones over 1-inch size. Backfill and compact in 6-inch lifts with three complete passes of a mechanical compaction device. Shape, clean, and fill sump with approved sump gravel. Compact in 8-inch layers to work rocks into a dense mass. Level gravel even with adjacent subgrade elevation required.

- C. Grate Frame: Set grate frame on natural ground or compacted subgrade material rather than on the gravel. Position frame at required elevation shown on Drawings for later installation of the grate.
- D. Elevation of edge of pad or gravel sump shall be the same as adjacent finished surface of edge of road, spur, or trail. Pads not meeting this requirement shall be removed by the Contractor and replaced at no cost to the Government. Backfill and fine grade around the outside of the pad.
- E. After hydrant pad base material is placed and compacted around hydrant area, anchor grate to the frame as shown on the Drawings. Prevent base material from entering sump area.

3.3 GRADING AND CLEANUP

- A. Remove all debris and material not utilized when complete according to Section 024100 "Waste Material Disposal."

END OF SECTION 221105

May 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 221300 - ONSITE WASTEWATER TREATMENT SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:
 - 1. Holding tanks
 - 2. Pipe and fittings, including cleanouts. Includes trench.
- B. Related Sections include the following:
 - 1. Section 024100 "Waste Material Disposal."
 - 2. Section 033020 "Concrete From Package Dry Mix For Minor Structures." 033000 "Cast-in-Place Concrete."
 - 3. Section 221100 "Water Distribution System."
 - 4. Section 221102 "Curb Valves and Boxes."
 - 5. Section 312000 "Earthwork."

1.2 DEFINITIONS

- A. ABS: Acrylonitrile-butadiene-styrene plastic.
- B. FRP: Fiberglass-reinforced plastic.
- C. GSP: Galvanized Steel Pipe.
- D. HDPE: High Density Polyethylene plastic.
- E. NPWL: Non-potable waterline (includes sewerline).
- F. PE: Polyethylene plastic.
- G. PVC: Polyvinyl chloride plastic.

1.3 SUBMITTALS

- A. Product Data: For the following:
 - 1. Concrete Holding Tanks (meeting New Mexico Environmental Department Regulation – NM Admin Code Code § 20.7.3.808 - HOLDING TANK REQUIREMENTS)
 - 2. Pipe and fittings, including cleanouts.
- B. Shop Drawings: Include access way openings, covers, pipe connections, and accessories for the following precast concrete structures:
 - 1. holding tanks.

- C. Coordination Drawings: Show piping, underground structures, and other utilities. Indicate size and invert elevations of piping and structures.
- D. Certificate: For holding tanks. Submit letter stating tank is approved by Health Department having jurisdiction (NMED).

1.4 QUALITY ASSURANCE

- A. All work is to be completed according to applicable Federal, State and Local codes.

1.5 PERFORMANCE REQUIREMENTS

- A. Gravity-Flow, Nonpressure, Drainage-Piping Pressure Rating: 10-foot head of water.

1.6 MEASUREMENT AND PAYMENT

- A. Payment will be as listed for the following items:
 - 1. Sewerline: Linear Feet of pipe of each type and size designated, including cleanouts and trench.
 - 2. Holding Tanks: Number of tanks of each type and size designated, access openings, vent pipe and vent and risers to grade, excavation, backfill and all work within 5 feet of the tank.
 - 3. Trailer Sewer Service Hookups: Number of trailer sewer service hookups installed and accepted, including pipe, concrete, fittings, accessories and all work with 5 feet of the trailer sewer service hookup.

PART 2 - PRODUCTS

2.1 DISTRIBUTION PIPES AND FITTINGS

- A. Sewer Pipe and Fittings: Gravity, non-pressurized.
 - 1. PVC: ASTM D 3034, SDR 35, nonperforated, for solvent-cement or elastomeric gasket joints.
 - a. Solvent Cement: ASTM D 2564.
 - b. Gaskets: ASTM F 477, elastomeric seal.

2.2 HOLDING TANKS

- A. Precast Concrete Holding Tanks: ASTM C 1227, single-chamber, with internal baffle, precast, reinforced-concrete tank with two risers and access ways.
 - 1. Design
 - a. For Installation in Non-Traffic Areas: Structural loading according to ASTM C 890.
 - b. For Installation in Traffic Areas: Structural loading for A-16 (HS20-44) traffic loading according to ASTM C 890.
 - c. Shall conform to New Mexico Environmental Department (NMED) requirements listed in Part 1.3 Submittals.

2. Coatings

- a. Exterior: ASTM D1227, Type IV, asbestos-free, fibered-asphalt based dampproofing compound, suitable for cold application. SealMastic Solvent Dampproofing, brush-on or spray grade, as manufactured by W.R. Meadows, Inc., or an approved equal.

3. Mastic Joint Sealant: Preformed flexible joint gasket meeting all requirements of ASTM C990. Sealant shall be RAM-NEK as manufactured by Henry Company, or approved equal.

B. Access Ways

1. Septic Tanks: 22-inch minimum diameter opening with reinforced-concrete access risers to grade and lockable, gasketed access lid with steel lift rings. Install two access ways per tank, centered over the inlet and outlet.
2. Design: Same structural loading requirements as the tank on which the manway is installed.

C. Resilient Connectors

1. ASTM C 923, of size required for piping, fitted into inlet and outlet openings.

2.3 NONPRESSURE-TYPE PIPE COUPLINGS

- A. Description: Comply with ASTM C 1173, elastomeric, sleeve-type, reducing or transition coupling, for joining underground nonpressure piping. Include ends of same sizes as piping to be joined and corrosion-resistant-metal tension band and tightening mechanism on each end.

1. Sleeve Materials:

- a. For Plastic Pipes: ASTM F 477, elastomeric seal or ASTM D 5926, PVC.
- b. For Dissimilar Pipes: ASTM D 5926, PVC or other material compatible with pipe materials being joined.

2.4 CONCRETE THRUST BLOCKS

- A. Concrete: Per Section 033020 "Concrete from Package Dry Mix for Minor Structures."

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and conditions for compliance with requirements and other conditions affecting performance of septic tank systems.
- B. Verify compatibility with and suitability of soil structure and materials.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 EARTHWORK

- A. Refer to Section 312000 “Earthwork” for excavating, trenching, bedding, and backfilling.
 - 1. Stockpile topsoil for reuse in finish grading without intermixing with other excavated material. Stockpile materials away from edge of excavation and do not store within drip line of remaining trees.
 - 2. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
- B. Excavating and Backfilling for Holding Tanks:
 - 1. Excavate sufficient width and length for tanks to depth determined by tank inlet elevation. Provide level bottom.
 - 2. Backfill with excavated soil.
 - 3. According to Section 312000 “Earthwork.”

3.3 PIPING INSTALLATION

- A. Construct the sewer system to the lines and grades shown or established in the field.
- B. Stake locations of sewerlines as shown on the Drawings. Sewerline location shall be approved by the Contracting Officer before installation.
- C. All sewerline shall be graded to drain.
- D. All pipe, fittings, and appurtenances shall be handled and installed in strict conformance to the manufacturer’s recommendations.
- E. Valves, and other regulating and controlling devices shall be installed in the line where called for on the drawings.
- F. The interior of all pipe, fittings, and other accessories shall be kept as free as possible from dirt and foreign matter at all times. Each piece of line as it is laid shall be cleaned of all debris. When the pipeline has become dirty on the inside during shipment or storage, it shall be swabbed out by drawing a damp swab through the line before placing in trench. Care shall be exercised to keep all joining surfaces clean. Any pipe with contaminated or damaged joining surfaces, which cannot be satisfactorily cleaned or repaired, shall be discarded. Under no circumstances shall pipe be installed in water, and no pipe shall be installed when trench conditions or the weather is unsuitable for such work. At all times when work is not in progress, all open ends of the pipe and fittings shall be securely closed to the satisfaction of the Contracting Officer so that no trench water, rodents, earth, or other substance will enter the pipe or fittings.
- G. Any section of pipe already installed and found to be defective shall be removed and replaced with new pipe at the Contractor’s expense.
- H. Water and Non-Potable/Sewer Line Crossings
 - 1. Potable and NPWL lines shall never be installed in the same trench.

2. If a new line crosses below an existing line, the existing line shall be supported to prevent settling.
3. Where waterlines intersect NPWLs or are within 10 horizontal feet of NPWLs, the following applies:
 - a. At least one of the pipes (either the waterline or NPWL) shall be constructed or reconstructed with 200 psi pipe or encased in a continuous sleeve with same rating. At crossing, extend 200 psi pipe or sleeve for a distance of 10 feet perpendicular distance from each side of the waterline. The waterline and NPWL pipes or sleeves shall be centered at the crossing so that the joints will be an equal distance and as far as possible from the crossing.
 - 1) Sleeves: Nominal sleeve diameter to be at least 3 inches larger than nominal diameter of pipe to be encased. Install end seals in annular space at each end of sleeves.
 - 2) In lieu of constructing or reconstructing the NPWL with pipe conforming to waterline requirements, the NPWL may be encased in concrete at least 6 inches thick around the pipe measured at the bell, for 10 feet each side of the waterline.
 - b. Construct according to applicable federal, state and local regulations. The Contractor is responsible for obtaining and following these regulations.
 - c. Where applicable, state and local regulations supersede the above requirements.
- I. Culvert Crossings: Where sewerline crosses a road culvert, adjust grade of sewerline to provide a minimum of 12 inches of separation between the sewerline and culvert, maintaining slope of sewerline to drain.
- J. A minimum of one-foot separation shall be maintained between electric and sewerlines in the same trench. Unless the electric conductors are in conduit the sewerline shall be a minimum of one foot below the electric conductors.
- K. Bury piping with depth of cover over top at least 18 inches, with top at least 12 inches below level of maximum frost penetration.
- L. Thrust Blocks: Install as indicated on the Drawings.
 1. Size: As shown on the Drawings.
 2. Placement: Place and form to give the full required support to the pipe. Install so pipe can later be removed from trench without damage to the pipe or the thrust block.
- M. Cutting and Handling of Pipe
 1. Cutting of pipe for closure pieces or for other reasons shall be done in a neat manner by methods, which will not damage the pipe. The pipe and fittings shall be handled in such a manner as to insure delivery and final placement in good, undamaged condition. Particular care shall be exercised not to injure the pipe surfaces or coatings. Damaged pipe shall not be used in the work.

N. Laying Pipe

1. No pipe shall be placed in the trench until the Contracting Officer has approved the trench and bedding.
2. Pipes shall be placed true to the lines and grades shown on the Drawings and as specified. Sewerlines shall be constructed to drain at a minimum grade of 2%. Sewerlines not meeting this requirement shall be removed by the Contactor and replaced at no cost to the Government.
3. Each section of the pipe shall rest upon the pipe bed for the full length of its barrel.
4. Backfill the pipe within 24 hours after installing the pipe in the trench, leaving the joints and one foot of pipe on each side of each joint exposed until testing is complete.
5. When work is not in progress, securely close open ends of pipe and fittings.
6. When high density polyethylene or solvent weld joint pipe is used, it shall be weaved in the trench to provide horizontal slack in the line for expansion and contraction.
7. High density polyethylene pipe shall be allowed to cool to within 20°F of the shaded trench bottom temperature before backfilling.
8. When pipes are installed in a structure or utility box, they shall be adequately supported.
9. Use proper fittings in order to follow the required line and grades and to avoid excessive curvature. Deflections from a straight line or grade, as required by vertical curves, horizontal curves, or offsets, shall not exceed the manufacturer's recommended maximum joint deflection or curvature for their pipe.
10. When more than one pipeline is in the same trench, the pipes shall be separated by at least eight horizontal inches, or as shown on the Drawings.
11. Bends that result in joint tension or permanent pipe deformation are prohibited.
12. Pipelines may be relocated slightly, if necessary, to avoid existing obstacles. However, the required slopes must be maintained.

O. Branch Connections

1. Branch connections to the main shall be made with wyes or sanitary tees.

P. Pipe Joints

1. Joints shall be made, using jointing materials and applied with the proper accessories, in accordance with the manufacturer's instructions. Completed joints shall be watertight, and capable of passing the required hydrostatic testing. If pipe and fittings are assembled with a lubricant, it must be nontoxic.
2. Dissimilar Joints: Connections between two dissimilar pipes or fittings shall be installed in manner recommended by the pipe manufacturer and approved by the Contracting Officer. Use adapters compatible with both piping materials, outer diameters and system working pressure.
3. Plastic Piping Solvent-Cemented Joints: Clean and dry joining surfaces. Join pipe and fittings according to the following:
 - a. ABS Piping: Join according to ASTM D 2235 appendices.

- b. PVC Pressure Piping: Join schedule number ASTM D 1785, PVC pipe and PVC socket fittings according to ASTM D 2672. Join other-than-schedule-number PVC pipe and socket fittings according to ASTM D 2855.
 - c. PVC Nonpressure Piping: Join according to ASTM D 2855.
 - d. PVC to ABS Nonpressure Transition Fittings: Join according to ASTM D 3138 Appendix.
- Q. Install piping by directional boring, tunneling, jacking, or combination, under streets and other obstructions that cannot be disturbed.

3.4 IDENTIFICATION

- A. According to Section 312000 "Earthwork."
- B. Wrap all buried, non-metallic, nonperforated piping with tracer wire.
- C. Use detectable warning tape over piping (including absorption-field piping), at outside edges of underground structures, and at outside edges of absorption fields.

3.5 PIPE SLEEVE INSTALLATION

- A. Where sewerline crosses road paving install sewerline in pipe sleeve. Extend pipe sleeve a minimum of three feet beyond edge of pavement.
- B. End Seals
 - 1. The annular space between the sleeve or casing and the installed pipe shall be plugged with an end seal to provide a positive seal against a hydrostatic head and prevent soil from entering the space.

3.6 HOLDING TANK INSTALLATION

- A. Install tank level and according to ASTM C 891.
- B. Coatings for Concrete Tanks
 - 1. Preparation of Surfaces: Apply to smooth, clean and dry surface. Repair damaged surface if necessary. Etch surface with muriatic acid (HCl) to remove any laitance.
 - 2. Application: Apply coating and follow safety instructions according to the manufacturer's recommendations.
 - a. Interior Coating: Apply two coats to all interior surfaces of the tank.
 - b. Exterior Coating: Apply coating only at temperatures above 40°F. Apply two coats (one prime and one finish) to all exterior surfaces of the tank. Apply coating at rate of 30 square feet per gallon per coat (min. thickness of 1/8 inch).
 - 3. Protect coating from damage during backfilling. Rocks larger than 2 inches shall not be placed against the exterior concrete wall.

C. Access Ways

1. Reinforced Concrete Riser: Install concrete risers from septic tank to grade allowing for cast iron manhole frame and lid. Use mastic joint sealant at tank and at any joints.
2. Install access way lid and frame to riser.
3. Set tops of frames and covers flush with finished surface of manholes that occur in pavements or gravel roadways. Set tops 3 inches above finished surface elsewhere, unless otherwise indicated.

D. Fill tank with water.

3.7 HYDROSTATIC TESTING FOR NON-PRESSURIZED (GRAVITY) SEWERLINE AND HOLDING TANKS

A. Sewerline: Test sanitary sewerage according to requirements of authorities having jurisdiction and the following:

1. Plastic Sewerlines

a. Gravity Sewer Test: Perform test using one of the following methods.

- 1) Gravity sewer tests shall consist of plugging the end of each sewerline section at the point of termination (manhole, septic tank, distribution box, drainfield, etc.), filling the sewerline with water, testing with at least 10-foot head of water, and maintaining such pressure for at least 1 hour. Recommended method is to install a 10-foot tall, vertical section of sewer pipe at the highest point of the sewerline section to be tested (at building cleanout, trailer sewer service connection, septic tank outlet, etc) and fill to the point of overflow.
- 2) Perform pressure test with water-filled piping according to the International Plumbing Code (latest edition) Section 312. All portions of the system shall be tested to a pressure of at least 10-foot head of water.

2. Tanks: After installation, but before backfilling, fill tank with water per manufacturer's instructions and let stand a minimum of 12 hours to check for leakage. If leak(s) occur, mark areas, drain the tank, and allow drying thoroughly per manufacturer's instructions. Grind areas down and repair with bonding agent and grout or per manufacturer's instructions. Recoat repaired area with appropriate surface coatings and repeat water test. All tanks shall be watertight before backfilling. If tanks fail to be leak proof after second repair, replace with new tank.

B. Leaks and loss in test pressure constitute defects that must be repaired.

C. Replace leaking piping using new materials and repeat testing until leakage is within allowances specified.

3.8 FIELD QUALITY CONTROL

A. Flushing Pipelines

1. After the pressure and leakage testing has been completed, all sewer pipes (including perforated piping) shall be thoroughly flushed.
2. Flushing shall remove all debris from the pipeline, including HDPE pipe shavings, gravel and any other material that may have accidentally entered the pipeline during construction.
3. If no outlets are installed at the end of the waterline, an outlet at the end of each lateral shall be provided. Outlets shall be large enough to develop a velocity in the waterlines of at least 2.5 feet per second. If necessary, use a booster pump or other method to obtain the necessary velocity.
4. Minimum velocities must be obtained in all mainlines, laterals and drainlines.
5. Ensure flushing outlets have adequate drainage and no damage or erosion will occur from the flushing process. Precautions should be taken to ensure flushing does not damage any portion of the drainfield. Install hoses and/or pipe extensions if necessary.

3.9 CLEANUP

- A. Clean up the area by removing the unused and waste materials. Disposal of waste materials shall be in accordance with Section 024100 “Waste Material Disposal.”

END OF SECTION 221300

May 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 260500 - COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Electrical equipment coordination and installation.
2. Grout.
3. Common electrical installation requirements.
4. Hangers and supports.

1.2 SUBMITTALS

A. Product Data:

1. For any substitutions for equipment referred to by name in Division 26 specifications or the drawings.

1.3 QUALITY ASSURANCE

- A. The installation shall conform to the 2020 Edition of the National Electrical Code (NFPA 70) and to the requirements specified herein.
- B. The Contractor shall perform all work necessary and required to accomplish the task as detailed on the drawings and discussed in the installation notes. All work shall be done by a state licensed electrician.

1.4 MEASUREMENT AND PAYMENT

- A. The work in this section, including all incidentals in other electrical sections, shall be measured and paid for by as shown in the Schedule of Items:
 1. Electrical Distribution System - Lump Sum including full compensation for all labor, materials, and incidentals necessary to complete the work as shown on the drawings including all wiring, trenching, bedding, backfill, and all other incidentals necessary for a functional system.

PART 2 - PRODUCTS

2.1 PRODUCTS REFERRED TO BY NAME

- A. The materials referred to by trade name, make, or catalog number on the drawings shall be regarded as establishing a minimum standard of quality; substitutions of equal or

greater quality can be made by submitting a data sheet of the proposed substituted item to the Contracting Officer, for approval.

2.2 GROUT

- A. Nonmetallic, Shrinkage-Resistant Grout: ASTM C 1107, factory-packaged, nonmetallic aggregate grout, noncorrosive, nonstaining, mixed with water to consistency suitable for application and a 30-minute working time.

PART 3 - EXECUTION

3.1 COMMON REQUIREMENTS FOR ELECTRICAL INSTALLATION

- A. Comply with NECA 1.
- B. Measure indicated mounting heights to bottom of unit for suspended items and to center of unit for wall-mounting items.
- C. Headroom Maintenance: If mounting heights or other location criteria are not indicated, arrange and install components and equipment to provide maximum possible headroom consistent with these requirements.
- D. Equipment: Install to facilitate service, maintenance, and repair or replacement of components of both electrical equipment and other nearby installations. Connect in such a way as to facilitate future disconnecting with minimum interference with other items in the vicinity.
- E. Right of Way: Give to piping systems installed at a required slope.

3.2 RACEWAY AND SIMILAR PENETRATIONS (NON-FIRE RATED, NOT SLEEVED)

- A. Concrete Slabs and Walls: Core-drilled holes.
 - 1. Fill oversized holes with grout to within 1/4-inch.
- B. Roof-Penetrations: Seal penetration of individual raceways with flexible boot-type flashing units applied in coordination with roofing work.
- C. Aboveground, Exterior-Wall Penetrations: Seal exterior opening around the raceway or cable, using a flexible, waterproofing, joint sealant appropriate for size, depth, and color to closely match the surrounding surface. Finish interior openings, filling opening and matching the exiting surface with appropriate materials and finish quality for the space.

3.3 HANGERS AND SUPPORTS

- A. Installation:
 - 1. Comply with NECA 1 and NECA 101 for installation requirements, except as specified in this article.

2. Separate dissimilar metals and metal products from contact with wood or cementitious materials by painting each metal surface in area of contact with a bituminous coating or by other permanent separation.
3. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor and fasten electrical items and their supports to building structural elements by the following methods, unless otherwise indicated or required by Code:
 - a. To Wood: Fasten with lag screws or through bolts.
 - b. To New Concrete: Bolt to concrete inserts.
 - c. To Existing Concrete: Expansion anchor fasteners.
4. Drill holes for expansion anchors in concrete at locations and to depths that avoid reinforcing bars.

END OF SECTION 260500

May 2024

USDA FOREST SERVICE, R3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 260519 - LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Copper building wire rated 600 V or less.
2. Connectors, splices, and terminations rated 600 V and less.

1.2 DEFINITIONS

- A. PV: Photovoltaic.
- B. RoHS: Restriction of Hazardous Substances.

1.3 SUBMITTALS

- A. Product Data: For each type of product.
- B. Field quality-control reports.

1.4 MEASUREMENT AND PAYMENT

- A. No separate measurement or payment will be made for work specified in this section. All work will be included in other items listed in the Schedule of Items.

PART 2 - PRODUCTS

2.1 COPPER BUILDING WIRE

- A. Description: Flexible, insulated, and uninsulated, drawn copper current-carrying conductor with an overall insulation layer or jacket, or both, rated 600 V or less.
- B. Standards:
 1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
- C. Conductors: Copper, complying with ASTM B 3 for bare annealed copper and with ASTM B 8 for stranded conductors.
- D. Conductor Insulation:
 1. Types RHW and RHW-2: Comply with UL 44.

2. Type THHN/THWN-2: Comply with UL 83.
 3. Type THW and Type THW-2: Comply with NEMA WC-70/ICEA S-95-658 and UL 83.
 4. Type UF: Comply with UL 83 and UL 493.
 5. Type USE-2: Comply with UL 854.
 6. Type XHHW-2: Comply with UL 44.
- E. Cable: Comply with NEMA WC 70/ICEA S-95-658 for:
1. Armored cable, Type AC.
 2. Metal-clad cable, Type MC.
 3. Nonmetallic-sheathed cable, Type NM with ground wire: Comply with UL 83 and UL 719.
 4. Type UF with ground wire.
 5. Type SO and similar cables with ground wire.
 6. Photovoltaic cable, Type PV.
 7. Type USE-2 with ground wire.
 8. Tray cable, Type TC-ER.
 9. Pump cable, Type RHW and THW.

2.2 CONNECTORS AND SPLICES

- A. Description: Factory-fabricated connectors and splices of size, ampacity rating, material, type, and class for application and service indicated.

PART 3 - EXECUTION

3.1 CONDUCTOR MATERIAL APPLICATIONS

- A. Circuits: Copper; solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.
1. Aluminum conductors are prohibited unless specifically called for on the drawings.
- B. Exposed to Direct Sunlight: Sunlight resistant.
- C. PV Circuits: Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.

3.2 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

- A. Service Entrance and Feeders: Type THHN/THWN-2 or XHHW, single conductors in raceway.
- B. Exposed Branch Circuits: Type THHN/THWN-2, single conductors in raceway.
- C. Branch Circuits Concealed in Concrete, below Slabs-on-Grade, and Underground: Type THHN/THWN-2, single conductors in raceway.

- D. Cord Drops and Portable Appliance Connections: Type SO, hard service cord with stainless-steel, wire-mesh, strain relief device at terminations to suit application.
- E. PV Circuits:
 - 1. Wiring on roofs or under PV modules shall be rated at 90 degrees C.
 - 2. Exposed Single Conductors: Type USE-2 or PV for PV source circuits rated at 600 V or less.
 - 3. DC source circuits entering buildings shall be installed in metal raceway or Type MC cable.

3.3 INSTALLATION OF CONDUCTORS AND CABLES

- A. Conceal cables in finished walls, ceilings, and floors unless otherwise indicated.
- B. Complete raceway installation between conductor and cable termination points according to Section 260533 "Raceways and Boxes for Electrical Systems" prior to pulling conductors and cables.
- C. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.
- D. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips that will not damage cables or raceway.
- E. Install exposed cables parallel and perpendicular to surfaces of exposed structural members and follow surface contours where possible.
- F. Support cables according to National Electrical Code (NEC).

3.4 CONNECTIONS

- A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A-486B.
- B. Make splices, terminations, and taps that are compatible with conductor material.
 - 1. Use oxide inhibitor in each splice, termination, and tap for aluminum conductors.
- C. Wiring at Outlets: Install conductor at each outlet, with at least 6 inches of slack.

3.5 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

- A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies.

END OF SECTION 260519

May 2024

USDA FOREST SERVICE, R3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 260526 - GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes grounding and bonding systems and equipment.

1.2 MEASUREMENT AND PAYMENT

- A. No separate measurement or payment will be made for work specified in this section. All work will be included in other items listed in the Schedule of Items.

PART 2 - PRODUCTS

2.1 SYSTEM DESCRIPTION

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. Comply with UL 467 for grounding and bonding materials and equipment.

2.2 CONDUCTORS

- A. Insulated Conductors: Copper or tinned-copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction.
- B. Bare Copper Conductors:
 - 1. Solid Conductors: ASTM B 3.
 - 2. Stranded Conductors: ASTM B 8.
 - 3. Tinned Conductors: ASTM B 33.

2.3 CONNECTORS

- A. Listed and labeled by an NRTL acceptable to authorities having jurisdiction for applications in which used and for specific types, sizes, and combinations of conductors and other items connected.
- B. Ground Rod Clamps: Mechanical type, copper or copper alloy.
- C. Lay-in Lug Connector: Mechanical type, tinned-copper rated for direct burial terminal with set screw and listed for use bonding to different metals.
- D. U-Bolt Clamps: Mechanical type, copper or copper alloy, terminal listed for direct burial.

- E. Water Pipe Clamps:
 - 1. Mechanical type, two pieces.
 - a. Material: Die-cast zinc alloy.
 - b. Listed for direct burial.

2.4 GROUNDING ELECTRODES

- A. Ground Rods: Copper-clad steel; 3/4 inch by 10 feet.

PART 3 - EXECUTION

3.1 APPLICATIONS

- A. Conductors: Install solid conductor for No. 8 AWG and smaller, and stranded conductors for No. 6 AWG and larger unless otherwise indicated.
- B. Underground Grounding Conductors: Install bare tinned-copper conductor.
 - 1. Bury at least 24 inches below grade.

3.2 EQUIPMENT GROUNDING

- A. Install insulated equipment grounding conductors with all feeders and branch circuits.
- B. Pole: Install grounding electrode and a separate insulated equipment grounding conductor in addition to grounding conductor installed with branch-circuit conductors.
- C. Arrays: Use oxide inhibitor in each termination and tap for aluminum mount equipment and module.

3.3 INSTALLATION

- A. Grounding Conductors: Route along shortest and straightest paths possible unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.
- B. Ground Rods: Drive rods until tops are 2 inches below finished floor or final grade unless otherwise indicated.
 - 1. Interconnect ground rods with grounding electrode conductor below grade and as otherwise indicated. Make connections without exposing steel or damaging coating if any.
- C. Grounding and Bonding for Piping:
 - 1. Metal Water Service Pipe: Install insulated copper grounding conductors, in conduit, from building's main service equipment, or grounding bus, to main metal water service entrances to building. Connect grounding conductors to main metal water service pipes; use a bolted clamp connector or bolt a lug-type connector to a pipe flange by using one of the lug bolts of the flange. Where a dielectric main

water fitting is installed, connect grounding conductor on street side of fitting. Bond metal grounding conductor conduit or sleeve to conductor at each end.

2. Well Casing: Install copper grounding conductors from main service equipment to well casing routing through and bonding well pump panel with submersible pump circuit. Connect grounding conductor to casing using a bolted lug connector.

D. Concrete-Encased Grounding Electrode (Ufer Ground): Fabricate rebar cages for pole wire tying rebar to all crossings and to at least two pole support bolts.

1. This may not meet NFPA 70 but is meant to supplement other grounding electrodes to be installed.
2. The utility building is slab on grade and does not have a footing.

END OF SECTION 260526

May 2024

USDA FOREST SERVICE, R3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 260533 - RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS

1. GENERAL

1.1 SUMMARY

A. Section Includes:

1. Metal conduits, tubing, and fittings.
2. Nonmetal conduits and fittings.
3. Boxes, enclosures, and cabinets.
4. Handholes and boxes for exterior underground cabling.

1.2 DEFINITIONS

- A. GRC: Galvanized rigid steel conduit.
- B. IMC: Intermediate metal conduit.

1.3 MEASUREMENT AND PAYMENT

- A. No separate measurement or payment will be made for work specified in this section. All work will be included in other items listed in the Schedule of Items.

PART 2 - PRODUCTS

2.1 METAL CONDUITS, TUBING, AND FITTINGS

- A. Listing and Labeling: Metal conduits, tubing, and fittings shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. GRC: Comply with ANSI C80.1 and UL 6.
- C. IMC: Comply with ANSI C80.6 and UL 1242.
- D. PVC-Coated Steel Conduit: PVC-coated rigid steel conduit or IMC.
1. Comply with NEMA RN 1.
 2. Coating Thickness: 0.040 inch, minimum.
- E. EMT: Comply with ANSI C80.3 and UL 797.
- F. LFMC: Flexible steel conduit with PVC jacket and complying with UL 360.
- G. Fittings for Metal Conduit: Comply with NEMA FB 1 and UL 514B.
1. Fittings for EMT:

- a. Material: Steel.
 - b. Type: compression.
- 2. Expansion Fittings: PVC or steel to match conduit type, complying with UL 651, rated for environmental conditions where installed, and including flexible external bonding jumper.
 - 3. Coating for Fittings for PVC-Coated Conduit: Minimum thickness of 0.040 inch (1 mm), with overlapping sleeves protecting threaded joints.

2.2 NONMETALLIC CONDUITS AND FITTINGS

- A. Listing and Labeling: Nonmetallic conduits, tubing, and fittings shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. RNC: Type EPC-40-PVC Insert type, complying with NEMA TC 2 and UL 651 unless otherwise indicated.
- C. LFNC: Comply with UL 1660.
- D. Rigid HDPE: Comply with UL 651A.
- E. Continuous HDPE: Comply with UL 651B.
- F. Fittings for RNC: Comply with NEMA TC 3; match to conduit and material.
- G. Fittings for LFNC: Comply with UL 514B.

2.3 BOXES, ENCLOSURES, AND CABINETS

- A. General Requirements for Boxes, Enclosures, and Cabinets: Boxes, enclosures, and cabinets installed in wet locations shall be listed for use in wet locations.
- B. Sheet Metal Outlet and Device Boxes: Comply with NEMA OS 1 and UL 514A.
- C. Cast-Metal Outlet and Device Boxes: Comply with NEMA FB 1, Type FD, with gasketed cover.
- D. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.
- E. Cast-Metal Access, Pull, and Junction Boxes: Comply with NEMA FB 1 and UL 1773, with gasketed cover.
- F. Box extensions used to accommodate new building finishes shall be of same material as recessed box.
- G. Hinged-Cover Enclosures: Comply with UL 50 and NEMA 250, Type 1, 3R, or 12 with continuous-hinge cover with flush latch unless otherwise indicated.
 - 1. Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.

2.4 HANDHOLES AND BOXES FOR EXTERIOR UNDERGROUND WIRING

A. General Requirements for Handholes and Boxes:

1. Boxes and handholes for use in underground systems shall be designed and identified as defined in NFPA 70, for intended location and application.
2. Boxes installed in wet areas shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
3. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to the following:
 - a. Armorcast Products Company.
 - b. Carson Industries LLC.
 - c. Oldcastle Precast, Inc.
 - d. Quazite: Hubbell Power Systems, Inc.; Hubbell Power Systems.
 - e. Synertech Moulded Products.
4. Standard: Comply with SCTE 77.
5. Configuration: Designed for flush burial with open bottom unless otherwise indicated.
6. Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure and handhole location.
7. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
8. Cover Legend: Molded lettering, "ELECTRIC."

B. Fiberglass Handholes and Boxes: Molded of fiberglass-reinforced polyester resin, with frame and covers of fiberglass.

PART 3 - EXECUTION

3.1 RACEWAY APPLICATION

A. Outdoors: Apply raceway products as specified below unless otherwise indicated:

1. Exposed Conduit: GRC or IMC.
 - a. EMT also allowed where UL listed raintight compression fittings are used and the raceway is 5-ft minimum above finished grade.
2. Underground Conduit within 5-ft of exposed grade penetration or building foundation penetration: PVC coated or wrapped GRC or IMC
3. Underground Conduit beyond 5-ft of exposed grade penetration or building foundation penetration: HDPE; or RNC, Type EPC-40-PVC direct buried.
4. Connections Between PV Modules: LFMC, LFNC, and approved cable or individual conductors.
5. Boxes and Enclosures, Aboveground: NEMA 250, Type 3R.
6. Application of Handholes and Boxes for Underground Wiring:
 - a. Handholes and Pull Boxes Subject to Light-Duty Pedestrian Traffic Only: Fiberglass-reinforced polyester resin, structurally tested according to SCTE 77 with 3000-lbf vertical loading.

- 1) Install a minimum of 10-feet from any roadway.
- B. Indoors: Apply raceway products as specified below unless otherwise indicated:
1. Exposed: EMT.
 2. Boxes and Enclosures: NEMA 250, Type 1.
- C. Raceway Fittings: Compatible with raceways and suitable for use and location.
1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings unless otherwise indicated. Comply with NEMA FB 2.10.
 2. PVC Externally Coated, Rigid Steel Conduits: Use only fittings listed for use with this type of conduit. Patch and seal all joints, nicks, and scrapes in PVC coating after installing conduits and fittings. Use sealant recommended by fitting manufacturer and apply in thickness and number of coats recommended by manufacturer.
 3. EMT: Use compression, steel fittings. Comply with NEMA FB 2.10.
 4. Flexible Conduit: Use only fittings listed for use with flexible conduit. Comply with NEMA FB 2.20.

3.2 INSTALLATION

- A. Comply with NECA 1 and NECA 101 for installation requirements except where requirements on Drawings or in this article are stricter. Comply with NECA 102 for aluminum conduits. Comply with NFPA 70 limitations for types of raceways allowed in specific occupancies and number of floors.
- B. Complete raceway installation before starting conductor installation.
- C. Arrange stub-ups so curved portions of bends are not visible above finished slab.
- D. Support within 12 inches of changes in direction.
- E. Install conduits parallel or perpendicular to building lines.
- F. Support conduit within 12 inches of enclosures to which attached.
- G. Coat field-cut threads on PVC-coated raceway with a corrosion-preventing conductive compound prior to assembly.
- H. Terminate threaded conduits into threaded hubs or with locknuts on inside and outside of boxes or cabinets. Install bushings on conduits up to 1-1/4-inch trade size and insulated throat metal bushings on 1-1/2-inch trade size and larger conduits terminated with locknuts. Install insulated throat metal grounding bushings on service conduits.
- I. Install raceways square to the enclosure and terminate at enclosures with locknuts.
- J. Cut conduit perpendicular to the length.
- K. Mounting heights of boxes are not individually indicated, give priority to ADA requirements.

- L. Fasten junction and pull boxes to or support from building structure. Do not support boxes by conduits.

3.3 INSTALLATION OF UNDERGROUND CONDUIT

A. Direct-Buried Conduit:

1. Trenches shall be 24 inches to top of conduit unless otherwise indicated. Trench depth for utility company controlled primary and secondary laterals shall meet utility company trench and conduit installation requirements.
2. Excavate trench bottom to provide firm and uniform support for conduit. Prepare trench bottom as specified in Section 312000 "Earthwork" for pipe less than 6 inches (150 mm) in nominal diameter.
3. Install backfill as specified in Section 312000 "Earthwork."
4. After installing conduit, backfill and compact. Start at tie-in point, and work toward end of conduit run, leaving conduit at end of run free to move with expansion and contraction as temperature changes during this process. Firmly hand tamp backfill around conduit to provide maximum supporting strength. After placing controlled backfill to within 12 inches of finished grade, make final conduit connection at end of run and complete backfilling with normal compaction as specified in Section 312000 "Earthwork."
5. Install manufactured conduit elbows for stub-ups at poles and towers and at buildings.
6. Underground Warning Tape: Comply with requirements in Section 260553 "Identification for Electrical Systems."

3.4 INSTALLATION OF UNDERGROUND HANDHOLES AND BOXES

- A. Install handholes and boxes level and plumb and with orientation and depth coordinated with connecting conduits to minimize bends and deflections required for proper entrances.
- B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1/2-inch sieve to No. 4 sieve and compacted to same density as adjacent undisturbed earth.
- C. Elevation: Set covers 1 inch above finished grade.

3.5 PROTECTION

- A. Protect coatings, finishes, and cabinets from damage and deterioration.
 1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
 2. Repair damage to PVC coatings or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 260533

May 2024

USDA FOREST SERVICE, R3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 260553 - IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:
1. Identification for conductors and communication and control cable.
 2. Underground-line warning tape.
 3. Instruction signs.

1.2 QUALITY ASSURANCE

- A. Comply with NFPA 70.

1.3 MEASUREMENT AND PAYMENT

- A. No separate measurement or payment will be made for work specified in this section. All work will be included in other items listed in the Schedule of Items.

PART 2 - PRODUCTS

2.1 CONDUCTOR AND COMMUNICATION- AND CONTROL-CABLE
IDENTIFICATION MATERIALS

- A. Marker Tapes: Vinyl or vinyl-cloth, self-adhesive wraparound type, with circuit identification legend machine printed by thermal transfer or equivalent process.

2.2 UNDERGROUND-LINE WARNING TAPE

- A. Description: Permanent, bright-colored, continuous-printed, polyethylene tape.
1. Not less than 3 inches (75 mm) wide by 4 mils (0.102 mm) thick.
 2. Compounded for permanent direct-burial service.
 3. Printed legend shall indicate type of underground line.

2.3 INSTRUCTION SIGNS

- A. Engraved, laminated acrylic or melamine plastic, minimum 1/16 inch (1.6 mm) thick for signs up to 20 sq. in. (129 sq. cm) and 1/8 inch (3.2 mm) thick for larger sizes.
1. Engraved legend with black letters on white face.
 2. Punched or drilled for mechanical fasteners.

PART 3 - EXECUTION

3.1 APPLICATION

- A. Conductor Identification: Identify as called for on the drawings.
- B. Locations of Underground Lines: Identify with underground-line warning tape for power, lighting, communication, and control wiring and optical fiber cable. Install underground-line warning tape for both direct-buried cables and cables in raceway.
- C. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting: Comply with 29 CFR 1910.145 and apply self-adhesive warning labels or baked-enamel warning signs. Identify system voltage with black letters on an orange background. Apply to exterior of door, cover, or other access.
 - 1. Equipment with Multiple Power or Control Sources: Apply to door or cover of equipment including, but not limited to, the following:
 - a. Power disconnect switch.
 - b. Controls with external control power connections.
 - 2. Equipment Requiring Workspace Clearance According to NFPA 70: Unless otherwise indicated, apply to door or cover of equipment in interior locations.
- D. Instruction Signs: Labels for pilot lights, chlorinator receptacle, disconnects, and others as shown on the drawings.

3.2 INSTALLATION

- A. Verify identity of each item before installing identification products.
- B. Location: Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment.
- C. Self-Adhesive Identification Products: Clean surfaces before application, using materials and methods recommended by manufacturer of identification device.
- D. Attach nonadhesive signs and plastic labels with screws and auxiliary hardware appropriate to the location and substrate.
- E. Underground-Line Warning Tape: During backfilling of trenches install continuous underground-line warning tape directly above line at 12 inches (150 to 200 mm) below finished grade. Use multiple tapes where width of multiple lines installed in a common trench exceeds 16 inches (400 mm) overall.

END OF SECTION 260553

May 2024

USDA FOREST SERVICE, R3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 262416 - OTHER ELECTRICAL TESTS AND CALIBRATIONS

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:

1. Float Operation Ranges.
2. System Testing.

1.2 QUALITY ASSURANCE

- A. Comply with NFPA 70.

1.3 COORDINATION

- A. Coordinate with the Contracting Officer for witnessing these tests.

1.4 MEASUREMENT AND PAYMENT

- A. No separate measurement or payment will be made for work specified in this section. All work will be included in other items listed in the Schedule of Items.

PART 2 - PRODUCTS

- A. Not applicable.

PART 3 - EXECUTION

3.1 STORAGE TANK FLOAT RANGES

- A. Verify that floats operate properly within the ranges noted on the drawings.

3.2 SYSTEM TESTING

- A. Verify normal pump operation and pump alternation from signals. Storage tank shall not overfill.

1. Signal from storage tank.
2. Signal to pump.

- B. Verify alarm operation.

1. Low tank level signal from storage tank.

- C. Any failures shall be corrected and then the system completely retested.

END OF SECTION 262416
May 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 262716 – WALK-IN ENCLOSURE

PART 1 - GENERAL

1.1 SUMMARY

- A. This work shall consist of furnishing and installing fiberglass walk-in enclosure, for housing electrical or mechanical equipment as shown on the drawings. Enclosure features shall include front removable, locked doors. Work also includes preparing foundation with earthen materials and constructing concrete foundation pad for new enclosure.

1.2 RELATED WORK

- A. The work shall be in accordance with the following subsidiary specifications. The subsidiary specifications are referred to in the text by the Section designation only.

- 1. Section 024100 Waste Material Disposal
- 2. Section 033000 Cast-In-Place Concrete
- 3. Section 311000 Clearing and Grubbing
- 4. Section 312000 Earthwork
- 5. Section 321204 Crushed Aggregate Base or Surface Coarse
- 6. Division 26 Electrical

1.3 SUBMITTALS

- A. Shop Drawings
- B. Manufacturer's Installation Instructions
- C. Manufacturer's Operation and Maintenance Instructions

1.4 MEASUREMENT AND PAYMENT

- A. Measurement and payment considered incidental to work outlined in Section 260500 "Common Work Results for Electrical". Work includes the preparation of foundation pad for furnished enclosure, including all utility stub ups into the interior of the enclosure and all associated clearing, excavation, aggregate base and backfill, compaction, grading and aggregate-surfaced aprons around the building necessary to complete the item. This also includes the construction of aprons built around the building, as shown on the drawings.

PART 2 - PRODUCTS

2.1 FIBERGLASS ENCLOSURE

- A. Power Design Inc. (Odell, Oregon; 541-354-3222): Model ME-1-72-2681-L1-MG, Dimensions 6'-0" x 6'-0" x 7'-9", as shown on the drawings, or an approved equal.

1. Enclosure (approved equal) shall be constructed to the following criteria:
 - a. Meet ANSI C57.12.28 "Pad Mounted Equipment Enclosure Integrity Standard".
 - b. Designed for mounting on concrete pad by means of internal flange completely around the base of the enclosure.
 - c. Fiberglass laminate shall be 1/4-inch nominal thickness except where local reinforcement is required.
 - d. Exterior shall be gel-coated to 0.014-inch nominal thickness and be manufacturer's standard dark green. The interior shall be gel-coated white.
 - e. Fiberglass shall not support combustion and be self-extinguishing.
 - f. Locking device shall provide for padlock provisions.
 - g. Provide minimum two (2) lifting lugs.
 - h. Provide stainless steel identification plate, affixed outside of the enclosure, which identifies the manufacturer, model number, and date of manufacture.
 - i. Provide two (2) louvers for adequate air flow and moisture dissipation.
 - j. Provide double door entry on one side of the enclosure and shall be split hinged type. Clear opening shall be 60" wide x 73" high with the bottom opening 6" off the ground. Provide 3-point latching and door stop rods.
 - k. Testing Standards:

1) Tensile Strength (ASTM D638)	11,120 psi
2) Flexural Strength (ASTM D790)	25,380 psi
3) Flexural M.O.E. (ASTM D790)	1.182 x 10 ⁶
4) Compression Strength (ASTM D695)	23,980 psi
5) Charpy Strength (ASTM D256)	4.6 ft-lbs./inch
6) Impact Resistance (ASTM D2444)	46 inch-lbs.
7) Water Absorption (ASTM D570)	0.092%
8) Salt Spray Test (ASTM B117)	1500 hours, 5% salt spray
9) Crosshatch Adhesion Test (ASTM D3359)	Method A
10) Humidity Test (ASTM D2247)	1000 hours, 45° C
11) Impact Test (ASTM D2794)	160 inch-lbs.

2.2 CONCRETE

- A. Shall comply with Section 033000 "Cast in-Place Concrete".

2.3 AGGREGATE BASE

- A. Shall comply with Section 321204 "Crushed Aggregate Base or Surface Course" Grading "C" and 1/2-inch minus gradation.

2.4 ELECTRICAL AND TELECOMMUNICATIONS LINE

- A. Shall be in accordance with the "Division 26 – Electrical" specifications.

PART 3 - EXECUTION

3.1 CLEARING AND GRUBBING

- A. Shall be performed in accordance with Section 311000 "Clearing and Grubbing".

3.2 CONCRETE MOUNTING PAD

- A. Import aggregate base or select borrow to build up the subgrade for the enclosure mounting pad. For placement and compaction requirements, reference Section 312000 (for earthwork) or Section 321204 (for aggregate base).
- B. Fiberglass mounting pad shall be reinforced concrete, two feet larger than the enclosure base dimensions, 4-inches thick, as shown on the drawings and in accordance with Section 033000. Place and compact aggregate base beneath mounting pad, in accordance with Section 321204 and as shown on the drawings.
- C. Provide stub ups for new utility connections planned for the interior of the enclosure. Stub-ups shall be rigid galvanized conduit or as shown on the drawings and shall be installed in accordance with local electrical codes.
 - 1. All interior electrical work, as shown on the E-sheets of the drawings, shall be evaluated under the Division 26 specifications.

3.3 ENCLOSURE INSTALLATION

- A. Install enclosure in accordance with manufacturer's instructions. Follow bolt pattern and use stainless steel concrete anchors and washers to secure enclosure to the mounting pad. Size anchors in accordance with manufacturer's instructions.

3.4 FINISH GRADING

- A. Provide aggregate-surfaced aprons around the building. Aprons shall be built to the lines and dimensions show on the drawings, utilizing a 1/2-inch minus gradation. For placement and compaction requirements, reference Section 312000 (for earthwork) or Section 321204 (for aggregate surfaced aprons).
- B. Finished slopes shall be left rough to accept seeding at the end of the project. Refer to the drawings for gradient of finished slopes.

END SECTION 262716

MAY 2022

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 311000 - CLEARING AND GRUBBING

PART 1 - GENERAL

1.1 SUMMARY

- A. This work shall consist of clearing, grubbing, trimming, removing, and disposing of or treatment of timber, construction slash, and debris. This work shall also include preservation of vegetation, and objects designated to remain from injury or defacement.

1.2 RELATED SECTIONS

- A. Section 024100 "Waste Material Disposal."

1.3 DEFINITIONS

- A. Areas to be Cleared and Grubbed - The limits of clearing and grubbing will be established by this specification, by other specification items, or on contract Drawings. The clearing and grubbing limits will normally coincide with the designated working limits; however, the Contracting Officer may also designate individual trees and snags outside the clearing limits for selective removal and disposal, or he may designate areas within the working limits where clearing and grubbing is not required or allowed within the provisions of this specification.
 - 1. Grading Limits - Area that is to be excavated or covered with additional materials during construction.
 - 2. Working Limits - Area consisting of the grading limits plus room for equipment to maneuver to perform the necessary clearing and grubbing. These limits, to be held to a minimum, will be designated for each project.
 - 3. Clearing Limits - Area consisting of the working limits plus any additional area to operate a boom or other above-ground clearance requirement.

1.4 MEASUREMENT AND PAYMENT

- A. The quantity to be measured shall be lump sum (LS), cleared and grubbed within the designated clearing limits indicated on the drawings or staking in the field by the Contracting Officer. Areas not shown on the drawings or not staked for clearing and grubbing will not be measured for payment.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION

3.1 CLEARING AND GRUBBING

- A. All trees, brush, shrubs, stumps, roots, and other vegetative material and debris within the designated clearing limits, shall be cleared, grubbed, removed, and disposed of except the following condition:
 - 1. Those items designated to remain by the Contract or Contracting Officer.
 - 2. Undisturbed stumps outside the roadway, trailway, and parking area, provided they do not extend more than 12 inches above the original ground (measured from the uphill side) and are not closer than 2 feet from the edge of road or parking area, are not in the cut or fill slopes, and they do not interfere with the placement or compaction of embankments.
- B. All roots over 3 inches in diameter within the roadbed/parking area area shall be grubbed to a minimum depth of 6 inches below subgrade. Roots over 3 inches in diameter protruding from the excavated slope shall be cut flush with the excavated slope surface.
- C. Clearing slash shall be treated in accordance with “Slash Treatment” paragraph below.
- D. Trees shall be felled within the clearing limits, usually towards the center, so as to prevent damage to the trees that are to be left standing. In some cases, controlled felling, cutting of trees into sections from the top downward, or other means shall be used when required to prevent damage to property, structures, traffic, and trees/vegetation to remain.
- E. Fire-dangerous dead trees or unstable live trees, designated by the Contracting Officer within the project area shall be cut off not more than 12 inches above the uphill ground line and utilized or removed.
- F. Branches on remaining trees or shrubs shall be trimmed to give a clear height of 14 feet above the roadbed unless otherwise shown on the Drawings. Tree limbs shall be trimmed as near flush with the trunk as practicable.

3.2 UTILIZATION OF TIMBER

- A. Merchantable timber is timber that meets Utilization Standards. Utilization Standards shall be trees larger than 8-inches in diameter measured 12-inches above grade.
- B. Utilization and Removal of Timber: Trees that meet “Merchantable Timber” standards shall be treated as follows:
 - 1. Disposal as Unmerchantable Timber. Timber on this project is not considered merchantable and shall be disposed of in accordance with “Slash Treatment” paragraph below.

3.3 PIONEER

- A. Pioneering operations shall be done in a manner that prevents undercutting of final excavation slopes, maintains materials within the road, trail, or parking limits, and controls runoff.

3.4 SLASH TREATMENT

- A. Treatment of construction slash larger than 3-inches in diameter and 3-feet in length shall be accomplished by one or more of the following methods:
 - 1. Scattering: Construction slash shall be scattered outside the clearing limits without damaging trees. All logs shall be limbed and placed away from trees, positioned so they will not roll, and not be placed on top of one another. Stumps shall be removed and disposed of in accordance with Section 024100. Other construction slash shall be limbed and scattered to reduce slash concentrations.
 - 2. Stacking of Unmerchantable Material: Logs not meeting utilization standards and other material that exceeds 4-inches in diameter shall be limbed, cut into 16-inch to 36-inch lengths, and stacked onsite at a location designated by the COR. Log stacked shall be stable and free of brush and soil.
- B. Construction slash less than 3 inches in diameter and 3 feet in length may be scattered within the clearing limits.
- C. No construction slash shall be deposited in lakes, meadows, streams, or streambeds. Construction slash that interferes with drainage structures shall be removed immediately.

3.5 CLEARING OR CLEARING AND GRUBBING REQUIREMENTS FOR VARIOUS ITEMS:

- A. Paths and Trails (Known as “Outdoor Recreation Access Route” per FSORAG document) - The path and trail clearing limits shall be 3 feet on each side of the tread centerline and extend to a height of 8 feet above the tread. Shrubs normally growing less than 2 feet tall when mature shall be left undisturbed where they occur outside the cut and fill slope areas. All branches, except short shrubs, which extend into the travel-way shall be removed by cutting them flush with the trunk of the tree or shrub.
- B. Roadways and Parking Areas - On roadways and parking areas, the area to be grubbed shall be an area between the cut and/or fill stakes. The clearing area will normally be an additional 4 feet on each side of the grubbed area and shall extend to a height of 14 feet above the finished roadway surface.
- C. Trenches for Water, Sewer, Electrical Lines, and other Underground Utilities - If no reference is made elsewhere, a clearing of 5 feet maximum on each side of centerline will be allowed.
- D. Septic Tanks – Clearing and grubbing limits shall be a minimum of 5 feet outside the edge of the tank.

3.6 TEMPORARY EROSION AND SEDIMENTATION CONTROL

- A. Provide temporary erosion and sedimentation control measures to prevent soil erosion and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways, according to Section 023701 “Sediment and Erosion Control Measures.”

- B. Inspect, repair, and maintain erosion and sedimentation control measures during construction until permanent vegetation has been established.
- C. Remove erosion and sedimentation controls and restore and stabilize areas disturbed during removal.

3.7 WASTE DISPOSAL

- A. Debris and refuse shall be disposed of in accordance with Section 024100 “Waste Material Disposal”.

END OF SECTION 311000

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 312000 - EARTHWORK

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:
 - 1. Sub-surface drainage.
- B. Related Sections include the following:
 - 1. Section 329206 "Seeding."

1.2 DEFINITIONS

- A. Backfill: Soil materials used to fill an excavation.
- B. Borrow or Select Borrow: Satisfactory soil material used for embankment, backfill, or fill construction that is either imported from off-site or excavated from designated locations at the site.
- C. Excavation: Removal of material encountered above subgrade elevations and to lines and dimensions indicated.
 - 1. Additional Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions as directed by Contracting Officer. Additional excavation and replacement material will be paid for according to Contract provisions for changes in the Work.
 - 2. Bulk Excavation: Excavations more than 10 feet (3m) in width and pits more than 30 feet (9m) in either length or width.
 - 3. Unauthorized Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions without direction of the Contracting Officer. Unauthorized excavation, as well as remedial work directed by Contracting Officer, shall be without additional compensation.
 - 4. Unclassified Excavation: Excavation to subgrade elevation and to lines and dimensions indicated regardless of the character of surface and subsurface conditions encountered, including rock, soil materials, and obstructions.
- D. Fill: Soil materials used to raise existing grades.
- E. Structures: Buildings, footings, foundations, retaining walls, slabs, tanks, curbs, mechanical and electrical appurtenances, or other man-made stationary features constructed above or below the ground surface.
- F. Subgrade: Surface or elevation remaining after completing excavation, or top surface of a fill or backfill immediately below subbase, aggregate base, drainage fill, initial or subsequent backfill, or topsoil materials.
- G. Utilities include on-site underground pipes, conduits, ducts, and cables, as well as underground services within building.

1.3 SUBMITTALS

- A. Contractor shall submit to the Contracting Officer for approval source of aggregates and backfill materials and certified sieve analysis. Materials from Government Sources are exempt from this requirement.

1.4 MEASUREMENT AND PAYMENT

- A. There will be no separate measurement or payment for work in this Section. Payment will be included in the contract unit price as shown on the Schedule of Items.

PART 2 - PRODUCTS

2.1 BACKFILL MATERIALS, GENERAL

- A. Excavated material may be processed and used for backfill if the Contractor can show compliance with the material specified herein to the satisfaction of the Contracting Officer. If excavated material is not sufficient to meet requirements, Contractor shall import needed material.
- B. Satisfactory Soils: ASTM D 2487 soil classification groups GW, GP, GM, SW, SP, and SM, or a combination of these group symbols; free of rock or gravel larger than 3 inches (75 mm) in any dimension, debris, waste frozen materials, vegetation, and other deleterious matter.
- C. Unsatisfactory Soils: ASTM D 2487 soil classification groups GC, SC, ML, MH, CL, CH, OL, OH, and PT, or a combination of these group symbols.
 - 1. Unsatisfactory soils also include satisfactory soils not maintained within 2 percent of optimum moisture content at time of compaction.
- D. Backfill and Fill: Satisfactory soil materials.
 - 1. Remove rocks over 8 inches in maximum dimension, ice or frozen earth, muck, debris, and earth with high void content.
 - 2. Remove rocks over 4 inches in maximum dimension for backfill placed within 12 inches of foundation.

2.2 BACKFILL MATERIALS, UTILITY TRENCHES

- A. Pipe Zone Material
 - 1. Backfill material for electrical cable and all types of pipe, except ductile iron pipe and galvanized steel pipe, shall consist of soil, sand, or fine granular material free of $\frac{3}{4}$ inch or larger stones, and free of organic material.
 - 2. Backfill material for all ductile iron or galvanized steel pipe shall consist of soil, sand, or rock smaller than two inches in largest dimension and free of organic material.
 - 3. Frozen material will not be allowed.
 - 4. Backfill material shall be trench-excavated material whenever it meets specification requirements. Whenever material meeting the requirements for pipe

zone backfill is not readily available from trench excavation, the Contractor will be required to import pipe zone material from a designated or approved source. Imported pipe zone material shall be paid for as such.

B. Above-Pipe-Zone Material

1. Backfill material shall be free from brush, perishable material, trash, rocks, or boulders larger than 6 inches in greatest dimension, or frozen material.
2. Backfill material shall be trench-excavated material whenever it meets specification requirements. Whenever trench excavated material contains less than 10 percent of oversized material, the Contractor will be required to remove rocks larger than 6 inches from the trench excavated material at no additional compensation and utilize it as backfill material. If, after all suitable trench excavated material has been used as backfill, the trench is not filled to the required grade, the Contractor shall delay his backfill operations until the Contracting Officer can obtain profile elevations of the top of the partially filled trench. These elevations shall be used in computing the cubic yards for which payment will be made for imported material. Whenever material meeting the specification requirements for backfill above the pipe zone is not available from trench excavation, the Contractor will be required to import material from a designated or approved source. Imported above-pipe-zone material shall be paid for as such.

C. Special Bedding – Imported

1. Special bedding material shall consist of rounded river gravel or crushed, free-draining material, meeting the following gradation, as determined by ASTM C 136 and ASTM C 117.

SPECIAL BEDDING GRADATION	
Sieve Designation Square Openings	Percentage by Weight Passing Sieve
1"	100
3/4"	90 – 100
1/2"	20 – 55
3/8"	0 – 15
No. 4	0 - 5

2.3 ENGINEERED FILL

- A. Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D 2940; with at least 90 percent passing a 1 1/2 inch sieve and not more than 12 percent passing a No. 200 sieve.

2.4 DRAINAGE FABRIC

- A. Nonwoven geotextile, specifically manufactured as a drainage geotextile; made from polyolefin's, polyesters, or polyamides; and with the following minimum properties determined according to ASTM D 4759 and referenced standard test methods:
1. Grab Tensile Strength: 110 lbf (490 N); ASTM D 4632.
 2. Tear Strength: 40 lbf (178 N); ASTM D 4533.
 3. Puncture Resistance: 50 lbf (222 N); ASTM D 4833.
 4. Water Flow Rate: 150 gpm per sq. ft. (100 L's per sq.m.); ASTM D 4491.

5. Apparent Opening Size: No. 50 (0.3 mm); ASTM D 4751.

PART 3 - EXECUTION

3.1 LOCATION, ALIGNMENT AND GRADE

- A. The location of all structures shall be staked out and grades established by the Contractor. Locations shall be approved by the Contracting Officer before excavation is started.

3.2 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout and other hazards created by earthwork operations.
- B. Protect subgrades and foundation soils against freezing temperatures or frost. Provide protective insulating materials as necessary.
- C. Provide erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.

3.3 DEWATERING

- A. Prevent surface water and ground water from entering excavations, from ponding on prepared subgrades, and from flooding Project site and surrounding area.
- B. Protect subgrades from softening, undermining, washout, and damage by rain or water accumulation.
1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.
 2. Install a dewatering system to keep subgrades dry and convey ground water away from excavations. Maintain until dewatering is no longer required.

3.4 EXPLOSIVES

- A. Do not use explosives.

3.5 EXCAVATION SUPPORT AND PROTECTION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards that could develop during excavation.
- B. Design, furnish, install, monitor, and maintain excavation support and protection system capable of supporting excavation sidewalls and resisting soil and hydrostatic pressures and superimposed and construction loads.
- C. The contractor shall meet State General Safety Orders and the provisions of the

Occupational Safety and Health Administration (OSHA) pertaining to excavation support and protection, including 29 CFR 1926 Subpart P.

- D. Walls of excavations 5 feet or more in depth shall be supported by shoring and bracing methods or the walls shall be sloped at one and a half to one.
- E. Remove excavation support and protection systems when construction has progressed sufficiently to support excavation and bear soil and hydrostatic pressures. Remove in stages to avoid disturbing underlying soils or damaging structures, pavements, facilities, and utilities.

3.6 EXCAVATION, GENERAL

- A. Unclassified Excavation: Excavation to subgrade elevations regardless of the character of surface and subsurface conditions encountered, including rock, soil materials, and obstructions. No changes in the Contract Sum or the Contract Time will be authorized for rock excavation or removal of obstructions.
 - 1. If excavated materials intended for fill and backfill include unsatisfactory soil materials and rock, replace with satisfactory soil materials.
- B. Topsoil shall be removed from the area to be excavated and from the area where excavated material will be piled, prior to excavation. Topsoil shall be stored as specified below.
- C. Maintain the excavations to guard against and prevent injury to employees and the public. Provide adequate shoring and bracing as required by OSHA and other local governing regulations.
- D. Excavations left open at the end of the working day shall be fenced to protect the public.

3.7 EXCAVATION FOR STRUCTURES

- A. Excavate to indicated elevations and dimensions within tolerance of plus or minus 1 inch. Extend excavations a sufficient distance from structures for placing and removing concrete formwork, for installing services and other construction, and for inspections.
 - 1. Excavations for Footings and Foundations: Do not disturb bottom of excavation. Excavate by hand to final grade just before placing concrete reinforcement. Trim bottoms to required lines and grades to leave solid base to receive other work.
 - 2. Excavation for Underground Tanks, Basins, Pump Houses and Utility Boxes: Excavate to elevations and dimensions indicated within a tolerance of plus or minus 1 inch. Do not disturb bottom of excavations intended for bearing surface.

3.8 EXCAVATION FOR WALKS AND PAVEMENTS

- A. Excavate surfaces under walks and pavements to indicated cross sections, elevations, and grades.

3.9 EXCAVATION FOR UTILITY TRENCHES

- A. All trench excavation shall conform, as near as possible, to the lines and grades

illustrated on the drawings.

- B. Classification of Excavation Material - Excavation will be unclassified as to materials and shall include all materials which are encountered in the required excavation. Any information that has been obtained by the USDA Forest Service concerning possible ground conditions is available at the Supervisor's Office for the Forest where the project is located to interested parties upon request.
- C. Unsatisfactory Material - During excavation, if material which does not meet the backfill requirements of Article 2 (such as structurally unstable material, solid rock, over-sized rock, angular or sharp rock), as determined by the Contracting Officer, is encountered at the grade line for the pipe or cable, the unsatisfactory material shall be removed to a minimum depth of 6 inches below the utility line. Trenching shall be performed by any acceptable method, excluding the use of explosives, as permitted by the Contract General Provisions. In addition to the General Provisions, the Contractor shall provide skilled blasting operators and precautions shall be taken to avoid damage to adjacent property.
- D. Trenching by Machine or by Hand - The use of trench digging machines will be permitted except in places where machines may cause damage to existing structures, utilities, or trees, in which case hand methods shall be employed. Areas specifically to be trenched by hand will be as indicated on the drawings and paid for separately. Machines shall be of the proper size to operate within the specified working limits. In areas being excavated by machine, any hand digging necessary to locate or cross utilities will not be paid for as hand trenching.
- E. Depth - Trench excavation shall provide a uniform (for all utilities) or gently changing (for all utilities except gravity flow sewer lines) flow line.
- F. Width of Trenches - The bottom width that will be used in arriving at pay quantities that are paid on the basis of volume shall be the design bottom width, as shown on the Schedule of Items, or as shown on the trench cross-section detail on the drawings. The width of trench allowed when computing excavation and/or backfill quantities shall be vertical lines for trenches less than 4 feet and for trenches greater than 4 feet shall be computed on 1/2 to 1 side slopes. In circumstances where trench sides will not stand or are not considered safe when sloped at 1/2 to 1, a slope will be determined in the field by the Contracting Officer for which pay quantities will be computed, and the slopes shall be laid back to the stable slope determined.
- G. The Contractor may excavate the trench narrower or wider than the design width shown on the drawings; however, the design width of the trench will be used to calculate the number of cubic yards of all excavated volume and volume of imported material that is paid for by unit volume.
- H. Any over excavation (width) performed by the Contractor for his convenience shall be at his own expense.
- I. Alignment and Grade - The location of all pipelines and structures will be staked out and grades established by the Contracting Officer before excavation is started. All trenches shall conform with the lines and grades illustrated on the drawings or staked on the ground. The Contractor shall set batter boards and shall establish grade lines and levels necessary for the work from dimensions and elevations shown on the drawings

and as established in the field. Any shifting or change from the indicated alignment and grade must receive prior approval by the Contracting Officer in writing. Alignment and grade shall also meet the requirements of Sections governing the utilities which are being installed in the trench.

3.10 APPROVAL OF SUBGRADE

- A. Notify Contracting Officer when excavations have reached required subgrade.
- B. If Contracting Officer determines that unsatisfactory soil is present, continue excavation and replace with compacted backfill or fill material as directed.
- C. Proof roll subgrade below the building slabs and pavements with heavy pneumatic-tired equipment to identify soft pockets and areas of excess yielding. Do not proof roll wet or saturated subgrades.
- D. Reconstruct subgrades damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, as directed by Contracting Officer.

3.11 UNAUTHORIZED EXCAVATION

- A. Fill unauthorized excavation under foundations or wall footings by extending bottom elevation of concrete foundation or footing to excavation bottom, without altering top elevation. Lean concrete fill may be used when approved by Contracting Officer.
 - 1. Fill unauthorized excavations under other construction or utility pipe as directed by Contracting Officer.

3.12 STORAGE OF SOIL MATERIALS

- A. Stockpile, borrow materials and satisfactory excavated soil materials. Stockpile soil materials without intermixing. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
 - 1. Stockpile soil materials away from edge of excavations. Do not store within drip line of remaining trees.
- B. Topsoil shall be kept separate from trench-excavated material by either stockpiling or by windrowing on the opposite side of the trench from which the trench excavated material will be placed. Topsoil will be placed in depressions on the obliterated road.

3.13 STRUCTURE BACKFILL

- A. Place and compact backfill in excavations promptly, but not before completing the following:
 - 1. Construction below finish grade including, where applicable, damp proofing, waterproofing, and perimeter insulation.
 - 2. Surveying locations of underground utilities for record documents.
 - 3. Inspecting and testing underground utilities and storage tanks.
 - 4. Removing concrete formwork.
 - 5. Removing trash and debris.
 - 6. Removing temporary shoring and bracing, and sheeting.

7. Installing permanent or temporary horizontal bracing on horizontally supported walls.
- B. Place and compact fills and backfills adjacent to structures in such a manner as to prevent wedging action or eccentric lodging upon or against the structures.
- C. Place backfill in horizontal layers not more than 12 inches thick with proper moisture content for the required degree of compaction. Flooding or puddling is not allowed. Compact each layer as specified. Backfill layers under concrete flatwork shall be not more than 6 inches thick.
- D. Do not place backfill against any concrete footings or structure without prior permission of the Contracting Officer and in no case less than 7 days after placement of concrete.
- E. Heavy equipment shall not be operated within four feet of any structure.
- F. Provide for anticipated settlement and shrinkage of the backfill and for the finished grades required.

3.14 UTILITY TRENCH BACKFILL

- A. Backfilling will be permitted only after all inspections of piping and/or cable have been performed and tests completed and the work to be covered has been approved by the Contracting Officer. Backfill which has been improperly placed and/or compacted shall be corrected, if directed by the Contracting Officer, by reopening the trench to the depth required to obtain proper compaction. Then the trench shall be refilled and compacted according to specifications.
- B. Place backfill on subgrades free of mud, frost, snow, or ice.
- C. Backfill at Pipe Zone
 1. Any backfill in trench bottom where over excavation was performed by the Contractor for his convenience, shall be brought back to the pipe grade indicated at his own expense. If the trench bottom is prepared in wet conditions, special bedding conforming to Article 2 shall be used if determined necessary by the Contracting Officer.
 2. The bottom of trenches shall be accurately graded to provide uniform bearing and support for each section of the pipe along its entire length, except for portions of the pipe sections where it is necessary to excavate for pipe joints. Depressions for joints shall be made in accordance with the recommendations of the manufacturers for the particular joint used. The bedding shall be a minimum of 6 inches in depth under the pipe and be of either special bedding or pipe zone material as conditions dictate. Trench bottom preparation shall be such that when final placement of pipe has been made, pipe will be true to line and grade. All adjustment to line and grade shall be made by scraping away or filling in with pipe zone material or special bedding material, as conditions dictate, under the body of the pipe and not by wedging or blocking.
 3. After pipe is placed as called for in applicable Sections governing the utilities being placed in the trench, pipe zone material shall be deposited in the trench uniformly on both sides of the pipe for the full width of the trench in 6-inch horizontal layers (loose measurement) and compacted from the bottom of the

- trench to a depth of 1 foot over the top of the pipe.
4. An exception to this is on water lines and sewer lines where the Contractor elects to hydrostatically pressure test the pipe. Joints, couplings, fittings, and valves shall then be left uncovered until after the pipe has been tested. After testing proves the pipe installation to be satisfactory, pipe zone material shall then be placed carefully and compacted around the joints, couplings, fittings, and valves to a depth of 1 foot above the pipe, after which the remainder of the trench shall be backfilled. On gravity flow sewer lines, the Contractor may elect to pressure test the pipe with air. In this case, the entire trench is to be properly backfilled prior to the acceptance test.
 5. Where electrical conduit is buried in the same trench as the waterline or sewer line, the backfill procedure for the conduit shall be performed as outlined in the preceding paragraphs. Location with respect to other utilities in a trench shall be as indicated on the trench cross-section detail as shown on the drawings.
 6. When an electrical conduit is buried singly in a trench, or if only conduit is buried in a trench, the bedding shall be a minimum of 2 inches in depth under the conduit. After the conduit is placed, pipe zone material shall be deposited in the trench uniformly for the full width of the trench and compacted from the bottom of the trench to a depth of 4 inches over the top of the conduit.
 7. Compaction and Testing: All compaction within the pipe zone (electrical conduit area is considered as pipe zone), shall meet the following: Material shall be compacted to not less than 95 percent of the maximum dry unit weight, as determined by AASHTO T 99, Method D, or ASTM D 698, Method D. Ascertain adequate compaction during the backfill operation by performing in-place density tests in accordance with one or more of the following standard test procedures: ASTM D 1556, D 2167, or D 2922, or AASHTO T 191, or T 205.

D. Backfill Above-Pipe-Zone

1. When shown on the drawings as being required, marking tape shall be installed eight inches below the ground surface and shall run the full length of the trenches.
2. Backfill in trenches in areas other than under roadways and parking areas shall be placed in horizontal layers 12 inches thick or less (loose measurement). Layers shall be compacted before the succeeding lift is placed with at least three passes of an approved mechanical compaction device.
3. Compaction and Testing: Backfill in trenches under roadways and parking areas shall be maintained, wetted, or dried to optimum moisture for maximum compaction, placed in the trench in horizontal layers not to exceed 6 inches in thickness (loose measurement), and compacted to not less than 95 percent of maximum dry unit weight, as determined by AASHTO T 99, Method D, or ASTM D 698, Method D. Ascertain adequate compaction during the backfill operation by performing in-place density tests in accordance with one or more of the following standard test procedures: ASTM D 1556, D 2167, or D 2922, or AASHTO T 191, or T 205.

E. Special Bedding – Imported

1. Special bedding shall be placed, as directed by the Contracting Officer, in trenches, as necessary, to provide a minimum of 6 inches firm bedding on which to set the pipe in areas where relatively unstable conditions exist, due to seeping

ground water or mud caused by ground water, or by water from any other source which cannot be diverted. After the material is placed in the trench, leveled, and consolidated, it shall be trimmed to proper sub grade and shaped to receive the pipe.

2. The Contractor shall construct restrictive sections (dams) in the special bedding material at least every 200 feet to minimize the possibility of excessive ground water flows undercutting the pipe. The dam shall extend across the entire width of the trench, be a minimum of 3 feet long, and shall extend to the top of the pipe zone material.

F. Imported Material

1. Any trench excavated material that can be transported less than 300 feet to other areas along the trench and used, in accordance with specifications, shall not be considered as imported material. When the Contractor is required to import material, it shall be from a designated or approved source.

G. Trenches in Embankments

1. When pipelines are to be placed in trenches excavated in embankments, the excavation of each trench shall be performed after the embankment has been constructed to an elevation at least 3 feet over the pipe or to finish grade, whichever is least.

H. Surface Restoration in Areas Other Than Roads

1. All surfaces shall be restored to the required grade (usually original ground line), mounded over or smoothed off as directed, and left in a uniform and neat condition, to the satisfaction of the Contracting Officer. Surface drainage shall be diverted so that it will not flow along a trench. In areas where natural revegetation is designated (no planting to be done), the Contractor shall scarify all disturbed or compacted areas and right-of-ways such that the surface of the ground is loose to a depth of at least one inch. In areas to be seeded, the area shall be prepared in accordance with Section 329206 "Seeding."

I. Warning Tape and Tracer Wire:

1. Warning Tape: Install directly above utilities, as shown on the Drawings.
2. Tracer Wire: Wrap all buried, non-metallic, nonperforated piping with tracer wire.
 - a. Conductivity Testing: Conductivity to be tested before and maintained while backfilling trench. After backfilling, the contractor shall perform a continuity test to the satisfaction of the Contracting Officer.
 - b. Access: Extend tracer wire at least 12-inches above grade at junctions, valves, hydrants and ends of new lines. Attach tracer wire to object or structure where possible. Fold or wrap remaining wire so it is accessible to owner, but not generally visible to the public.

3.15 FILL

- A. Preparation: Remove vegetation, topsoil, debris, unsatisfactory soil materials,

obstructions, and deleterious materials from ground surface before placing fills.

- B. Plow, scarify, bench, or break up sloped surfaces steeper than 1 vertical to 4 horizontal so fill material will bond with existing material.
- C. Place and compact fill material in layers to required elevations as follows:
 - 1. Under grass and planted areas, use satisfactory soil material.
 - 2. Under sidewalks, concrete slabs and pavements, use satisfactory soil material.
 - 3. Under building slabs, use engineered fill.
 - 4. Under utility buildings, use engineered fill or fill according to Section 133400 "Precast Concrete Utility Building."
 - 5. Under footings and foundations, use engineered fill.

3.16 MOISTURE CONTROL

- A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill layer before compaction to within 2 percent of optimum moisture content.
 - 1. Do not place backfill or fill material on surfaces that are muddy, frozen, or contain frost or ice.
 - 2. Remove and replace, or scarify and air-dry, otherwise satisfactory soil material that exceeds optimum moisture content by 2 percent and is too wet to compact to specified dry unit weight.

3.17 COMPACTION OF BACKFILLS AND FILLS

- A. The minimum degree of compaction required shall be a percent of the maximum laboratory density obtained by the standard proctor test AASHTO T99 or ASTM D698. The in-place field density shall be determined by AASHTO T238 or ASTM D2922. The minimum compaction requirements are:
 - 1. For Utility Trenches, see "Utility Trench Backfill" article above.
 - 2. Under structures, water and septic tanks, utility boxes, building slabs, steps, and pavements, scarify and recompact top 12 inches of existing subgrade and each layer of backfill or fill material at 95 percent.
 - 3. Under walkways, scarify and recompact top 6 inches below subgrade and compact each layer of backfill or fill material at 90 percent.
 - 4. Under lawn or unpaved areas, scarify and recompact top 6 inches below subgrade and compact each layer of backfill or fill material at 85 percent.

3.18 GRADING

- A. General: Uniformly grade areas to a smooth surface, free from irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.
 - 1. Provide a smooth transition between adjacent existing grades and new grades.
 - 2. Cut out soft spots, fill low spots, and trim high spots to comply with required surface tolerances.
- B. Site Grading: Slope grades to direct water away from buildings and to prevent ponding.

Finish subgrades to required elevations within the following tolerances:

1. Lawn or Unpaved Areas: Plus or minus 1 inch (25 mm).
 2. Walks: Plus or minus 1 inch (25 mm) unless variation will not meet FSORAG Section 2.0.
 3. Pavements: Plus or minus 1/2 inch (13 mm).
- C. Grading inside Building Lines: Finish subgrade to a tolerance of 1/2 inch (13 mm) when tested with a 10-foot (3-m) straightedge.

3.19 FIELD QUALITY CONTROL

- A. Testing Agency: Contractor will engage a qualified independent geotechnical engineering testing agency to perform field quality control testing.
- B. Allow testing agency to inspect and test subgrades and each fill or backfill layer. Proceed with subsequent earthwork only after test results for previously completed work complies with requirements.
- C. Testing agency will test compaction of soils in place according to AASHTO T310. Tests will be performed at the following locations and frequencies:
1. Minimum testing requirements are listed in Table 014100-1 of Specification Section 014100 "Quality Control" and as listed herein.
 2. Moisture-Density Relationship: One test for each soil type encountered.
 3. Utility Lines:
 - a. Pipe Zone Material: One test per 600 linear feet of trench at locations designated by the Contracting Officer.
 - b. Above Pipe Zone Material: One test per 600 linear feet of trench at locations designated by the Contracting Officer.
- D. When testing agency reports that subgrades, fills or backfills have not achieved degree of compaction specified, scarify and moisten or aerate, or remove and replace soil to depth required; recompact and retest until specified compaction is obtained.
- E. Excessive settlement or other evidence of improper backfill shall be corrected by reopening the trench or excavation to the depth required for proper compaction and then shall be refilled and satisfactorily compacted.
- F. The correction and retesting of unacceptable work shall be paid by the Contractor at no expense to the Government.

3.20 PROTECTION

- A. Protecting Graded Areas: Protect newly graded areas from traffic, freezing and erosion. Keep free of trash and debris.
- B. Repair and reestablish grades to specify tolerances where completed or partially completed surfaces become eroded, rutted, settled, or where they lose compaction due to subsequent construction operations or weather conditions.
1. Scarify or remove and replace soil material to depth as directed by Contracting Officer, reshape and recompact.

- C. Where settling occurs before Project correction period elapses, remove finished surfacing, backfill with additional soil material, compact, and reconstruct surfacing.
 - 1. Restore appearance, quality, and condition of finished surfacing to match adjacent work, and eliminate evidence of restoration to the greatest extent possible.

3.21 SURFACE FINISH

- A. In unpaved areas all surfaces shall be restored to the original ground line or elevations shown on the drawings and left in a uniform and neat condition. Any stockpiled topsoil shall be smoothly distributed over disturbed areas to elevations shown on the drawings.
- B. In paved areas, apply surface treatment as specified and shown on the drawings.

3.22 WASTE MATERIAL

- A. According to Section 024100 "Waste Material Disposal."

END OF SECTION 312000

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 312100 - PROJECT SITE PREPARATION AND GRADING

PART 1 - GENERAL

1.1 This Section includes the following:

- A. Sediment and erosion control measures, shaping, compacting, excavating, and/or filling to the established sub grade as shown on the drawings, details and as staked. Such items covered include but are not limited to construction of subgrade for roads, parking areas, pathways, subsurface drainage structures, and grading around constructed features.
- B. This section also includes the disposal of waste materials generated.

1.2 RELATED SECTIONS:

- A. The following Sections contain requirements that relate to this Section:
 - 1. Section 023701 "Sediment and Erosion Control Measures."
 - 2. Section 024100 "Waste Material Disposal" for the loading, handling, hauling, and placing of excess excavation material, and unsuitable excavation material.
 - 3. Section 311000 "Clearing and Grubbing" for site clearing, grubbing, trimming, disposing of or treatment of timber, slash, and construction debris, and for protection of existing trees and plantings.
 - 4. Section 312000 "Earthwork" for preparing subgrades for slabs-on-grade, walks, pavements, lawns, and plantings, and for excavating and backfilling for buildings, structures, utility trenches, and buried mechanical and electrical utilities.
 - 5. Section 312225 "Excavation and Embankment" for construction of roads, parking areas, and drainage features.

1.3 REFERENCES

- A. The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.
 - 1. Standard Moisture Density Test, AASHTO T-99, Method C or D.
 - 2. Density of Soil In-Place by the Sand-Cone Method, AASHTO T191
 - 3. Density of Soil In-Place by the Drive Cylinder Method, AASHTO T204
 - 4. Density of Soil In-Place by the Rubber-Balloon Method, AASHTO T205
 - 5. Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft³ (600 kN-m/m³)), ASTM D698
 - 6. Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method, ASTM D1556
 - 7. Standard Test Method for Density and Unit Weight of Soil in Place by the Rubber Balloon Method, ASTM D2167
 - 8. Standard Classification of Industrial Fluid Lubricants by Viscosity System, ASTM D2422

1.4 METHOD OF MEASUREMENT AND PAYMENT:

- A. Project Site Preparation and Grading: Lump Sum for Project Site Preparation and Grading and shall include sediment and erosion control, removal and disposal of existing fencing, shaping, earthwork, and excavation and/or embankment (including compacting), and all other incidentals necessary to complete the work as shown on the drawing details and as staked. This item shall also include hauling excess excavation, boulders and clearing and grubbing waste to the areas shown on the drawings.

PART 2 - PRODUCTS

PART 3 - EXECUTION

3.1 PREPARATION

- A. Stripping Sod and Other Organic Materials: Sod, pine needles, and soil heavy in organic materials (topsoil) shall be stripped to mineral soil prior to any excavation or placement of fill material. Stripped material shall be stockpiled on site and placed on disturbed areas and cut/fill slopes upon completion of grading operations. Excess stripped material shall be disposed of in accordance with Section 024100 "Waste Material Disposal".
- B. Clearing and Grubbing
 - 1. All clearing and grubbing shall be performed in advance of beginning excavation and grading operations in accordance with Section 311000 "Clearing and Grubbing".
 - 2. Care shall also be taken to protect and preserve trees and plantings not marked for removal.
 - 3. During all phases of construction the contractor shall confine all of his operations within the working limits as defined in Section 311000 "Clearing and Grubbing".
- C. Sediment and Erosion Control Measures: All sediment and erosion control measures shall be placed prior to any ground disturbance in accordance with Section 023701 "Sediment and Erosion Control Measures."

3.2 GRADING

- A. Graded areas shall be constructed of shallow cuts and fills using native excavated soil and/or select borrow. Some light excavation shall be performed to remove protruding boulders and surface irregularities and at other locations as staked by the Contracting Officer.

3.3 EARTHWORK

- A. Shall be completed in accordance with Section 312000 "Earthwork" for the construction of subgrades for slabs-on-grade, sidewalks, and for excavating and backfilling for buildings, culverts, and structures.
- B. Shall be completed in accordance with Section 312225 "Excavation and Embankment"

for construction of Roads, Parking Areas, and Drainage Features.

3.4 CLEANUP

- A. The Contractor shall confine his operation to that area within the staked fill or cut line of existing roadways, parking areas, paths, etc. No equipment shall be moved over, operated from, or be parked outside this staked fill or cut line, clearing limit, or existing roadway. No materials shall be moved over, stored, or obtained from outside the staked cut or fill lines except by written permission of the Contracting Officer. No rocks, stumps, limbs, debris, soil, or other material shall be left as a windrow or accumulation along the toe of fills, top of cuts, or adjacent areas.

END OF SECTION 312100

May 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 312225 - EXCAVATION & EMBANKMENT

PART 1 - GENERAL

1.1 SUMMARY

- A. This work consists of excavating material and constructing embankments. This work includes furnishing, hauling, stockpiling, placing, disposing, sloping, shaping, compacting, and finishing earthen and rocky material; drainage excavation; and excavation and disposal of unsuitable material.

1.2 DEFINITION

- A. Excavation - All material excavated from within the right-of-way or easement areas that is not included under other pay items listed in the Schedule of Items. Roadway excavation includes all material encountered regardless of its nature or characteristics.
- B. Borrow or Select Borrow - Material used for embankment construction that is obtained from within the roadway prism where shown on the Grading Plan Drawings. Additional sources of borrow excavation will be subject to approval in advance by the Contracting Officer.
- C. Embankment Construction. Embankment construction consists of placing and compacting roadway or borrow excavation. This work includes:
 - 1. Preparing foundation for embankment;
 - 2. Constructing roadway embankments;
 - 3. Benching for side-hill embankments;
 - 4. Constructing dikes, ramps, mounds, and berms, and
 - 5. Backfilling subexcavated areas, holes, pits, and other depressions.
- D. Suitable Material - Granular material conforming to ASTM D 2487 soil classification groups GW, GP, GM, SW, SP, and SM, or a combination of these group symbols; free of rock or gravel larger than 24 inches in any dimension, debris, waste frozen materials, vegetation, and other deleterious matter.
- E. Unsuitable Material - ASTM D 2487 soil classification groups GC, SC, ML, MH, CL, CH, OL, OH, and PT, or a combination of these group symbols.

1.3 MEASUREMENT AND PAYMENT

- A. There will be no separate measurement or payment for work in this section. Payment will be included at the contract unit price for other items shown on the Schedule of Items (See "Project Site Preparation and Grading").

PART 2 - PRODUCTS

NOT APPLICABLE

PART 3 - EXECUTION

3.1 CLEARING & GRUBBING

- A. Clearing and Grubbing shall be completed in accordance with specification Section 311000 "Clearing and Grubbing".
- B. Clearing and grubbing shall be accomplished before excavation or embankment placement begins. Pioneering of roads, slash disposal, and grubbing of stumps may proceed concurrently with clearing and grubbing operation when approved by the Contracting Officer. Excavation and placement operations shall be conducted so slash material to be treated will not be incorporated in the roadway.

3.2 ROADWAY EXCAVATION AND EMBANKMENT SURFACE PREPARATION

- A. General: Do not disturb material and vegetation outside the construction limits. Incorporate only suitable material into embankments. At the end of each day's operations, shape to drain and compact the work area to a uniform cross-section. Eliminate all ruts and low spots that could hold water.
- B. Rock Cuts: Excavate rock cuts to 6 inches below subgrade within the roadbed limits. Backfill to subgrade with suitable material.
- C. Earth Cuts: Scarify earth cuts to 6 inches below subgrade within the roadbed limits.
- D. Embankments Surface Preparation: Remove topsoil and break up the ground surface to a minimum of 6 inches by plowing or scarifying.
- E. All material shall be compacted according to "Embankment Placing Methods" paragraph in Part 3.

3.3 PIONEERING

- A. Pioneering operations for the top of excavation slopes, toe of embankments, or pioneer road construction shall prevent undercutting of the final excavation slope, depositing of materials outside of the roadway limits, and any restriction of drainage.

3.4 UTILIZATION OF EXCAVATED MATERIALS

- A. All suitable, excavated material shall be used in the construction of embankments, subgrades, shoulders, slopes, bedding, and backfill for structures and for other purposes as shown on the Drawings.
 - 1. Excess Excavation:
 - a. Designed excess excavation shall be disposed of as shown on the Drawings.
 - 2. Rock for Slope Protection:
 - a. Excavated rock suitable for protection of embankments may be conserved and used in lieu of a DESIGNATED materials source.
 - 3. Conserving Material:

- a. Material encountered in the excavation, suitable for cushion, road finishing, or other purposes, may be conserved and utilized instead of materials from DESIGNATED sources. Excessively wet material that is otherwise suitable for embankment shall be field drained and dried before placement.
- 4. Excavation of Unsuitable Material:
 - a. Unsuitable material shall be excavated and disposed of as shown on the Drawings or on site as directed by the Contracting Officer (CO). Excavated areas shall be backfilled with suitable material when necessary to complete the work. Frozen material shall not be placed in embankments. **Rocks that are too large to be incorporated into the embankment shall be broken for incorporation into the embankment.**

3.5 DRAINAGE EXCAVATION

- A. Drainage excavation shall include construction of side ditches, minor channel changes, inlet and outlet ditches, furrow ditches, ditches constructed along the road but beyond the roadway limits, and other minor earth drainage structures as shown on the Drawings. Excavated material shall be utilized in accordance with “Utilization of Excavated Materials” paragraph above.

3.6 FINISHING ROADBED

- A. For roads receiving aggregate base or surface course, only rocks that do not protrude above the subgrade more than one-third of the depth of the base or surface course, or 3 inches, whichever is less, may remain in place.
- B. For unsurfaced roads, unless otherwise shown on the Drawings, the top 4 inches below the finished road surface shall not contain rocks larger than 4 inches in greatest dimension. Oversize material shall be removed, reduced to acceptable size, or covered by importing suitable material approved by the Contracting Officer.
- C. The subgrade shall be visibly moist during shaping and dressing. Low sections, holes, cracks, or depressions shall be brought to grade with suitable material approved by the Contracting Officer. Final compaction of the subgrade shall meet the requirements of the embankment placing method specified.

3.7 SNOW REMOVAL

- A. Snow or ice shall not be incorporated in the embankment. Snow shall be removed in advance of the work and deposited beyond the roadway limits in a manner that will not cause resource damage or waste material.

3.8 FINISHING SLOPES

- A. Finished slopes shall conform reasonably to the lines staked on the ground or shown on the Drawings. The finished slope shall be left in a roughened condition to facilitate the establishment of vegetative growth. The finish associated with template and stringline or hand-raking methods will not be allowed. Loose rock, loose debris, and other loose material, each of which is larger than 6 inches in diameter, shall be removed from the

slope unless otherwise shown on the Drawings.

- B. The tops of excavations, excluding areas of solid rock, shall be blended with the adjacent terrain by rounding where shown on the Drawings. Decomposed rock that may be cut without blasting or ripping shall be rounded. Earth overlying rock shall be rounded above the rock.
- C. No blasting shall be permitted if hard rock is encountered.

3.9 OVERBUILDING & LANDSCAPE & STREAM PROTECTION

- A. Unless otherwise agreed to by the Contracting Officer, excavation or embankment material shall be confined within the roadway limits to avoid overbuilding and to protect the landscape and streams.

3.10 SUBGRADE TREATMENTS

- A. Subgrade treatment shall consist of soil modification by admixing aggregates, placing geotextiles, fiber mat, wood corduroy, rock blanket, or other similar materials. The construction and material requirements for the type of subgrade treatment will be shown on the Drawings.

3.11 EARTH BERMS

- A. Permanent earth berms shall be constructed along the shoulder of the traveled way at locations shown on the Drawings. Material used in the construction of berms shall be well graded with no rocks having a dimension greater than one-fourth the height of the berm.
- B. Acceptable material for the berm may be windrowed as the roadbed is constructed. When the local material is not acceptable, material shall be imported from approved sources. Material used for berm construction shall contain no frozen material, roots, sod, or other deleterious material. Material shall not be wasted over the embankment slope.
- C. Compaction shall be accomplished by operating the spreading equipment over the full section of the berm.

3.12 EMBANKMENT PLACING METHODS

- A. All Methods: When an embankment is to be placed across swampy ground and removal of unsuitable material or subgrade treatment is not required, the lower part of the embankment shall be constructed in a single layer to the minimum depth necessary to support construction equipment.
- B. Specific Methods: All embankments shall be placed by **Method 4** unless otherwise noted.
 - 1. Method 1. Side Casting and End Dumping: Embankment may be placed by side casting and end dumping. Where material containing a large amount of rock is used to construct embankments, working smaller rocks and fines in with the larger rocks and fines to fill the voids shall provide a solid embankment.

2. Method 2. Layer Placement: Surfaces steeper than a ratio of 3 horizontal to 1 vertical (3:1) upon which embankment is to be placed, shall be roughened or stepped to provide permanent bonding of new and old materials.
 - a. Embankment shall be layer placed, except over rock surfaces, in which case material may be placed by end dumping to the minimum depth needed for operation of spreading equipment. Each embankment layer shall be leveled and smoothed before placement of subsequent layers. Hauling and spreading equipment shall be operated uniformly over the full width of each layer.
 - b. Suitable material shall be placed in layers no more than 12 inches thick, except when the material contains rock more than 9 inches in diameter, in which case layers may be of sufficient thickness to accommodate the material involved. No layer shall exceed 24 inches before compaction.
 - c. Placing individual rocks or boulders greater than 24 inches will be permitted provided the embankment would accommodate them. Such rocks and boulders shall be at least 6 inches below subgrade. They shall be carefully distributed and the voids filled with finer material to form a dense and compacted mass.
 - d. Where material containing large amounts of rock is used to construct embankments, the layers may be of sufficient thickness to accommodate the material involved. A solid embankment with adequate compaction shall be constructed by working smaller rock and fines in with the larger rocks to fill the voids and by operating hauling and spreading equipment uniformly over the full width of each layer as the embankment is constructed.
 - e. Material shall be at moisture content suitable to obtain a mass that will not visibly deflect under the load of the hauling and spreading equipment. Excessively wet material shall be handled in accordance with Subsection 3.3.1.3.
3. Method 3. Layer Placement (Roller Compaction): Embankments shall be placed as specified in Method 2. Placement shall be in horizontal layers not exceeding 12 inches prior to compaction except when the material contains rock more than 9 inches in diameter, in which case layers may be of sufficient thickness to accommodate the material involved. Compaction equipment shall be operated over the full width of each layer until visible deformation of the layer ceases or in the case of the sheepfoot roller, the roller "walks out" of the layer. At least three complete passes shall be made.
 - a. Compaction equipment shall be capable of obtaining compaction requirements without detrimentally affecting the compacted material. The compacting units may be of any type, provided they are capable of compacting each lift of material as specified and meet the minimum requirements contained herein. Minimum requirements for rollers are as follows:
 - 1) Tamping or sheepfoot rollers shall meet the following minimum requirements:
 - a) Diameter - 40"
 - b) Completely filled with liquid.
 - 2) Grid rollers shall be capable of exerting a force of 250 pounds per

- inch of width of roller drum.
- 3) Steel-wheel rollers, other than vibratory, shall be capable of exerting a force of not less than 250 pounds per inch of width of the compression roll or rolls.
 - 4) Vibratory steel-wheel rollers shall have a minimum weight of 6 tons. The compactor shall be equipped with amplitude and frequency controls and specifically designed to compact the material on which it is used.
 - 5) Pneumatic-tire rollers shall have smooth tread tires of equal size that provide a uniform compacting pressure for the full width of the roller and shall meet the following minimum requirements:
 - 6) Towed rollers:
 - a) Ballast density shall exceed 100 pounds per cubic foot;
 - b) Volume of ballast shall exceed 7 cubic feet per tire;
 - c) Tire pressure shall exceed 45 pounds per square inch
 - 7) Self propelled rollers: Shall exert a force that exceeds 200 pounds per linear inch of rolling width.
4. Method 4. Controlled Compaction: Embankments shall be placed as specified in Method 2, except earth embankments shall be placed in horizontal layers not exceeding 12 inches (loose measure) and compacted. Material shall be at moisture content suitable for attaining the required compaction. Embankments and the top 1-foot of excavation sections shall be compacted to at least 95 percent of the maximum density as determined by AASHTO T 99, Method C or D.
- a. The density of the embankment material will be determined during the progress of the work in accordance with AASHTO T 191, T 205 or T 238; T 217, T 239, or T 255. Corrections for coarse particles will be made in accordance with AASHTO T 99, Note 7.
 - b. Density requirements will not apply to portions of rock embankments that cannot be tested in accordance with approved methods. When this condition exists, compaction shall be provided by working smaller rocks and fines in with the larger rocks to fill the voids and by operating equipment over the embankment materials.
5. Method 5. Controlled Compaction Using Density Control Strips. The embankment placement requirements for Method 4 shall apply for this method except that compaction shall be performed as shown below:
- a. To determine target density, a control strip shall be constructed at the beginning of work on each type of material to be compacted. Each control strip, constructed to acceptable density and surface tolerances, shall remain in place and become a section of the completed roadway. Unacceptable control strips shall be corrected or removed and replaced at the contractor's expense. A control strip shall have an area of approximately 400 square yards and shall be of the same depth specified for the construction of the course that it represents.
 - b. The materials used in the construction of the control strip shall meet the specification requirements. They shall be furnished from the same source

- and shall be of the same type and moisture content used in the remainder of the course represented by the control strip.
- c. The base upon which a control strip is to be constructed will be approved by the Contracting Officer before placing control strip material.
 - d. The equipment used in the construction of the control strip will be approved by the Contracting Officer and shall be of the same type and weight as that to be used on the remainder of the course represented by the control strip.
 - e. Compaction of control strips shall commence immediately after the course has been placed to the specified thickness and shall be continuous and uniform over the entire surface. Compaction of the control strip shall be continued until no discernable increase in density can be obtained by additional compactive effort.
 - f. Upon completion of the compaction, the mean density of the control strip will be determined by averaging the results of 10 nuclear density tests taken at randomly selected sites within the control strip. The mean density of the control strip shall be the target density for the remainder of the course that it represents.
 - g. If the mean density of the control strip is less than 95 percent of the maximum density as determined in the laboratory compacted specimens, the Contracting Officer may request the construction of another control strip. The test procedure used to establish the maximum density will be shown on the Drawings.
 - h. A new control strip may also be requested by the Engineer or by the contractor when:
 - i. A change in the material or job mix formula is made.
 - j. Ten days of production have been accepted without construction of a new control strip.
 - k. There is reason to believe that a control strip density is not representative of the material being placed.
 - l. The specified course shall be compacted to at least 95 percent of the target density. A portable nuclear moisture-density test device in accordance with AASHTO T 238 and T 239 will test density.
 - m. Where portions of rock embankment are constructed that cannot be tested in accordance with approved methods, each layer shall be rolled full width with the same number of passes as the adjacent embankment containing material represented by a control strip.
6. Method 6. Special Project Controlled Compaction: Embankments shall be placed and compacted to at least 90 percent of the maximum density determined by AASHTO T 180, Method C or D, except that compaction of not less than 95 percent of AASHTO T 180, Method C or D, shall be obtained for a minimum depth of 1 foot below subgrade for the width of the roadbed in both excavation and embankment sections.
- a. The density will be determined during the work in accordance with AASHTO T 191, T 205 or T 238; T 217, T 239 or T 255. Corrections for coarse particles will be made in accordance with AASHTO T 99, Note 7.

- A. The tolerance class shall be “C” for roads, spurs and parking areas and “E” for all pathways. Roadway ditches shall be constructed to flow in the direction shown on the Drawings.
- B. Deviations shall be uniform in the direction of change for a distance of 200 feet or more along the project centerline.

TOLERANCE CLASS **

Item	A	B	C	D	E	F	G	H	I	J
Roadbed Width (Ft)	+0.5	+0.5	+1	+1	+1	+1	+1	+1.5	+1	+2
Subgrade Elevation (Ft)	±0.1	±0.2	±0.2	±0.5	±0.5	±1	±1	±1.5	±2	±3
Centerline Alignment (Ft)	0.2	0.2	0.5	0.5	1	1	1	1.5	2	3
Slopes, Excavation and Embankment (Percent Slope)*	±3	±5	±5	±5	±5	±5	±10	±10	±10	±10

**Maximum allowable deviation from construction stakes and drawings.

*Maximum allowable deviation from staked slope measured from slope stakes or hinge points.

END OF SECTION 312225

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 321204 - CRUSHED AGGREGATE BASE OR SURFACE COURSE

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes furnishing, hauling and placing one or more courses of aggregate base or surface course material on roadways, parking areas, picnic shelter pads, paths and sidewalks. In addition, may include furnishing, hauling, and placing crushed aggregate for bedding and backfill.

1.2 SUBMITTALS

- A. Aggregate source, gradation, and material properties.
 - 1. Submit target values within the gradation ranges shown in Table 321204-1 and /or 321204-2 for the required grading. After reviewing the Contractor's proposed target values the CO will determine the final values for the gradation and notify the Contractor in writing.
- B. Compaction density test results and proctor.

1.3 MEASUREMENT AND PAYMENT

- A. Crushed Aggregate: Number of Cubic Yards of material supplied and installed measured in place to the lines and grades shown on the Drawings. Crushed aggregate shall be used for the construction of roads, parking areas, pullouts, paths, under concrete slabs and other items as shown on the Drawings.
- B. Restrooms, Bulletin Board, and Fee Tube Gathering Areas: Crushed aggregate required for these items shall be evaluated by the total square feet (SF) furnished and placed.
- C. Family Unit and Tent Pad: Crushed aggregate required for family unit and tent pads are considered incidental to that item and will be paid for separately as shown in the Schedule of Items.
- D. 3-foot-wide passageway around gate: Crushed aggregate required for passageway shall be considered incidental to work associated with the gate (Section 323116) and will be paid for separately as shown in the Schedule of Items.
- E. No adjustment in a contract unit price will be made for variations in quantity due to differences in the specific gravity or moisture content.

PART 2 - PRODUCTS

2.1 AGGREGATES

- A. General: Furnish aggregate Subbase, Base, or Surface Courses meeting the gradation ranges shown in Table 321204-1 and Table 321204-2. Aggregate grade selection shall

be as shown on the Drawings and in the Schedule of Items.

- B. Materials shall be obtained from an approved source. Furnish aggregates that consist of hard, durable particles or fragments of crushed stone, crushed slag, or crushed gravel meeting the appropriate gradation and conforming to the following:

1. Los Angeles abrasion, AASHTO T 96.....% max
2. Sodium sulfate soundness loss (five cycles), AASHTO T 104.....% max
3. Durability index, AASHTO T 210.....35 min
4. Fractured faces, ASTM D 5821 (Subbase or Base)50% min
5. Fractured faces, ASTM D 5821 (Surface Course)75% min
6. Free from organic matter and lumps or balls of clay.

- C. Obtain the aggregate gradation by crushing, screening, and blending processes as necessary.

2.2 AGGREGATE GRADATION

Table 321204-1, Crushed Aggregate Grading Requirements for Subbase and Base.

Percent Passing (AASHTO T27 and T11)					
Sieve	Grading A (Subbase)	Grading B (Subbase)	Grading C (Base)	Grading D (Base)	Grading E (Base)
2 1/2 -inch	100				
2-inch	97-100	100	100		
1-1/2-inch		97-100			
1-inch	65-79 (6)		80-100 (6)	100	
3/4-inch			60-94 (6)	86-100 (6)	100
1/2-inch	45-59 (7)				
3/8-inch			40-69 (6)	51-82 (6)	62-90 (6)
No. 4	28-42 (6)	40-60 (8)	31-54 (6)	36-64 (6)	36-74 (6)
No. 40	9-17 (4)			12-26 (4)	12-26 (4)
No. 200	4-8 (3)	4-12 (4)	4-7 (3)	4-7 (3)	4-7 (3)

() The value in the parentheses is the allowable deviation (+ / -) from the target values.
Liquid Limit, AASHTO T89 = 25 max. Plastic Limit, AASHTO T-90 = nonplastic.

Table 321204-2, Crushed Aggregate Grading Requirements for Surface Course.

Percent Passing (AASHTO T27 and T11)		
Sieve	Grading F (Surface Course)	Grading G (Surface Course)
1-1/2-inch	100	
1-inch	97-100	100
3/4-inch	76-89 (6)	97-100
1/2-inch		
3/8-inch	56-68 (6)	70-80 (6)
No. 4	43-53 (7)	51-63 (7)
No. 8		
No. 16	23-32 (6)	28-39 (6)
No. 40	15-23 (5)	19-27 (5)
No. 200	10-16 (4)	10-16 (4)

() The value in the parentheses is the allowable deviation (+ / -) from the target values.
 Liquid Limit, AASHTO T 89 = 35 max, Plastic Index, AASHTO T90 = 2 to 9 if percent passing the No. 200 sieve is less than 12% and less than 2 if the percent passing the No. 200 sieve is greater than 12%.

If the plasticity index (PI) is greater than 0, the TV range for the No. 200 sieve size is 8-12 (4).

2.3 AGGREGATE FOR ACCESSIBLE AREAS, TENT PADS & FAMILY UNITS AND PATHWAYS

- A. Furnish material 100 percent passing the 1/2-inch sieve, in accordance with AASHTO T27 and AASHTO T11.
- B. Decomposed Granite shall be an acceptable product, meeting the following requirements:
 1. Decomposed granite shall be any granitoid igneous rock which has been weathered in place and which has as principal constituent granular fragments of quartz and feldspar.
 - a. It may also contain fragments of granitic rock not yet broken down into the component minerals.
 - b. This material shall remain stable when saturated with water.
 - c. Particles larger than 2 inches, which will not be broken in the process of rolling and tamping during construction, shall be used.
 2. Decomposed granite shall conform to the following requirements:
 - a. 100% shall pass the 1/2" sieve.
 - b. Not more than 20% shall pass the No. 200 sieve.
 - c. The P.I. of the material passing the No. 200 sieve shall not be less than 3 nor greater than 10.

PART 3 - EXECUTION

3.1 GENERAL

- A. Verify that subgrade is dry and in suitable condition, locate areas that are unstable or that require further compaction.
- B. Proceed with aggregate placement only after unsatisfactory conditions have been corrected and subgrade is approved in writing by the Contracting Officer (CO).

3.2 PREPARATION OF SUBGRADE

- A. The subgrade shall be prepared in accordance with requirements of other specifications sections.
- B. The subgrade shall conform to the lines and grades shown on the Drawings. Suitable material shall be utilized in the preparation of the subgrade. When embankment or fill is necessary, subgrade shall be placed in compacted layers not exceeding 6 inches. Unless specified otherwise, subgrade shall be compacted to 95 percent of AASHTO T 99, method C or D.
- C. Suitable material for subgrade shall be granular material or fine compatible soil free of excess moisture, muck, frozen lumps, roots, sod, and other deleterious material. Remove all rock particles and hard earth clods larger than 3 inches in the longest dimension.

3.3 MIXING AND SPREADING

- A. Mix the aggregate and adjust the moisture content to obtain uniform moisture. Spread and shape the mixture on the prepared surface in a uniform layer not to exceed 6 inches in compacted thickness.
- B. Route hauling equipment uniformly over the full width of the surface to minimize rutting or uneven compaction.

3.4 COMPACTING

- A. Compact each layer of aggregate full width. Compact each layer to a density of at least 95 percent of the maximum density as determined by AASHTO T 99 method C or D.
 - 1. Provide compaction density testing at the following intervals:
 - a. One test every 300 square yards of aggregate placed.
 - b. One test for each family unit or concrete slab

3.5 SURFACE AND CONSTRUCTION TOLERANCES

- A. Shape the surface to the required template and as staked. Surface shall be graded and shaped smooth to within ½-inch in 10 feet.
- B. Maintain the aggregate course to the correct lines, grade, and cross-section by blading,

watering, and rolling until placement of the next course.

- C. Upon completion of full placement and after haul trucks have completed their haul across section of the road, the road shall be finish bladed, watered, and rolled.
- D. Aggregate shall be placed as shown on the drawings and as staked. Tolerance for thickness of aggregate shall be $\pm \frac{1}{2}$ -inch and for width shall be + 1-foot.

3.6 ACCEPTANCE

- A. Aggregate shall be accepted following placement when shown to meet material quality, gradation, compaction requirements, required depth and width, and finish blading.

END OF SECTION 321204

August 2020

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 321206 - LOOSE ROCK RIPRAP

PART 1 - GENERAL

1.1 SUMMARY

- A. This item shall consist of furnishing and placing a protective covering of stone on designated slopes or in fills or dikes, in accordance with these specifications, in conformity with the drawings and to the lines and grades established.

1.2 SUBMITTALS

- A. Product Data for geotextile fabric.

1.3 MEASUREMENT AND PAYMENT

- A. The quantity to be measured shall be the number of cubic yards (CY), measured in place, completed, and accepted. The limiting dimensions shall not exceed those shown on the drawings or established. No separate measurement shall be made for the geotextile fabric material placed beneath the riprap when shown on the drawings, and rather shall be included in the measurement for the loose rock riprap.

PART 2 - PRODUCTS

- 2.1 STONES - The stones used for this work shall be durable, angular, field, or quarry stones which are sound, hard, and free from laminations, fractures, or other structural defects. They shall be of such quality that they will not disintegrate on exposure to water or weathering.

- A. Stones used shall be such that at least 60 percent of the individual stones shall have a volume of 8 cubic feet or more. Not more than 10 percent of the stone shall have a perimeter of less than 24 inches when measured on the smallest section.

- B. Gradation Requirements for Riprap

Class	% of Rock Equal or Smaller by Count, D _x	Range of Intermediate Dimensions, ⁽²⁾ inches (millimeters)	Range of Rock Mass, ⁽³⁾ pounds (kilograms)
0	100	6 – 8 (150 – 200)	17 – 41 (8 – 19)
	85	5 – 6 (150 – 150)	10 – 17 (5 – 8)
	50	2 – 5 (50 – 125)	0.6 – 10 (0.3 – 5)
	15	0 – 2 (0 – 50)	0 – 0.6 (0 – 0.3)
1	100	9 – 15 (230 – 380)	59 – 270 (27 – 120)
	85	7 – 11 (180 – 280)	28 – 110 (13 – 50)
	50	5 – 8 (130 – 200)	10 – 42 (5 – 19)
	15	3 – 6 (80 – 150)	2 – 18 (1 – 8)

2	100	15 – 21 (380 – 530)	270 – 750 (120 – 340)
	85	11 – 15 (280 – 380)	110 – 270 (50 – 120)
	50	8 – 11 (200 – 280)	42 – 110 (19 – 50)
	15	6 – 8 (130 – 200)	10 – 42 (6 – 19)
3	100	21 – 27 (530 – 690)	750 – 1600 (340 – 730)
	85	15 – 19 (380 – 480)	270 – 560 (120 – 250)
	50	11 – 14 (280 – 360)	110 – 220 (50 – 100)
	15	8 – 10 (200 – 250)	42 – 81 (19 – 37)
4	100	27 – 33 (690 – 840)	1600 – 2900 (730 – 1300)
	85	19 – 23 (480 – 580)	560 – 990 (250 – 450)
	50	14 – 17 (360 – 430)	220 – 400 (100 – 180)
	15	9 – 12 (230 – 300)	59 – 140 (27 – 64)
5	100	33 – 39 (840 – 990)	2900 – 4850 (1300 – 2200)
	85	23 – 28 (580 – 710)	990 – 1800 (450 – 820)
	50	17 – 20 (430 – 510)	400 – 650 (180 – 290)
	15	11 – 15 (280 – 380)	110 – 270 (50 – 120)
6	100	39 – 45 (990 – 1140)	4850 – 7400 (2200 – 3350)
	85	28 – 32 (710 – 810)	1800 – 2650 (820 – 1200)
	50	20 – 23 (510 – 580)	650 – 990 (290 – 450)
	15	13 – 17 (330 – 430)	180 – 400 (82 – 180)

(1) Gradation includes spalls and rock fragments to provide a stable, dense mass.

(2) The intermediate dimension is the longest straight-line distance across the rock that is perpendicular to the rock's longest axis on the rock face with the largest projection plane.

(3) Rock mass is based on a specific gravity of 2.65 and 85 percent of the cubic volume as calculated using the intermediate dimension.

(4) Furnish rock with breadth and thickness at least one-third its length.

2.2 SOURCE - Riprap shall be obtained from a commercial source. Color and shape shall match closely the native rock onsite.

2.3 GEOTEXTILE FABRIC

A. Shall be Type IV. Mirafi No. 1100N nonwoven, as manufactured by Mirafi, Inc. of Charlotte, North Carolina, or an approved equal.

PART 3 - EXECUTION

3.1 FOUNDATION - The slope or area upon which the riprap is to be placed shall be shaped to the required lines and grades. The surface shall be roughened to provide a surface to which the base stones will key and be firmly bedded. Foundation trenches shall be excavated at the toe of the slope or area to receive the base stones and provide a secure footing. These trenches shall be of sufficient width and extend a minimum of 2 feet below existing ground or the bed of the stream or wash. Slopes and trenches shall be approved before the placing of riprap is begun.

3.2 PLACING RIPRAP

A. The stones shall be placed or dumped on the approved slope and in the trench to form the cross section desired. They shall be manipulated sufficiently to secure a

roughly regular surface and mass stability. All stones shall be firmly keyed or bedded. Insofar as possible, the larger stones shall be placed at the bottom.

- B. The voids between the larger stones shall be filled with smaller stones. This shall be done in such a manner that they do not prevent the interlocking of the larger stones. The finished surfaces which will remain exposed, shall present a broken joint surface on which voids are reasonably well filled.
- C. When the thickness of the riprap is not shown on the drawings, it shall be at least 36 inches measured perpendicular to the slope.
- D. When the rock riprap is completed, the area shall be cleaned up by removing all debris and material not used. Material excavated from foundation trenches, etc., shall be satisfactorily disposed of.

END OF SECTION 321206

April 2023

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 321460 – TRAILS AND PATHS

PART 1 - GENERAL

1.1 SUMMARY

- A. This work shall consist of constructing trails and paths in close conformity to the dimensions, lines and grades AS SHOWN ON THE DRAWINGS or STAKED ON THE GROUND.
- B. Work shall include staking, clearing, excavation and embankment, and crushed aggregate surfacing (paths only).
- C. Trails and Paths are colloquially known as “Outdoor Recreation Access Routes” in the FSORAG document. Design and construction shall meet the appropriate sections in the FSORAG.

1.2 RELATED WORK SPECIFIED ELSEWHERE

- A. Clearing and Grubbing – Section 311000.
- B. Aggregate Surface Course – Section 321204

1.3 MEASUREMENT AND PAYMENT

- A. Payment for paths shall be linear feet (LF) of 5-foot-wide aggregate surfaced paths as measured along the centerline of the path. Payment shall include shaping and compacting subgrade and placement and compaction of aggregate surface course as shown on the drawings.
- B. Clearing and grubbing for paths shall be included under pay item 311000 “clearing and grubbing”.

PART 2 - PRODUCTS

2.1 SOURCE

- A. Materials shall be obtained from a commercial source. The contractor shall submit test results for aggregate gradation and a Certificate of Compliance that states gradation meets contract requirements in accordance with Section 321204.
 - 1. 1/2-inch gradation.

PART 3 - EXECUTION

3.1 STAKING

- A. Trail centerline location (and elevation if required) shall be staked by the Contractor Officer where SHOWN ON THE DRAWINGS.

3.2 SITE PREPARATION AND GRADING

- A. Before commencing work, thoroughly remove all debris, vegetation, and rock 2 inch and larger.
- B. All areas within the limit of the work shall be carefully and accurately graded and shaped to the elevations shown.
- C. All noticeable depressions shall be filled, and all mounds graded down so that the entire area presents a uniformly even surface.
- D. All low areas shall be filled in and the trail area shall be slightly above the surrounding area after compaction to prevent water from standing on the trail area.
- E. Clearing limits shall be AS SHOWN ON THE DRAWINGS.
- F. Felling, cutting, and trimming methods shall not cause bark damage to standing trees.
- G. Stump holes shall be backfilled with compacted mineral soil or borrow material from designated source.

3.3 EXCAVATION AND EMBANKMENT

- A. Excavation and embankment shall be accomplished to meet the lines and grades AS SHOWN ON THE DRAWINGS or DESIGNATED ON THE GROUND.
- B. Deviations from the trail/path alignment will be permitted to avoid or minimize disturbance to the physical terrain.
- C. Mineral soil within the spread of existing trees that are to remain shall not be disturbed.
- D. All excavated material shall be conserved and used in construction of the trailbed:
 - 1. Rocks, which protrude above the trailbed, shall be removed and cavities backfilled with compacted mineral soil.
 - 2. Rocks shall be laid in a continuous row along the down slope side of the trail.
 - 3. The row shall be bedded to be stable and placed to ensure no blockage of drainage.
- E. Embankment shall be constructed of compacted mineral soil or borrow material from designated sources.
 - 1. All disturbed soil within the trailbed area shall be recompacted to provide a soil bed.

3.4 CRUSHED AGGREGATE BASE

- A. The aggregate shall be spread in a uniform layer, with no segregation of size, and to a loose depth that shall have the required thickness when compacted.
- B. If the required compacted depth of any aggregate base or surface course exceeds 6 inches, it shall be placed in two or more layers of approximately equal thickness. If the nominal maximum particle size exceeds 3 inches, the aggregate shall be placed in layers that do not exceed twice the maximum size of the aggregate size specified.
- C. Hauling equipment shall be operated over the surface of the previously constructed layer in a dispersed manner to minimize rutting or uneven compaction.

3.5 COMPACTION

- A. Paths: Aggregate shall be compacted using pneumatic vibrators or rollers. Compact each layer of aggregate full width. Compact each layer to a density of at least 95 percent of the maximum density as determined by AASHTO T 99 method C or D.

3.6 THICKNESS REQUIREMENTS

- A. The thickness of the compacted aggregate shall not vary more than 1/2 inch for aggregates with a nominal maximum particle size of 1 inch or less, nor more than 1 inch for aggregates with a nominal maximum particle size greater than 1 inch from the thickness SHOWN ON THE DRAWINGS.
- B. The compacted thickness shall not be consistently above or below the specified thickness.

END OF SECTION 321460
April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 322100 – FAMILY UNIT AND TENT PAD

PART 1 - GENERAL

1.1 SUMMARY

- A. This work shall consist of construction of family units and tent pads. It includes furnishing surfacing material, clearing, excavation and constructing the pads in accordance with these specifications and details as SHOWN IN THE DRAWINGS and in compliance with applicable sections of the FSORAG.

1.2 MEASUREMENT AND PAYMENT

- A. Family Units, Day Use Pads & Tent Pads: Payment shall be number of each (EA) Family unit/Day Use Pad/Tent Pad installed and accepted including aggregate surfacing.
 - 1. Site Furniture for family units and day use pads shall be measured and paid for separately according to specification 129300 Site Furniture.
 - 2. Site preparation (grading, excavating & filling) for family units, day use pads and tent pads shall be measured and paid for according to specification 312100 Project Site Preparation and Grading.

PART 2 - PRODUCTS

2.1 SURFACE MATERIAL FOR TENT PADS

- A. Source:
 - 1. Materials shall be obtained from a commercial source. The contractor shall submit test results for aggregate gradation and a Certificate of Compliance that states gradation meets contract requirements.
 - 2. Sampling and testing of the material before incorporation into the work shall be in accordance with AASHTO requirements and shall occur as follows:
 - a. For commercially produced aggregates: at the producer's plant or stockpile.
 - b. These test results shall not preclude later sampling and testing for final acceptance after final processing of the material.
 - c. Crushing, screening, and mixing plants shall be equipped with sampling devices.
 - d. The contractor shall take additional samples as required to validate the certification.
- B. Crushed Aggregate Surface Course for Family Unit Pads and Tent Pads shall be 1/2-inch minus gradation, in accordance with Section 321204, and shall remain stable when saturated with water.

2.2 TREATED TIMBER / LUMBER

- A. Treated lumber shall be pre-cut and pre-drilled Douglas Fir No. 2 Grade or Hem Fir No. 2 Grade, as graded by the Western Wood Products Association (WWPA).
 - 1. Size: Treated Header – 6” x 6”.

PART 3 - EXECUTION

3.1 SITE CLEANING

- A. General:
 - 1. Remove trees, shrubs, grass and other vegetation, improvements, or obstructions interfering with installation of new construction.
 - 2. Remove such items elsewhere on site or premises as specifically indicated.
 - 3. Removal includes digging out stumps and roots.
 - 4. Carefully and cleanly cut roots and branches of trees indicated to be left standing, where such roots and branches obstruct new construction.
 - 5. Clearing shall be kept to a minimum within the specified working limits and care exercised to not damage the root system of adjacent trees or shrubbery.
- B. Topsoil:
 - 1. Topsoil is defined as friable clay loam surface soil found in a depth of not less than 4". Satisfactory topsoil is reasonably free of subsoil, clay lumps, stones, and other objects over 2" in diameter, and without weeds, roots, and other objectionable material.
 - 2. Strip topsoil to whatever depths encountered in a manner to prevent intermingling with underlying subsoil or other objectionable material.
 - 3. Where trees are indicated to be left standing, stop topsoil stripping a sufficient distance to prevent damage to main root system.
 - 4. Stockpile topsoil in storage piles in areas shown, or where approved.
 - 5. Construct storage piles to freely drain surface water.
 - 6. Dispose of unsuitable or excess topsoil same as waste material, herein specified.
- C. Clearing and Grubbing:
 - 1. Clear site of trees, shrubs and other vegetation, except for those indicated to be left standing.
 - 2. Completely remove stumps, roots, and other debris protruding through ground surface.
 - 3. Use only hand methods for grubbing inside drip line of trees indicated to be left standing.
 - 4. Fill depressions caused by clearing and grubbing operations with satisfactory soil material, unless further excavation or earthwork is indicated.
 - a. Place fill material in horizontal layers not exceeding 6" loose depth, and thoroughly compact to a density equal to adjacent original ground.
- D. Removal of Improvements: Remove existing above-grade and below-grade improvements necessary to permit construction, and other work as indicated.

- E. Preparation of Subgrade shall be in accordance with Specification 321204.
- F. Mixing, Spreading and Compacting Aggregate Surface Course for Day Use and Family Unit Pads shall be in accordance with Specification 321204.

3.2 TREATED LUMBER

- A. Timber to be laid on a continuous solid soil surface.
- B. All timbers will be of full length (no splicing, except when necessary).
- C. All timber length, lap joints and holes will be cut and drilled, and then C.C.A. treated.
- D. All joints shall be lapped and pinned.
- E. All timber corners shall be cross-braced.
- F. If tent pad is located directly adjacent to the campground unit the common timber border shall be eliminated.

3.3 SURFACING MATERIAL

- A. The surfacing material shall be spread in a uniform layer, with no segregation of size, and to a loose depth that shall have the required when compacted.
- B. Removing all debris and material not utilized shall clean up the area.

3.4 FINISH GRADING

- A. All surfaces and slopes shall be shaped to blend with the original ground line, mounded over or smoothed off, hand raked, and left in a uniform and neat condition. Surface drainage shall be diverted so that it will not enter into the area. See drawings for details.

3.5 CLEANUP

- A. After finish grading has been completed, the disturbed area shall be finished to present as near a natural appearance as possible and cleaned up by removing all debris and materials not utilized. Cleanup shall include disposal of waste materials in accordance with Section 024100. Stockpiled topsoil shall be smoothly distributed over disturbed areas and hand raked to blend with ground line.

END OF SECTION 024100
MAY 2024

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS
SECTION 322205 - BARRIER ROCKS

PART 1 - GENERAL

1.1 SUMMARY

- A. This item shall consist of collecting, hauling, and placing barrier rocks. They shall be placed and installed in conformity with the details as shown on the barrier rock drawing detail, at the locations staked by the contracting Officer, (CO).

1.2 MEASUREMENT AND PAYMENT

- A. Payment will be made for each (EA) commercially sourced barrier rock, furnished, installed, and accepted.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Rock: The rock used for this work shall be angular or rounded, sound, hard, and free from laminations, fractures, or other structural defects. They shall be of such quality that they will not disintegrate on exposure to water or weathering. Color shall match rocks native to the project site.
- B. Rocks used shall have a volume of greater than 15 cubic feet (approximately 2500 pounds).
- C. Source: Commercial Source.

PART 3 - EXECUTION

3.1 PROTECTION OF TREES AND SHRUBS

- A. All work under this item shall be done in such a manner that trees and larger vegetation are not damaged.

3.2 EXCAVATION

- A. All barrier rock shall be placed according to detailed drawings.
- B. All excavated material 3 inches and smaller shall be used for backfill of barrier rock.
- C. All material not suitable for backfill shall be segregated. Stockpiled, and disposed of at the disposal site.
- D. Excavation shall be of such size and depth that 30 to 50 percent by volume of the placed barrier rock is set below finished grade or natural ground as the specific location may dictate.

3.3 PLACING OF ROCK BARRIERS

- A. Rock barriers shall be placed according to detailed spacing shown on drawings.
- B. Fifty to 70 percent of rock volume shall be visible when work is complete.
- C. Vary the placement of barrier rock in size, height, shape and spacing, to avoid uniformity. Irregular placement is encouraged; no straight-lines.

3.4 COMPACTION

- A. Accepted segregated backfill material used in placing the barrier rock shall be placed by hand and compacted with a minimum of three passes of an approved mechanical tamper.

3.5 CLEANUP AND EXCESS MATERIAL DISPOSAL

- A. After the barrier rocks have been placed and the backfill completed, the general area shall be cleaned up by removing all debris and material not utilized.
- B. Any excess excavated material that can be transported less than 300 feet within the total project area and used in accordance with other project specifications will not be paid for separately. For disposal of excavated material in excess of total project needs or material not meeting other specifications, the Contractor shall see Section 024100, "Waste Material Disposal."

END OF SECTION 322205

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 322622 - PRECAST CONCRETE CURB STOP

PART 1 - GENERAL

1.1 SUMMARY

- A. This item shall consist of furnishing and installing pre-cast concrete curb stops in accordance with these specifications and details shown on drawings in conformity with the typical layout and dimensions to the grade established.

1.2 MEASUREMENT AND PAYMENT

- A. Curb Stop: Payment will be made for each (EA) type of pre-cast concrete curb stop (including concrete, reinforcement, delivery, labor, and anchor pins) installed and accepted and as listed in the Schedule of Items.

PART 2 - PRODUCTS

2.1 PRECAST CONCRETE CURBS

- A. Precast Curb Stops - The precast concrete curb stops shall be the size and shape as shown on the drawings. The concrete shall be reinforced as shown on the drawings and shall be 4,000 psi (28 day), 6.5 bag mix, with 6 percent entrained air.
- B. Before any of the precast sections are placed in the work, the Contractor shall furnish to the Contracting Officer a "certificate confirming compliance for reinforcement and strength" issued by the Pre-cast manufacturer.

PART 3 - EXECUTION

3.1 INSPECTION

- A. The Forest Service shall have the right to inspect the precast curb stops either at the precast plant or after delivery. Units delivered to the project site shall be rejected if they have spalls, surface hairline cracks exceeding 12 inches in any single square foot area, or any cracks exceeding 1/16 inch width. All curb stops that are damaged through improper handling or placing shall be rejected.

3.2 BEDDING

- A. When the curbing is to be installed on a bituminous surfaced area or aggregate surfaced area, the curb stops shall be set on top of the surfacing and no bedding will be required.
- B. When the curbing is to be installed on a crushed aggregate or native surfaced area, the base on which the curb units are to set shall be compacted by a minimum of three

passes with a vibratory compactor to a firm, even surface. All soft or unsuitable material shall be removed and replaced with material suitable to the Contracting Officer.

3.3 PLACING AND ANCHORING

- A. The pre-cast curb stops shall be placed or set as illustrated on the drawings or as directed by the Contracting Officer. This may be in a continuous line or as cords of a curve with or without openings between. They shall, when set in final location, set solidly on the bedding or surfacing.
- B. When lifting wires have been cast in the bottom of the sections, they shall be removed. The wires may be removed by cutting reasonably flush so that they do not interfere with the section setting solidly.
- C. The pre-cast sections shall be anchored in place by driving steel pins or dowels flush with the top through the two holes provided. These pins shall be 3/4 inch round or No. 6 deformed reinforcing steel, 24 inches long.

3.4 CLEANUP

- A. When the installation of the curbing has been completed, the area shall be cleaned up by removing all debris and material not utilized.

END OF SECTION 322622

December 2019

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 323116 - GATES FOR ROADS

PART 1 - GENERAL

1.1 SUMMARY

- A. This item shall consist of gates of various types, constructed by fabricating pipe and installing in accordance with these specifications and details shown on the drawings at the locations designated, in conformity with the line and grade established.

1.2 MEASUREMENT AND PAYMENT

- A. Gates: Number of gates of each size designated, including materials, fabrication, earthwork, installation, concrete, backfill, signs, incidentals and all other work within 10 feet of the gate, including 3-foot wide passageways along outside post as shown on the drawings.

PART 2 - PRODUCTS

2.1 PIPE

- A. Pipe used to fabricate the posts and gates shall be standard weight black steel conforming to the requirements of ASTM A53. The entire gate shall be painted as shown on the drawings.

2.2 CONCRETE

- A. The concrete used in the bases shall be as required by Section 033000.

2.3 SIGNS AND OBJECT MARKERS

- A. Signs and object markers shall conform to the Manual of Uniform Traffic Control Devices (MUTCD) and shall be furnished and installed by the Contractor.

2.4 VANDAL-PROOF HARDWARE

- A. Signs and object markers shall be installed using vandal-proof 5/16" diameter one-way bolts with breakaway nuts as supplied by Walker Safety & Signing Co., Idaho Falls, Idaho, phone No. (208) 529-2723 or an approved equal

2.5 LOCKS

- A. Padlocks will be furnished by the Forest Service.

PART 3 - EXECUTION

3.1 FABRICATING GATES AND POSTS

- A. The gates, receiving sockets, and posts shall be fabricated in accordance with the dimensions and details shown on the drawings. Ends of pipe members shall be coped to approximately fit and welded, using a continuous fillet weld. All exposed pipe ends shall be ground smooth. Paint all exposed metal surfaces with one coat of Benjamin Moore Retardo metal primer and two coats of Benjamin Moore brown Impervo No. 20059 or approved equal.

3.2 ERECTION

- A. Neat excavation shall be made to receive the concrete base and socket or post. In boulder formation, individual rocks may intrude into the neat excavation, but they shall be firmly imbedded. Protruding rock surfaces shall be cleaned and dampened to improve bonding to the concrete.
- B. The concrete shall be thoroughly tamped while being placed to insure a firm bearing against the sides of the excavation. The top-exposed portion of each concrete base shall be formed so that the base will appear relatively circular. The top surface of the concrete shall be wood float finished and conical shaped for drainage. The concrete shall be covered with suitable material and kept moist for 3 days or a liquid membrane of approved curing material may be applied.
- C. The receiving sockets or posts, when the concrete has been placed, shall be plumb and at the desired elevation.
- D. The gates, entrances and service road, shall rotate freely and be plumb, level, and in line.

3.3 SIGN AND OBJECT MARKERS

- A. Install signs and object markers using vandal-proof hardware. Use 4 bolts for each sign and each type 3 object marker. Adhere type two object markers directly to metal plates.

3.4 GRADING AND CLEANUP

- A. The area around the unit shall be finish graded to the desired contour. When complete, the general area shall be cleaned up by removing all debris and material not utilized.
- B. Construct 3-foot-wide passageway where shown on the drawings. Place ½-inch minus aggregate gradation for finished surface of passageway. Refer to Section 321204 for gradation details.

END OF SECTION 322716

May 2019

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 329206 - SEEDING

PART 1 - GENERAL

1.1 SUMMARY

- A. This item shall consist of broadcast seeding designated areas using specified seed mixtures with fertilizer.
- B. The areas to be seeded shall be all cut slopes, fill slopes and other disturbed areas.
- C. Related Specifications include the following:
 - 1. Federal Specifications JJJ-S-181

1.2 MEASUREMENT AND PAYMENT

- A. Measurement and payment shall be by acres seeded in accordance with spread rates determined by seed mix. Payment shall include fertilizer.

PART 2 - PRODUCTS

2.1 SEED

- A. The seed mix and rate shall be determined through contact and coordination with the local NRCS office or Local Forest Service Botanist.
- B. All seed shall conform to the requirements of Federal Specifications JJJ-S-181; State Laws; and U.S. Department of Agriculture Rules and Regulations under the Federal Seed Act; in effect on the date of invitation of bids. Seed which has become wet, moldy or otherwise damaged in transit or storage will not be accepted.
- C. Seed shall be furnished separately or in mixture in standard containers clearly marked or labeled with (1) Seed name, (2) Lot number, (3) Net weight, (4) Percentage of purity and of germination and hard seed, (5) Percentage of weed seed content (must be less than 0.3%), and (6) all seed shall be certified as noxious weed-free seed for all Western States. The Contractor shall furnish the Contracting Officer duplicate signed copies of a statement by the vendor certifying that each lot of seed has been tested by a recognized laboratory for seed testing within 6 months of date of delivery. This statement shall include: (1) Name and address of laboratory, (2) Date of test, (3) Lot number of each kind of seed, and (4) Results of tests as to name percentages of purity and of germination, and percentage of weed content, for each kind of seed furnished, and in case of a mixture, the proportions of each kind of seed.

2.2 FERTILIZER

- A. The fertilizer shall be 16% total nitrogen, 16% available phosphoric acid, and 16% total water-soluble potash applied at the rate of 400 pounds per acre.
- B. Fertilizer shall be dry, free-flowing type suitable for application with broadcast seeding. It shall be a standard commercial fertilizer supplied separately or in mixtures in standard containers with name, weight, and guaranteed analysis of contents clearly marked. Fertilizer which has become wet or otherwise damaged in transit or storage will not be accepted.

PART 3 - EXECUTION

3.1 GENERAL

- A. The specified seed mixture, with fertilizer, shall be uniformly spread on the designated areas to the density in pounds of live seed per acre as specified.
- B. Each area or suitable section of the area to be seeded shall be seeded as soon as the grading and finishing work have been completed and the area prepared and approved for seeding. Seeding shall follow the finishing work as closely as feasible and if possible before the ground has become packed or hardened. No regard shall be given to the season of the year except that no seeding shall be done during windy weather or when the ground is excessively wet or deeply frozen.

3.2 PREPARATION OF SEEDING AREA

- A. Cut slopes, fill slopes, islands, embankments or other areas to be seeded shall be shaped and finished as specified under the Sections involved. The area, where necessary, shall then be hand raked or otherwise worked such that the surface is loose to a depth of at least one inch. Each area shall be approved for seeding by the Contracting Officer before seed is applied.

3.3 SEEDING

- A. The seed or seed mixtures, with or without fertilizer, shall be accurately proportioned as stipulated and thoroughly mixed. They shall be remixed as necessary so that a uniform mixture will result as each loading of the seeder is made.
- B. Seed, with fertilizer, shall be applied with a rotary hand seeder or other approved type commercial seeder or by an agreed upon method. All portions of the area shall be uniformly covered to the required density.
- C. Immediately after seeding the contractor shall rake in the seed such that the seed becomes integrated with the soil.

3.4 MAINTENANCE OF SEEDED AREA

- A. The Contractor will not be required to maintain an area which has been satisfactorily seeded except that he shall protect the area against traffic by warning signs or barricades or other methods approved by the Contracting Officer.

- B. When a seeded area has become damaged by storm or otherwise prior to final acceptance of the project, the Contracting Officer may order the area reworked. The damage shall then be repaired as directed and the area reseeded.

END OF SECTION 329206

April 2021

USDA FOREST SERVICE, R-3
HOPEWELL LAKE SITE IMPROVEMENTS

SECTION 332228 - CULVERT PIPE AND PIPE ARCHES

PART 1 - GENERAL

1.1 SUMMARY

- A. This work shall consist of furnishing and installing metal pipe and pipe appurtenances, including all excavation, bedding, and backfilling required to complete the work.

1.2 DEFINITIONS

- A. GSP: Galvanized Steel Pipe.
- B. HDPE: High Density Polyethylene Plastic Pipe.
- C. PVC: Polyvinyl Chloride Plastic Pipe.
- D. CMP: Corrugated Metal Pipe.

1.3 SUBMITTALS

- A. Provide fabricator's certification that the sheet and pipe fabrication are in accordance with AASHTO M 36, M 196, and M245, as applicable. Submit the certification before installing the pipe.

1.4 COORDINATION

- A. The lengths and locations of individual pipe SHOWN ON THE DRAWING are approximate. Do not order pipe until culvert locations are DESIGNATED ON THE GROUND and a written list of the correct lengths is approved by the Contracting Officer.

1.5 MEASUREMENT AND PAYMENT

- A. Linear Feet of pipe of each type and size designated, including, coupling bands, fittings, trenching, bedding, backfill and all other work needed to complete the item. When ends of a round pipe culvert are cut on a skew or slope, measurement will be the average of the top and bottom centerline lengths. When ends of a pipe arch culvert are cut on a slope, measurement will be the bottom centerline length.

PART 2 - PRODUCTS

2.1 STEEL PIPE AND PIPE ARCHES

- A. Corrugated Iron or Steel Pipe and Pipe Arches

1. Riveted Pipe and Pipe Arches: AASHTO M36.
2. Welded Pipe and Pipe Arches: Resistance spot welded, AASHTO M36.
3. Helical Pipe: Non-perforated helically corrugated pipe with continuous lock or welded seams, AASHTO M36.
4. Coupling Bands: AASHTO M36.
5. Special Sections (Elbows, tees, wyes, etc.): Same thickness as the conduit to which they are joined, AASHTO M36.
6. Flared End Sections: AASHTO M36. Fabricated as shown on the Drawings. Minor variations may be accepted to permit the use of the manufacturer's standard methods of fabrication.

B. Corrugated Iron or Steel Pipe For Underdrains

1. Plain Galvanized: AASHTO M36.
2. Precoated: AASHTO M245.
3. End Sections: AASHTO M218

C. Galvanized Steel End Sections: AASHTO M218.

2.2 BEDDING AND BACKFILL

- A. Bedding and Backfill Material: According to Section 312000 "Earthwork."

2.3 MATERIAL COMPATIBILITY

- A. The materials used in each pipe installation shall be compatible with each other to prevent electrolysis or physical failure. Either annular or helical pipe corrugations will be acceptable, but each pipe installation shall consist of only one class of corrugation.

PART 3 - EXECUTION

3.1 POLLUTION CONTROL

- A. Water pollution and stream degradation shall be controlled.
- B. Pipe which is installed in or which will affect streams shall be installed only during those periods SHOWN ON THE DRAWINGS.

3.2 EXCAVATION

- A. Excavation shall be according to Section 312000 "Earthwork" and the following:
1. Construct the width of trenches in natural ground to permit satisfactory joining of the culvert sections and thorough tamping of the bedding material under and around the culvert. Excavate trenches to a minimum width equal to the culvert diameter plus 2 feet.
 2. Construct trenches for culverts being placed in embankments to a width of one diameter, plus one diameter on each side.

3. Excavate unsuitable foundation material below the invert of the culvert to an approximate depth of 2 feet and a width of at least the culvert diameter plus 4 feet. Remove rock, hardpan, or other unyielding material below the foundation grade for a depth of at least 1 foot and a width of at least 2 feet greater than the outside width of the culvert.
4. Excavate to foundation grade without unduly disturbing the trench or foundation surface. Foundation grade is the elevation at the bottom of any bedding of the installation of the structure.
5. All suitable excavated material shall be utilized as backfill or embankment. No excavated material shall be placed in live streams. All surplus material shall be disposed of according to Section 024100 "Waste Material Disposal."

3.3 BEDDING

- A. The material shall be select mineral soil and shall have the moisture content needed for effective compacting. Bedding shall be readily compactable material free of frozen lumps, chunks of highly plastic clay, rocks larger than 3 inches in diameter or other objectionable material.
- B. Bed the pipe to a depth of not less than 10 percent of its total height. The foundation surface, after excavation, shall be compacted in accordance with paragraph 3.7 and shaped to fit the pipe.
- C. The completed bedding shall have a longitudinal camber when SHOWN ON THE DRAWINGS.
- D. The bedding material shall be selected mineral soil meeting the requirements for backfill in paragraph 3.7.

3.4 LAYING PIPE

- A. No pipe shall be placed in service until a suitable outlet is provided.
- B. Paved or partially lined pipe shall be laid so the longitudinal centerline of the paved segment coincides with the flowline.
- C. Elliptical pipe shall be placed with the major axis within 5 degrees of a vertical plane through the longitudinal axis of the pipe.
- D. The final installed alignment of all pipe shall have no sag and no point shall vary from a straight line drawn from inlet to outlet by more than 2 percent horizontally and vertically of the culvert length or 1 foot, whichever is less, unless otherwise SHOWN ON THE DRAWINGS.
- E. Helically corrugated lock-seam pipe shall be installed with the seam at the inlet end placed below the horizontal centerline.
- F. Longitudinal laps on riveted or spot-welded pipe shall be positioned at any location between 45 degrees above or below horizontal.
- G. Place outside circumferential laps facing upstream.

- H. Damaged spelter coating caused by welding, field cutting, or mishandling shall be cleaned and painted as specified in AASHTO M36.

3.5 JOINING PIPES

- A. Pipe shall be firmly joined by form-fitting coupling bands. End sections shall be attached to pipe by connecting bands or other means as recommended by the manufacturer. Rubber gaskets shall be installed at each joint to form a watertight connection when SHOWN ON THE DRAWINGS. Dimpled bands shall not be used when the slope of the pipe is greater than 25 percent.
- B. The coupling bands shall meet the strength requirements of field joints for Non-Erodible Soil Condition--Special Joint Type according to Division 2, Section 23 of the "Standard Specification for Highway Bridges" by AASHTO.

3.6 SHOP ELONGATION

- A. When SHOWN ON THE DRAWINGS, the vertical diameter of round pipe shall be increased 5 percent by shop elongation.

3.7 BACKFILLING

- A. Pipe meeting any of the following conditions shall not be placed or back-filled until the excavation and foundation have been approved by the Contracting Officer:
 - 1. Embankment height greater than 10 feet at subgrade centerline.
 - 2. Installation in a live stream.
 - 3. Round pipe with a diameter of 48 inches or greater.
 - 4. Pipe arches with a span of 50 inches or greater.
- B. After the bedding is prepared and the pipe is placed, backfill material shall be placed in layers not exceeding 6 inches loose thickness and compacted under the haunches and alongside the pipe.
- C. The material shall have the moisture content needed for effective compacting. Backfill shall be readily compactable material free of frozen lumps, chunks of highly plastic clay, or other objectionable material. Rocks larger than 3 inches in diameter shall not be used within 1 foot of the pipe. Backfill shall be compacted without damaging or displacing the pipe. The density shall be that of the surrounding embankment or 95 percent of the maximum density as determined by AASHTO T 99 Method C or D, whichever is greater. Backfilling and compacting shall be continued until the backfill is 12 inches above the top of the pipe. After being bedded and backfilled pipe shall be protected by an adequate cover of embankment before heavy equipment is permitted to cross during roadway construction. Pipe distorted more than 5 percent of nominal dimension, ruptured, or broken shall be replaced at the expense of the Contractor.

END OF SECTION 332228

April 2021